CHD Study Book

Module 5 - Congenital Heart Disease

Full lessons (C1 to C7) + interactive-visual diagrams + mock exam

C1 - Circulation and cyanosis: fetal circulation, the neonatal transition, the cyanosis threshold, ductal-dependent lesions and the DECREASED vs INCREASED pulmonary-flow framework (full lesson)

> Visual: `study/visuals/C1-circulation-cyanosis.html` (an interactive fetal-circulation diagram where clicking each shunt animates blood through it and shows its route + oxygen saturation, a first-breath transition card that animates the PVR drop and the shunts snapping shut, a cyanosis-threshold slider showing why anaemic patients cyanose late and polycythaemic patients cyanose early, a ductal-dependent + PGE1 card showing PGE1 holding the duct open vs indomethacin slamming it shut, and the master DECREASED-vs-INCREASED pulmonary-flow splitter). Directly tested (Master Exam): **Q12** in fetal circulation which vessels carry the HIGHEST oxygenated blood (answer **D, the umbilical VEIN**, which carries oxygenated blood from the placenta TO the fetus; the umbilical arteries carry deoxygenated blood AWAY). **Q6** a marked cyanotic defect with DECREASED pulmonary vascularity is treated first with (answer **C, Prostaglandin E1**, which keeps the ductus arteriosus OPEN so blood can still reach the lungs; indomethacin would CLOSE it). This lesson also builds the master framework (cyanotic CHD splits into decreased vs increased pulmonary blood flow) that C2 and C3 sit on top of. The **2018 congenital mock's written Q20** (list the components required to calculate PVR) is also owned here, see concept 6.

Step 0. The big theory: it is all one oxygen-delivery story

Congenital heart disease makes sense only if you first know the plumbing the baby is BORN with and how that plumbing is supposed to switch over at the first breath. Everything in this lesson hangs off five linked ideas, in order:

- **The fetus does not use its lungs.** The placenta is the oxygenator, so fetal blood is routed to BYPASS the lungs through three shunts (ductus venosus, foramen ovale, ductus arteriosus). The single most oxygenated vessel is the one coming FROM the placenta: the **umbilical vein**.
- **At birth the switch flips.** The first breath opens the lungs, pulmonary vascular resistance (PVR) collapses, the placenta is removed, and the three shunts close. Fetal parallel plumbing becomes adult series plumbing.
- **Cyanosis is an absolute number, not a percentage.** Blue is visible once there are about **3 to 5 g/dL of desaturated (reduced) haemoglobin**. That is why an anaemic baby can be dangerously hypoxic and still look pink, and a polycythaemic baby looks blue early.
- **Some hearts cannot survive the duct closing.** If the only path to the lungs (or to the body) runs through the ductus arteriosus, closing it is lethal, so you keep it open with **prostaglandin E1 (PGE1)**. Indomethacin does the exact opposite (it closes the duct).
- **Cyanotic disease sorts into two buckets by lung blood flow.** Either too LITTLE blood reaches the lungs (obstruction, oligaemic CXR) or too MUCH (mixing/transposition, plethoric CXR). This one split is the master map that C2 (decreased) and C3 (increased) are built on.

Hook: **"Placenta oxygenates, three shunts bypass the lungs, first breath drops the PVR and shuts the shunts, blue needs 5 g/dL of dead haemoglobin, keep the duct open with PGE1, and cyanotic CHD is either too little or too much lung blood flow."**

Step 1. The map: the five ideas of C1

These numbers are the SAME as the concept numbers in Step 2 (map row 1 = concept 1, and so on).

#	Idea	The one-line mechanism	Exam anchor
1	**Fetal circulation and the three shunts**	Placenta oxygenates, blood bypasses the lungs via **ductus venosus** (liver bypass), **foramen ovale** (RA to LA), **ductus arteriosus** (pulmonary artery to aorta). Most oxygenated vessel = **umbilical vein**	**Q12 = D (umbilical vein)** carries the highest oxygenated blood; umbilical arteries carry deoxygenated blood back to the placenta
2	**Fetal-to-neonatal transition**	First breath opens the lungs, **PVR falls**, placenta removed so **SVR rises**, LA pressure > RA pressure closes the **foramen ovale**, rising O2 and falling prostaglandins close the **ductus arteriosus**	The slow further PVR fall over 6 to 8 weeks is why left-to-right shunts present late (foreshadows C4)
3	**The cyanosis threshold**	Visible cyanosis needs about **3 to 5 g/dL of desaturated Hb** (an absolute amount). **Anaemic = cyanose LATE** (can be hypoxic and pink), **polycythaemic = cyanose EARLY**	Central cyanosis is unreliable in anaemia; a pink hypoxic baby is not reassured by colour
4	**Ductal-dependent lesions and PGE1**	If the lungs (or the body) can only be reached through the duct, keep it OPEN with **PGE1 (alprostadil)**. **Indomethacin / ibuprofen CLOSE** the duct (prostaglandin blockers)	**Q6 = C (PGE1)** for a cyanotic defect with DECREASED pulmonary flow; indomethacin is the impostor (it would kill the baby)
5	**The master framework: decreased vs increased pulmonary flow**	Cyanotic CHD splits by CXR lung fields: **DECREASED** flow (obstruction, oligaemic, ductal-dependent, needs PGE1) vs **INCREASED** flow (mixing/transposition, plethoric)	This is the spine of the whole module: C2 = decreased-flow lesions, C3 = increased-flow lesions
6	**PVR and how it is calculated**	**PVR = (mean PA pressure – mean LA / wedge pressure) / pulmonary flow (Qp)** = transpulmonary gradient / Qp, in **Wood units** (normal ~1 to 3; **> 3 = high**). Three measurements: mean PA pressure, mean LA (or PCWP / wedge), pulmonary flow Qp (by Fick)	**2018 mock Q20 (written):** list the components to calculate PVR. Also the operability number behind Glenn/Fontan (C6, Q2) and shunt closure

The one rule that ties it together: trace the oxygen. In the fetus the oxygen enters at the umbilical vein and is steered to the brain; at birth the lungs take over and the shunts close; if a heart cannot deliver oxygenated blood after birth it is either a duct problem (open it with PGE1) or a flow-distribution problem (decreased vs increased lung flow).

Step 2. The concepts you must know (numbered to match the map in Step 1: concept N = map row N)

1) Fetal circulation and the three shunts (owns Q12)

- **Synonym:** prenatal circulation; the fetal shunts (ductus venosus, foramen ovale, ductus arteriosus).
- **The core fact first (the exam point):** the **umbilical VEIN carries the most oxygenated blood in the whole fetus** (roughly 80 to 85% saturated), because it is the vessel bringing freshly oxygenated blood FROM the placenta TO the fetus. The **umbilical ARTERIES carry the LEAST oxygenated (deoxygenated) blood**, taking it from the fetus back to the placenta to be re-oxygenated. This is the reverse of the adult naming logic, where "artery" usually means oxygenated, so it is a classic trap.
- **Reads as (trace one red cell, this is the whole circuit):** 1. Oxygenated blood leaves the placenta in the **umbilical vein**. 2. It bypasses the fetal liver through the **ductus venosus**, going straight into the inferior vena cava (IVC), so the best blood is not wasted on the liver. 3. It enters the **right atrium**. Because the fetal lungs are useless, this well-oxygenated IVC stream is shunted right-to-left across the **foramen ovale** into the LEFT atrium, then to the left ventricle and out the **aorta**. This preferentially feeds the **coronaries and the brain** (the upper body) with the best available blood. 4. Deoxygenated blood returning from the head (superior vena cava) goes right atrium to right ventricle to pulmonary artery. But the lungs have very

HIGH resistance (collapsed, fluid-filled), so this blood does not go to the lungs. Instead it is shunted through the **ductus arteriosus** from the pulmonary artery into the DESCENDING aorta, supplying the lower body and then draining back down the **umbilical arteries** to the placenta.

- **The oxygen gradient to memorise (high to low):** umbilical vein (highest) > ductus venosus / IVC stream to the brain > ascending aorta (upper body) > descending aorta after the ductus arteriosus dumps in (lower body) > umbilical arteries (lowest). The brain gets better blood than the legs by design.
- **Why the two shunts matter:** the **foramen ovale** delivers the good blood to the brain; the **ductus arteriosus** offloads the deoxygenated right-heart blood away from the high-resistance lungs. Both exist ONLY because the fetal lungs are bypassed.
- **The fates after birth (what each shunt becomes, high-yield):** foramen ovale becomes the **fossa ovalis**; ductus arteriosus becomes the **ligamentum arteriosum**; ductus venosus becomes the **ligamentum venosum**; umbilical vein becomes the **ligamentum teres** (round ligament of the liver); umbilical arteries become the **medial umbilical ligaments**.
- **Example / exam anchor: Master Exam Q12** asks which vessels carry the highest oxygenated blood, with distractors aorta (mixed), umbilical arteries (deoxygenated), vitelline veins (drain the yolk sac, not the highest), and cardinal veins (drain the deoxygenated body). Answer = **D, the umbilical VEIN**.
- **Hook:** "Umbilical **VEIN** = best blood in from placenta; umbilical **arteries** = worst blood out. Three shunts: **ductus venosus** (skip the liver), **foramen ovale** (RA to LA, feed the brain), **ductus arteriosus** (pulmonary artery to aorta, skip the lungs)."

2) Fetal-to-neonatal transition: the first breath flips the switch

- **Synonym:** birth transition; the switch from parallel (fetal) to series (adult) circulation.
- **Reads as (the causal chain, in order):** 1. **First breath** inflates the lungs. Alveolar oxygen rises and the lungs go from collapsed and fluid-filled to open and air-filled. 2. That rising oxygen and lung expansion **relax the pulmonary arterioles**, so **pulmonary vascular resistance (PVR) falls sharply**. Pulmonary blood flow surges up. 3. **Clamping the cord removes the placenta**, which was a huge low-resistance circuit, so **systemic vascular resistance (SVR) rises**. 4. More blood now returns from the lungs to the LEFT atrium, so **LA pressure rises above RA pressure**. That pressure reversal pushes the flap of the **foramen ovale shut** (functional closure). It seals anatomically over months to become the fossa ovalis. 5. The **ductus arteriosus closes** for two reasons: the **rising arterial oxygen** constricts its muscular wall, and the **placental prostaglandins (PGE2) that kept it open are gone**. Functional closure happens over the first 24 to 72 hours; anatomical closure over weeks. Remember these two levers (oxygen up, prostaglandins down) because they are exactly what concept 4 manipulates. 6. The **ductus venosus closes** once umbilical flow stops.
- **The slow part that matters clinically:** PVR does not reach adult levels instantly. It keeps falling over about **6 to 8 weeks** as the thick fetal pulmonary arterioles remodel and thin out. This is why babies with a big left-to-right shunt (VSD, PDA) are often fine at birth and only get symptomatic at 4 to 8 weeks, once PVR has dropped enough to let a large left-to-right shunt develop. (You will use this directly in C4.)
- **Why it matters:** the two things you can pharmacologically grab are the SAME two levers that close the duct. Give oxygen or an NSAID and the duct closes; give prostaglandin and it stays open. Concept 4 is just this chain, run backwards on purpose.
- **Example / exam anchor:** a term newborn who is pink and stable at birth but collapses at day 2 to 3 as "the duct closes" is the classic story of a **ductal-dependent lesion** unmasking (leads straight into concept 4).
- **Hook:** "First breath: **PVR falls**, cord clamp: **SVR rises**, lung return floods the LA so the **foramen ovale slams shut**, and **oxygen up + prostaglandins down close the ductus arteriosus**. PVR keeps drifting down for 6 to 8 weeks."

3) The cyanosis threshold: about 3 to 5 g/dL of desaturated haemoglobin

- **Synonym:** the clinical threshold for visible (central) cyanosis.
- **The core fact:** cyanosis (the blue colour) becomes visible when there is roughly **3 to 5 g/dL of desaturated (reduced, deoxygenated) haemoglobin** in the blood. Crucially this is an **ABSOLUTE amount of grams, not a percentage**. Colour tracks the raw quantity of blue haemoglobin, not the saturation number.
- **Reads as (why the total haemoglobin flips the timing):** whether you reach that 5 g/dL of blue haemoglobin depends on how much haemoglobin you started with.
- **Polycythaemia (high Hb) = cyanose EARLY.** With Hb 20 g/dL you only need to desaturate 25% to already have 5 g/dL of reduced Hb, so the patient looks blue at a relatively HIGH oxygen saturation (around 75%). A polycythaemic patient looks worse than they are.
- **Anaemia (low Hb) = cyanose LATE, or never.** With Hb 6 g/dL you would have to desaturate more than 80% to accumulate 5 g/dL of reduced Hb, which means a saturation near 17%. In practice a severely anaemic patient can be profoundly hypoxic and still look PINK, because there is simply not enough haemoglobin to make 5 g/dL of it blue. A pink anaemic patient is NOT reassuring.
- Normal Hb (about 15 g/dL) needs roughly one third desaturation, so cyanosis appears around a saturation of 67%.
- **Why it matters (the trap):** never rule out hypoxia by colour in an anaemic patient, and do not overcall severity in a polycythaemic one. Central cyanosis is a specific but INsensitive sign, and its sensitivity depends entirely on the haemoglobin level. This is also why chronic cyanotic CHD patients become polycythaemic (the marrow makes more red cells to carry oxygen), which then makes them cyanose even more visibly.
- **Example / exam anchor:** the classic teaching point is the pair "anaemic patient cyanoses late, polycythaemic patient cyanoses early," and the number to quote is **3 to 5 g/dL of reduced haemoglobin** (an absolute amount).
- **Hook:** "Blue needs about **5 g/dL of DEAD (reduced) haemoglobin**, a raw amount not a percentage. **Lots of Hb (polycythaemia) = blue early; little Hb (anaemia) = blue late or never** (hypoxic but pink)."

4) Ductal-dependent lesions and prostaglandin E1 (owns Q6)

- **Synonym:** duct-dependent circulation; PGE1 = alprostadil.
- **The core idea:** in some defects the baby survives only because the **ductus arteriosus is still open**, so when the duct closes on day 1 to 3 (concept 2) the baby crashes. There are two flavours:
- **Ductal-dependent PULMONARY circulation:** the route to the LUNGS is blocked (pulmonary atresia, critical pulmonary stenosis, tricuspid atresia, severe tetralogy). The only way blood reaches the lungs is aorta to ductus arteriosus to pulmonary artery. Duct closes, lung blood flow vanishes, **severe cyanosis**. These are the DECREASED-pulmonary-flow lesions of concept 5.
- **Ductal-dependent SYSTEMIC circulation:** the route to the BODY is blocked (hypoplastic left heart, critical coarctation, critical aortic stenosis, interrupted aortic arch). The body is fed backwards from the pulmonary artery through the duct into the descending aorta. Duct closes, the body loses its blood supply, **shock and collapse** (grey, absent femoral pulses, acidosis).
- **Reads as (why PGE1 works, straight off concept 2):** the two levers that CLOSE the duct after birth are rising oxygen and falling prostaglandins. So to keep it OPEN you replace the prostaglandin: **prostaglandin E1 (PGE1, alprostadil) keeps the ductus arteriosus patent**, restoring pulmonary (or systemic) blood flow and buying time until surgery or catheter intervention. It is the emergency drug for a sick, ductal-dependent newborn.
- **The mirror-image drug (the trap): indomethacin and ibuprofen are prostaglandin-synthesis inhibitors (NSAIDs) that CLOSE the duct.** They are used deliberately to close a persistent PDA in a

PREMATURE infant. Giving indomethacin to a ductal-dependent baby would slam the duct shut and be fatal. It is the exact opposite of what you want, which is why it is the wrong-answer bait in Q6.

- **Why the other Q6 options are wrong: digoxin** treats heart failure but does nothing to the duct; **epinephrine** is an inotrope/pressor and does not open the duct. Only **PGE1** addresses the actual problem (the closing duct).
- **Example / exam anchor: Master Exam Q6.** An infant with a marked cyanotic defect and DECREASED pulmonary vascularity has a ductal-dependent PULMONARY circulation, so the first treatment is **C, prostaglandin E1** (keep the duct open, maintain lung blood flow). Indomethacin (B) is the impostor.
- **Hook:** "Duct closing kills a ductal-dependent baby. **PGE1 (alprostadil) = KEEP the duct OPEN** (replace the prostaglandin). **Indomethacin / ibuprofen = CLOSE the duct** (block the prostaglandin, used for PDA in preemies). PGE1 saves, indomethacin kills."

5) The master framework: cyanotic CHD splits into DECREASED vs INCREASED pulmonary blood flow

- **Synonym:** the pulmonary-vascularity approach to cyanotic CHD; oligaemic vs plethoric lung fields.
- **Reads as:** every cyanotic newborn has deoxygenated blood reaching the aorta, but you sort the lesions by ONE question answered on the chest X-ray: **is there too little or too much blood going to the lungs?**
- **DECREASED pulmonary blood flow (oligaemic, dark lung fields):** something OBSTRUCTS flow to the lungs, so less blood gets oxygenated. These are typically **ductal-dependent for pulmonary flow and need PGE1** (concept 4). Lesions: **tetralogy of Fallot, tricuspid atresia, pulmonary atresia, critical pulmonary stenosis, severe Ebstein**. This whole bucket is **C2**.
- **INCREASED pulmonary blood flow (plethoric, congested lung fields):** blood reaches the lungs fine (often too much) but the OXYGENATED and deoxygenated blood are not being separated properly, because of transposition or complete mixing. Lesions: **D-transposition of the great arteries (D-TGA), total anomalous pulmonary venous return (TAPVR), truncus arteriosus, single ventricle, double-outlet right ventricle**. This whole bucket is **C3**.
- **Why it matters:** this single split is the exam's spine for the cyanotic lesions. Given a blue baby, the very first sorting move is "what do the lung fields show?" Decreased points you toward obstructed, PGE1-dependent lesions (TOF and friends); increased points you toward mixing/transposition lesions (TGA and friends). It also connects straight back: the DECREASED bucket IS the ductal-dependent-pulmonary group that Q6 is testing.
- **The cyanotic lesions are "the 5 Ts" (numbered by the number in the T):** 1 = **Truncus** arteriosus (one trunk), 2 = **Transposition** (D-TGA, two arteries swapped), 3 = **Tricuspid atresia** (three cusps), 4 = **Tetralogy** of Fallot (four features), 5 = **TAPVR** (total anomalous pulmonary venous return). Then split them by flow: Tricuspid atresia and Tetralogy sit in the DECREASED bucket (obstruction); Truncus, Transposition and TAPVR sit in the INCREASED bucket (mixing).
- **What each of the 5 Ts actually does (brief placeholder, full mechanism comes in C2 and C3):** every cyanotic lesion gets blue blood into the aorta, either by a BLOCK to the lungs that forces a right-to-left shunt (decreased flow) or by misrouted / mixed plumbing with no block (increased flow).
- **Tricuspid atresia (decreased):** no tricuspid valve, so the RA cannot reach the RV or PA. Blood detours RA to ASD to LA to LV, and the lungs are under-supplied (often duct-dependent). Dark lungs.
- **Tetralogy of Fallot (decreased):** four parts, pulmonary stenosis (RVOT block) + VSD + overriding aorta + RVH. The RVOT block shunts blue blood right-to-left across the VSD into the aorta. Dark lungs.
- **Truncus arteriosus (increased):** one common great vessel over a VSD instead of a separate aorta and PA, so red and blue mix in the trunk and flood the lungs. Congested lungs.
- **Transposition (D-TGA) (increased):** the aorta comes off the RV and the PA off the LV (swapped), giving two parallel loops that survive only with mixing (ASD / VSD / PDA). Congested lungs.

- **TAPVR (increased):** the pulmonary veins drain to the wrong side (into the right heart, not the LA), so oxygenated blood returns to the RA and mixes, and an ASD is needed to feed the body. Congested lungs.
- **The shortcut:** the two with a pulmonary / RVOT obstruction (Tricuspid atresia, Tetralogy) = decreased flow; the three that are a wiring / mixing problem with no obstruction (Truncus, Transposition, TAPVR) = increased flow.
- **Example / exam anchor:** the framing "cyanotic defect with DECREASED pulmonary vascularity" in Q6 is exactly this framework. It tells you the lesion is an obstructed, ductal-dependent-pulmonary one, which is why the answer is PGE1.
- **Hook:** "Blue baby, look at the lung fields. **DECREASED (dark) = obstruction, ductal-dependent, PGE1** (TOF, tricuspid atresia, pulmonary atresia) = C2. **INCREASED (congested) = mixing/transposition** (TGA, TAPVR, truncus, single ventricle) = C3. The 5 Ts: 1 Truncus, 2 Transposition, 3 Tricuspid atresia, 4 Tetralogy, 5 TAPVR."

6) Pulmonary vascular resistance (PVR) and how it is calculated (owns the 2018 written Q20)

- **Synonym:** PVR; pulmonary arteriolar resistance; Wood units; the transpulmonary gradient.
- **Why it earns its own concept:** PVR is the single number that decides **operability** across the whole CHD module. Whether you can safely CLOSE a shunt, and whether a Glenn or Fontan will work, both hinge on PVR being low enough. The 2018 congenital mock asks it directly as a **written question: "the components required for calculation of PVR."**
- **The formula (memorise the shape):** resistance = the pressure drop across a circuit divided by the flow through it. For the lungs that is **PVR = (mean pulmonary artery pressure – mean left atrial pressure) / pulmonary blood flow (Qp)** = **transpulmonary pressure gradient (TPG) / Qp**.
- **The three components you must be able to list (this IS the exam answer):** 1. **Mean pulmonary artery pressure (mPAP)** — measured directly at right-heart catheterisation. 2. **Mean left atrial pressure (mLAP)**, or its stand-in the **pulmonary capillary wedge pressure (PCWP)** (LV end-diastolic pressure can substitute). mPAP minus this is the **transpulmonary gradient**, the true pressure lost across the lung vasculature. 3. **Pulmonary blood flow (Qp)** — the cardiac output through the lungs, obtained by the **Fick principle** (oxygen consumption VO2 divided by the pulmonary arteriovenous oxygen content difference) or by **thermodilution**.
- **Units: Wood units (mmHg/L/min)**; multiply by **80** to convert to dyn·s·cm^{-5}. Normal PVR is about **1 to 3 Wood units**; **> 3 Wood units is raised**.
- **Why it matters (it recurs all over the module):** a high PVR means the lungs are a high-resistance circuit, so a passive (Glenn/Fontan) connection cannot push blood through, and closing a long-standing left-to-right shunt risks fixed pulmonary hypertension (Eisenmenger). This is exactly the number behind **Q2's Glenn prerequisites** (mean PA pressure around 14, **PVR < 3 Wood units**, a low LV end-diastolic pressure), which you meet again in C6.
- **Hook:** "**PVR = (mean PA – mean LA/wedge) / Qp**, in **Wood units** (> 3 = high). Three numbers: **mean PA pressure, mean LA (or wedge), and pulmonary flow Qp (by Fick).** It is the operability number for shunt closure and for Glenn/Fontan."

⚑ Treated differently (the exceptions to flag)

- **Umbilical VEIN, not artery, carries the best blood.** In the fetus the naming flips: the vein brings oxygenated blood from the placenta, the arteries take deoxygenated blood back. Do not fall for "arteries are oxygenated." This is the whole of **Q12 (answer = umbilical vein)**.
- **Cyanosis is an absolute number, not a saturation.** 3 to 5 g/dL of reduced haemoglobin makes blue, so the SAME hypoxia looks blue in a polycythaemic patient and pink in an anaemic one. An anaemic patient can be severely hypoxic and NOT cyanosed. Never clear hypoxia by colour without knowing the Hb.

- **PGE1 and indomethacin are exact opposites on the duct.** PGE1 OPENS/holds the duct (for ductal-dependent CHD); indomethacin/ibuprofen CLOSE it (to shut a PDA in a preterm baby). Same organ, opposite drugs. Q6 bait is indomethacin.
- **Ductal-dependent PULMONARY vs SYSTEMIC present differently.** Duct closing in a ductal-dependent-pulmonary lesion (obstructed lung flow) gives sudden CYANOSIS; in a ductal-dependent-systemic lesion (obstructed body flow, e.g. hypoplastic left heart, critical coarctation) it gives SHOCK with absent femoral pulses and acidosis. Both are rescued by PGE1, but the presentation differs.
- **A late-presenting shunt is not a stable heart, it is a falling PVR.** Left-to-right shunt babies (VSD, PDA) are often asymptomatic at birth and only decompensate at 4 to 8 weeks because PVR keeps falling for 6 to 8 weeks after birth, progressively increasing the left-to-right flow. Do not read the quiet first days as a normal heart.
- **Tricuspid atresia is a "T" but sits in the DECREASED bucket.** The 5 Ts are all cyanotic, but the flow split is not by the mnemonic: Tricuspid atresia and Tetralogy are DECREASED-flow (obstruction), while Truncus, Transposition and TAPVR are INCREASED-flow (mixing). Sort by flow, not by the letter.

Mnemonics to lock in

- **The whole lesson in one line: "Placenta oxygenates, three shunts bypass the lungs, first breath drops PVR and shuts the shunts, blue needs 5 g/dL of dead Hb, PGE1 keeps the duct open, cyanotic CHD is too little or too much lung flow."**
- **The three fetal shunts (in flow order): "Ductus VENOSUS skips the liver, foramen OVALE feeds the brain (RA to LA), ductus ARTERIOSUS skips the lungs (pulmonary artery to aorta)."**
- **Fetal oxygen naming trap: "Umbilical VEIN = rich (from placenta). Umbilical ARTERIES = poor (to placenta)."**
- **Shunt fates after birth: "Foramen ovale to fossa ovalis, ductus arteriosus to ligamentum arteriosum, ductus venosus to ligamentum venosum, umbilical vein to ligamentum teres, umbilical arteries to medial umbilical ligaments."**
- **Birth transition: "First breath: PVR DOWN. Cord clamp: SVR UP. LA pressure up shuts the FORAMEN OVALE. Oxygen UP + prostaglandins DOWN shut the DUCTUS ARTERIOSUS."**
- **Cyanosis threshold: "5 g/dL of reduced Hb = blue. Polycythaemia = blue EARLY. Anaemia = blue LATE (hypoxic but pink)."**
- **The duct drugs: "PGE1 = Prostaglandin, keeps it PATENT (open). Indomethacin = INhibitor, closes it. Open saves a ductal-dependent baby; close is for a preemie PDA."**
- **Master framework: "Blue baby, check lung fields. DECREASED = obstruction, ductal-dependent, PGE1 (C2). INCREASED = mixing/transposition (C3)."**
- **The 5 Ts (by the number in the T): "1 Truncus, 2 Transposition, 3 Tricuspid atresia, 4 Tetralogy, 5 TAPVR."**
- **PVR calculation (2018 Q20): "PVR = (mean PA – mean LA/wedge) / Qp, in Wood units (> 3 = high). Three inputs: mean PA pressure, mean LA (or wedge) pressure, pulmonary flow Qp (by Fick)."**

🧪 Quiz yourself (commit your answer, key is sealed at the bottom)

Q12. In fetal circulation, which vessels carry the highest oxygenated blood? A. Aorta B. Umbilical arteries C. Vitelline veins D. Umbilical veins E. Cardinal veins

Q6. An infant with a marked cyanotic congenital heart defect with DECREASED pulmonary vascularity should be initially treated with: A. Digoxin B. Indomethacin C. Prostaglandin E1 D. Epinephrine

Q-A (fetal shunts). Which fetal shunt carries relatively oxygenated blood from the right atrium into the left atrium, directing the best-oxygenated blood toward the brain? A. Ductus venosus B. Foramen ovale C. Ductus arteriosus D. Umbilical vein E. Coronary sinus

Q-B (transition / PVR). Immediately after the first breath, the main haemodynamic change that promotes closure of the foramen ovale is: A. A rise in pulmonary vascular resistance B. A fall in systemic vascular resistance C. A fall in pulmonary vascular resistance that raises left atrial pressure above right atrial pressure D. A rise in placental blood flow E. A fall in arterial oxygen tension

Q-C (cyanosis threshold). Compared with a patient who has a normal haemoglobin, a severely ANAEMIC patient with the same degree of hypoxaemia will: A. Appear cyanosed at a higher oxygen saturation B. Appear cyanosed at the same oxygen saturation C. Appear cyanosed only at a much lower oxygen saturation, or not at all D. Never be hypoxic E. Have more reduced haemoglobin per litre

Q-D (PGE1 vs indomethacin). In a term neonate with pulmonary atresia and a closing duct, giving indomethacin instead of prostaglandin E1 would: A. Have no effect on the duct B. Keep the ductus arteriosus open C. Close the ductus arteriosus and worsen the cyanosis D. Raise the pulmonary blood flow E. Treat the underlying atresia

Q-E (PVR calculation, 2018 mock Q20, written). Write down the components (measurements) required to calculate pulmonary vascular resistance, the formula that combines them, and the units.

🔒 Answer key (only after you have committed)

- **Q12 = D (Umbilical veins).** The umbilical VEIN carries the most oxygenated blood in the fetus (about 80 to 85% saturated) because it brings freshly oxygenated blood FROM the placenta TO the fetus. The umbilical ARTERIES (B) carry the LEAST oxygenated blood, taking it back to the placenta. The aorta (A) is mixed, the vitelline veins (C) drain the yolk sac, and the cardinal veins (E) drain the deoxygenated body. The fetal naming is the reverse of adult intuition, and that reversal is the trap.
- **Q6 = C (Prostaglandin E1).** A marked cyanotic defect with DECREASED pulmonary vascularity is an obstructed, ductal-dependent-PULMONARY lesion (pulmonary atresia, critical pulmonary stenosis, tricuspid atresia, severe TOF): the only path to the lungs is through the ductus arteriosus. **PGE1 (alprostadil) keeps the duct OPEN**, maintaining pulmonary blood flow until definitive repair. **Indomethacin (B) would CLOSE the duct** (it blocks prostaglandin synthesis) and be fatal, so it is the impostor. Digoxin (A, heart failure) and epinephrine (D, inotrope) do nothing to the duct.
- **Q-A = B (Foramen ovale).** The foramen ovale shunts the relatively oxygenated IVC/ductus-venosus stream from the RIGHT atrium into the LEFT atrium, so the best blood goes left ventricle to aorta to the coronaries and BRAIN. The ductus venosus (A) bypasses the liver before the heart; the ductus arteriosus (C) shunts deoxygenated pulmonary-artery blood into the descending aorta (away from the lungs); the umbilical vein (D) is upstream, coming from the placenta.
- **Q-B = C (Fall in PVR that raises LA pressure above RA pressure).** The first breath opens the lungs and drops pulmonary vascular resistance, so pulmonary blood flow surges and pulmonary venous return to the LEFT atrium rises. LA pressure now exceeds RA pressure, pushing the flap of the foramen ovale shut. A does the opposite, E (falling oxygen) would keep the duct open not close the foramen, and removing the placenta RAISES SVR (so B is wrong as stated).

- **Q-C = C (Cyanosed only at a much lower saturation, or not at all).** Cyanosis needs an ABSOLUTE 3 to 5 g/dL of reduced haemoglobin. A severely anaemic patient has little total haemoglobin, so even profound hypoxaemia may not produce 5 g/dL of reduced Hb, and they stay pink (cyanose late or never). The polycythaemic patient is the opposite (cyanose early, at a higher saturation). E is wrong: the anaemic patient has LESS reduced Hb per litre, which is exactly why they cyanose late.
- **Q-D = C (Close the ductus arteriosus and worsen the cyanosis).** Indomethacin is a prostaglandin-synthesis inhibitor, so it CLOSES the duct. In a ductal-dependent-pulmonary lesion like pulmonary atresia, the duct is the only route to the lungs, so closing it removes pulmonary blood flow and worsens the cyanosis (potentially fatal). PGE1 is the correct drug: it keeps the duct open. Indomethacin is used to close a PDA in PREMATURE infants, the opposite clinical setting.
- **Q-E (PVR calculation). PVR = (mean pulmonary artery pressure – mean left atrial pressure) / pulmonary blood flow (Qp)** = transpulmonary gradient / Qp. The measurements required: **(1) mean pulmonary artery pressure**, **(2) mean left atrial pressure (or the pulmonary capillary wedge pressure as its surrogate)**, and **(3) pulmonary blood flow Qp** (by the Fick principle, VO2 divided by the pulmonary arteriovenous O2 content difference, or by thermodilution). Units are **Wood units (mmHg/L/min)** (multiply by 80 for $dyn \cdot s \cdot cm^{-5}$); normal is about 1 to 3, and **> 3 Wood units is raised**. This is the same PVR threshold that gates Glenn/Fontan operability (Q2) and shunt closure.

Diagrams in the visual (hand-built inline SVG, nothing embedded from the web)

1. **Interactive fetal-circulation schematic:** a hand-drawn SVG of the placenta, umbilical vein/arteries, liver with the ductus venosus, the two atria with the foramen ovale, the ventricles, and the pulmonary artery to aorta ductus arteriosus. Clicking each shunt highlights its route, prints its oxygen saturation, and animates a blood dot travelling through that shunt inline. Colour codes oxygen saturation (red = high, purple = mixed, blue = low). No web image is used, so nothing needs source verification.
2. **First-breath transition animation:** a two-bar PVR/SVR meter that, on clicking "take the first breath", drops the PVR bar and raises the SVR bar while the foramen ovale and ductus arteriosus visibly snap shut. Pure CSS/JS, self-contained.
3. **Cyanosis-threshold slider:** a haemoglobin slider (anaemic to polycythaemic) that computes the oxygen saturation at which 5 g/dL of reduced haemoglobin is reached, showing the patient turning blue early (polycythaemia) or staying pink despite hypoxia (anaemia).
4. **Ductal-dependent + PGE1 card:** a duct that opens under PGE1 (prostaglandin replaced) and closes under indomethacin/oxygen (prostaglandin blocked), with the pulmonary blood-flow response.
5. **Master-framework splitter:** the DECREASED vs INCREASED pulmonary-flow buckets with their lesion lists, tied back to C2 and C3.

C1 - Interactive visual (rendered): diagrams & schematics

The live, toggleable version is the HTML file on the CHD hub. Below is a static render of its default view; open the hub to step through the other toggle states.

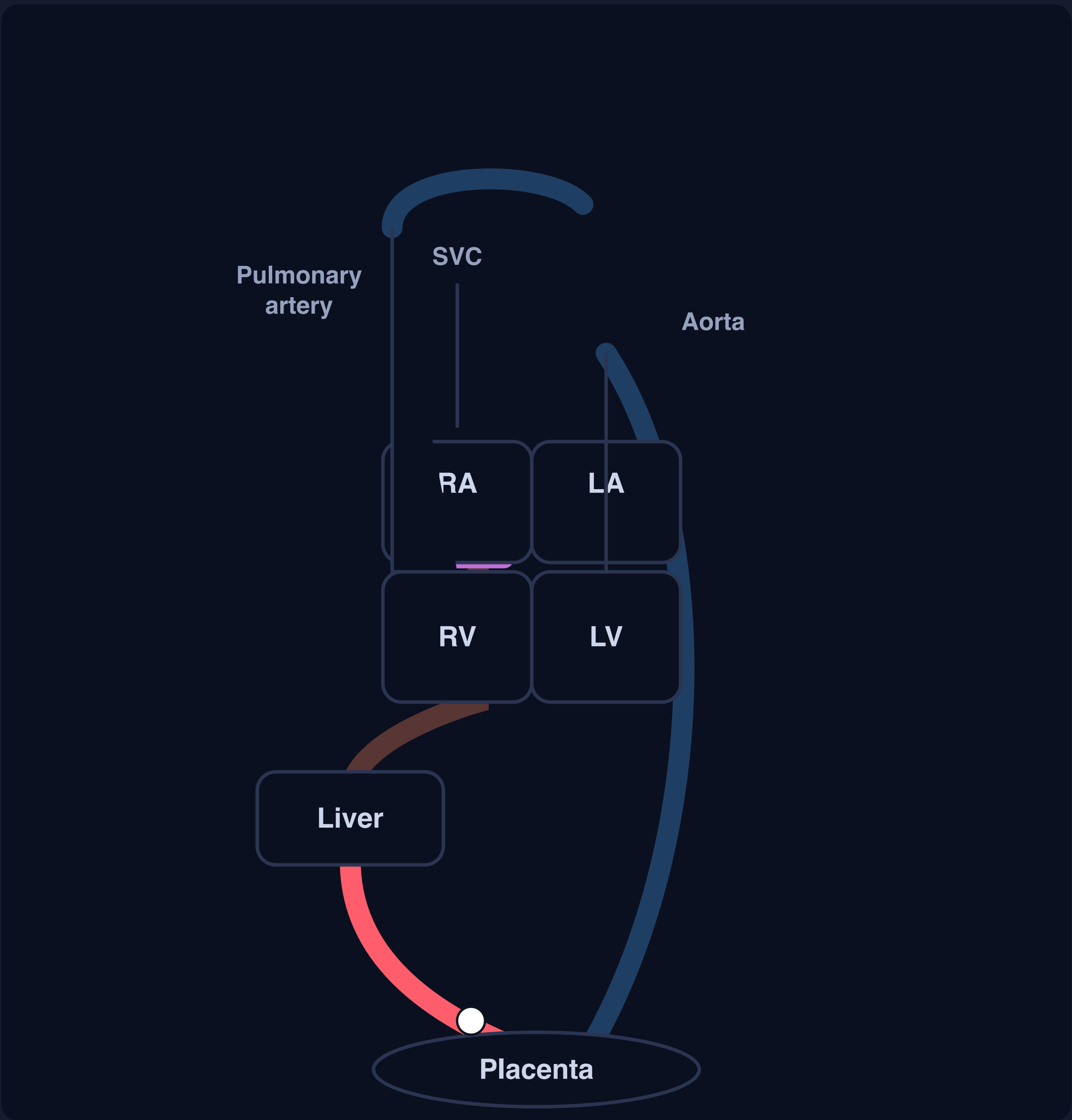
C1 - Circulation and cyanosis

Fetal shunts, the first-breath transition, the cyanosis threshold, and PGE1. Exam anchors: Q12 the umbilical VEIN carries the highest oxygenated blood; Q6 a cyanotic defect with DECREASED pulmonary flow is treated first with PGE1 (keeps the duct open), never indomethacin.

1. FETAL CIRCULATION: CLICK A SHUNT TO TRACE IT (ROUTE + O2 SATURATION ANIMATE INLINE)

Umbilical vein (in from placenta) | Ductus venosus (skip liver) | Foramen ovale (RA to LA) | Ductus arteriosus (skip lungs)

Umbilical arteries (out to placenta)



Colour = oxygen saturation: red high, purple mixed, blue low. Hand-drawn schematic, nothing embedded from the web.

Umbilical vein

~ 80 to 85% (HIGHEST)

The most oxygenated vessel in the fetus. Carries fresh oxygenated blood FROM the placenta TO the fetus. This is Q12: the answer is the umbilical VEIN.

Becomes the **ligamentum teres** after birth.

Q12: the umbilical VEIN carries the HIGHEST oxygenated blood (from placenta to fetus). The umbilical arteries carry the LOWEST (deoxygenated, back to placenta). Aorta = mixed, vitelline veins = yolk sac, cardinal veins = deoxygenated body.

2. THE FIRST BREATH: WATCH PVR DROP AND THE SHUNTS SNAP SHUT

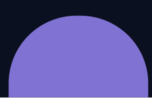
Take the first breath | Reset to fetal state

Foramen ovale: OPEN | Ductus arteriosus: OPEN | Ductus venosus: OPEN

Fetal state: lungs collapsed, PVR high, most blood bypasses the lungs. Click "first breath" to run the switch.

Chain: first breath **drops PVR**, cord clamp **raises SVR**, lung return raises LA pressure above RA so the **foramen ovale shuts**, and **rising O2 + falling prostaglandins shut the ductus arteriosus**. PVR keeps drifting down for **6 to 8 weeks** (why left-to-right shunts present late).

3. THE CYANOSIS THRESHOLD: WHY ANAEMIC PATIENTS CYANOSE LATE, POLYCYTHAEMIC EARLY



Cyanoses at SpO2 ~67%

Total haemoglobin: **15** g/dL

anaemic 4 — 22 polycythaemic

Cyanosis appears once SpO2 falls to about 67% (that is where reduced Hb = 5 g/dL). This patient cyanoses **around this saturation**.

> Cyanosis needs an **absolute 3 to 5 g/dL of reduced (blue) haemoglobin**, not a saturation. Formula: reduced Hb = Hb x (1 - SpO2). Solve for 5 g/dL: **SpO2 = 1 - 5/Hb**.

> **Polycythaemia = blue EARLY** (reaches 5 g/dL of reduced Hb at a high saturation). **Anaemia = blue LATE or never** (too little Hb to make 5 g/dL blue, so hypoxic but PINK). Never clear hypoxia by colour without knowing the Hb.

4. DUCTAL-DEPENDENT LESION: PGE1 OPENS THE DUCT, INDOMETHACIN CLOSES IT

Give PGE1 (replace prostaglandin) | **Give indomethacin / O2 (block prostaglandin)**

Duct OPEN (PGE1)

PGE1 (alprostadil) replaces the prostaglandin that keeps the duct open. Blood flows aorta to duct to pulmonary artery, so the lungs are perfused.

The emergency drug for a sick ductal-dependent newborn. Buys time until surgery.

Lungs perfused: cyanosis relieved

> **Q6: cyanotic defect + DECREASED pulmonary flow = ductal-dependent-pulmonary lesion. First treatment = Prostaglandin E1 (keep the duct OPEN). Indomethacin is the impostor (it CLOSES the duct, used for PDA in preemies). Digoxin and epinephrine do nothing to the duct.**

5. MASTER FRAMEWORK: CYANOTIC CHD SPLITS BY PULMONARY BLOOD FLOW (SPINE OF C2 AND C3)

Blue baby: look at the chest X-ray lung fields and sort into ONE of two buckets.

DECREASED (oligaemic, dark) = C2

Too little lung blood flow: OBSTRUCTION

Blocked path to the lungs, usually **ductal-dependent, needs PGE1**.

- Tetralogy of Fallot
- Tricuspid atresia
- Pulmonary atresia
- Critical pulmonary stenosis
- Severe Ebstein anomaly

INCREASED (plethoric, congested) = C3

Too much lung blood flow: MIXING / TRANSPOSITION

Blood reaches the lungs fine but is not separated from deoxygenated blood.

- D-transposition of the great arteries (D-TGA)
- Total anomalous pulmonary venous return (TAPVR)
- Truncus arteriosus
- Single ventricle
- Double-outlet right ventricle (DORV)

> **The 5 Ts (numbered by the number in the T):** 1 Truncus, 2 Transposition (D-TGA), 3 Tricuspid atresia, 4 Tetralogy, 5 TAPVR. Sort by FLOW not by letter: Tricuspid atresia and Tetralogy are DECREASED (obstruction); Truncus, Transposition and TAPVR are INCREASED (mixing).

What each of the 5 Ts does (brief placeholder, full mechanism comes in C2 and C3): every cyanotic lesion gets blue blood into the aorta, either by a BLOCK to the lungs that forces a right-to-left shunt (decreased) or by misrouted / mixed plumbing with no block (increased).
Tricuspid atresia (dec): no tricuspid valve, the RA cannot reach the RV or PA; blood detours RA to ASD to LA to LV and the lungs are under-supplied.
Tetralogy of Fallot (dec): pulmonary stenosis (RVOT block) + VSD + overriding aorta + RVH; the block shunts blue blood right-to-left across the VSD into the aorta.
Truncus arteriosus (inc): one common great vessel over a VSD instead of a separate aorta and PA; red and blue mix in the trunk and flood the lungs.
Transposition, D-TGA (inc): aorta off the RV and PA off the LV (swapped); two parallel loops that survive only with mixing (ASD / VSD / PDA).
TAPVR (inc): the pulmonary veins drain to the wrong side (into the right heart, not the LA); oxygenated blood returns to the RA and mixes, and an ASD is needed to feed the body.
Shortcut: the two with a pulmonary / RVOT obstruction (Tricuspid atresia, Tetralogy) = decreased; the three wiring / mixing problems (Truncus, Transposition, TAPVR) = increased.

C1 visual - Module 5 CHD. All diagrams hand-drawn inline SVG, fully self-contained, no external assets.

C2 - Cyanotic CHD with DECREASED pulmonary blood flow: the physiology of right-heart outflow obstruction, tetralogy of Fallot (the anchor) and its tet spell, tricuspid atresia and pulmonary atresia (full lesson)

> Visual: `study/visuals/C2-cyanotic-decreased-flow.html` (the DECREASED-flow physiology splitter that carries over from C1, an interactive tetralogy-of-Fallot heart where clicking each of the 4 components highlights it on the anatomy and explains it, a hand-drawn boot-shaped-heart chest-X-ray schematic with the upturned apex and concave pulmonary bay, a tet-spell animation where SVR drops so the right-to-left shunt grows and the baby turns bluer and clicking a treatment reverses the cycle inline, and a TOF vs tricuspid-atresia axis card contrasting right axis with left axis). This is the DECREASED-pulmonary-flow branch of the master framework built in C1 (cyanotic CHD splits into DECREASED vs INCREASED lung blood flow). Directly tested (Master Exam): **Q8** a tet spell is treated with all of the following EXCEPT (answer **D, digoxin**, because a positive inotrope increases contractility and WORSENS the dynamic right-ventricular-outflow obstruction; everything else raises SVR or relaxes the infundibulum). **Q19** a 6-hour-old neonate, cyanosis worse with crying, ECG axis 120 degrees (right axis deviation) with RV prominence, boot-shaped heart on CXR (answer **A, tetralogy of Fallot**; the right axis is what nails TOF over tricuspid atresia, which classically has LEFT axis). This lesson also ties back to C1's PGE1: the severest of these lesions are ductal-dependent for pulmonary flow.

Step 0. The big theory: block the road to the lungs and blood turns right

In C1 you learned the one sorting question for a blue baby: what do the lung fields show? This lesson is the whole DECREASED-flow (dark, oligaemic lung field) bucket, and every lesion in it is one physiological story. Everything hangs off five linked ideas, in order:

- **Obstruct the right heart outflow and the blood has to go somewhere.** If the path from the right ventricle to the lungs is narrowed or blocked (infundibular or valvar pulmonary stenosis, pulmonary atresia, or no functioning right ventricle at all in tricuspid atresia), right-heart pressure rises above left-heart pressure and deoxygenated blood is pushed the wrong way, **right-to-left across a VSD or the foramen ovale**, straight into the aorta. That right-to-left shunt is what makes the baby blue, and the blocked lung path is what makes the lung fields dark.
- **Tetralogy of Fallot is the anchor, and it is ONE lesion, not four.** A single embryological error (the outflow septum deviates anteriorly) produces all four features at once: a **VSD**, an **overriding aorta**, **right-ventricular outflow (infundibular) obstruction**, and **right-ventricular hypertrophy**. The obstruction plus the VSD is the machine that makes the right-to-left shunt.
- **The tet spell is dynamic, not fixed.** The infundibular muscle can SPASM, and systemic resistance can FALL (crying, feeding, waking, dehydration). Both deepen the right-to-left shunt suddenly, so the baby goes acutely, dangerously blue. The fix is the mirror image: **raise SVR and relax the infundibulum**. Anything that increases contractility (digoxin) makes the dynamic obstruction WORSE, which is the whole trap in Q8.
- **Squatting and the knee-chest position are free SVR.** A toddler with TOF squats, and you hold an infant knees-to-chest, for the same reason: it **kinks the femoral arteries and raises systemic vascular**

resistance, so the aortic pressure now exceeds right-heart pressure and the shunt reverses back toward left-to-right. Boring physiology, life-saving.

- **The severest of these lesions are ductal-dependent (back to C1).** Pulmonary atresia, critical pulmonary stenosis and severe TOF can only reach the lungs through the ductus arteriosus, so when the duct closes on day 1 to 3 the baby crashes. **PGE1 keeps the duct open** (C1 concept 4); indomethacin would be fatal.

Hook: **"Block the road to the lungs, pressure backs up, blood shunts right-to-left across the VSD into the aorta, and the baby is blue with dark lungs. TOF is one error making four features. A tet spell is dynamic obstruction plus falling SVR, so you RAISE SVR (squat, knee-chest, phenylephrine) and RELAX the infundibulum (O2, morphine, propranolol), never digoxin. The worst of these are duct-dependent, so keep the duct open with PGE1."**

Step 1. The map: the five ideas of C2

These numbers are the SAME as the concept numbers in Step 2 (map row 1 = concept 1, and so on).

#	Idea	The one-line mechanism	Exam anchor
1	**The DECREASED-flow physiology**	Right-heart outflow obstruction raises right-side pressure above left, so blood shunts **right-to-left across a VSD / foramen ovale** into the aorta. Result: **cyanosis + oligaemic (dark) lung fields**. Severest are **ductal-dependent, need PGE1**	This is the C1 DECREASED bucket. The lesions are **TOF, tricuspid atresia, pulmonary atresia, critical pulmonary stenosis, severe Ebstein**. Ductal-dependent ones tie back to **Q6 (PGE1)**
2	**Tetralogy of Fallot (the anchor)**	One embryological error (anterior outflow-septum deviation) gives **4 features: VSD, overriding aorta, RVOT / pulmonary stenosis, RVH**. On tests: **boot-shaped heart, decreased pulmonary vascularity, right axis deviation (RAD) + RVH**	**Q19 = A (TOF)**: cyanosis worse with crying, RAD (axis 120), RV prominence, boot-shaped heart with dark lungs. RAD is what nails TOF over tricuspid atresia
3	**The tet spell and its management**	A hypercyanotic spell = **dynamic infundibular spasm + a fall in SVR**, deepening the right-to-left shunt. Fix = **raise SVR** (knee-chest / squat, phenylephrine, fluids) and **relax the infundibulum / break the cycle** (O2, morphine, propranolol, bicarbonate)	**Q8 = D (digoxin) is the EXCEPTION**: a positive inotrope increases contractility and WORSENS the dynamic RVOT obstruction. Everything else on the list helps
4	**Tricuspid atresia (the axis discriminator)**	No tricuspid valve, so the RV is hypoplastic and blood MUST cross an ASD (right-to-left) then a VSD to reach the small RV / lungs. Obligatory mixing + decreased lung flow. Classic **LEFT axis deviation (LAD)**	The single discriminator vs TOF in Q19: **tricuspid atresia = LEFT axis; TOF = RIGHT axis**. LAD in a cyanotic decreased-flow baby points to tricuspid atresia
5	**Pulmonary atresia**	No outlet at all from RV to pulmonary artery, so lung flow is **totally ductal-dependent**. Presents with severe cyanosis in the first hours to days as the duct closes	The purest ductal-dependent-pulmonary lesion: **PGE1 immediately** (C1 Q6 logic). With an intact ventricular septum the RV is tiny; the CXR shows dark lung fields

The one rule that ties it together: in every DECREASED-flow lesion the road to the lungs is obstructed, so deoxygenated blood is pushed right-to-left into the aorta (blue baby, dark lungs). Grade the obstruction: dynamic and spell-prone in TOF (raise SVR, relax the infundibulum), fixed and absolute in pulmonary atresia (open the duct with PGE1), and read the ECG axis to separate TOF (right) from tricuspid atresia (left).

Step 2. The concepts you must know (numbered to match the map in Step 1: concept N = map row N)

1) The DECREASED-flow physiology: obstruct the lungs, shunt right-to-left, dark lung fields

- **Synonym:** cyanotic CHD with decreased (oligaemic) pulmonary blood flow; the obstruction / right-to-left group; ductal-dependent-pulmonary lesions.

- **The core idea first:** these lesions all share one mechanism. Something obstructs the flow from the right heart to the lungs, so **right-side pressure climbs above left-side pressure**, and deoxygenated blood takes the low-resistance escape route: **right-to-left across a VSD or the foramen ovale, directly into the systemic circulation (the aorta)**. Two consequences fall out: the aorta now carries deoxygenated blood so the baby is **cyanosed**, and less blood is reaching the lungs so the chest X-ray shows **oligaemic (dark, dark) lung fields**. This is the C1 DECREASED bucket made concrete.
- **Reads as (the causal chain):** 1. Fixed or dynamic **obstruction** to right-ventricular outflow (infundibular narrowing, valvar pulmonary stenosis, pulmonary atresia, or an atretic tricuspid valve so blood cannot even enter the RV). 2. Right-heart **pressure rises** until it exceeds left-heart pressure. 3. Deoxygenated blood is driven **right-to-left across the VSD / foramen ovale** into the aorta. 4. **Cyanosis** (deoxygenated blood in the systemic arteries) plus **decreased pulmonary vascularity** (less blood reaching the lungs to be oxygenated).
- **Why it matters (the ductal-dependent tie to C1):** when the obstruction is near-total (pulmonary atresia, critical pulmonary stenosis, severe TOF), the ONLY way blood reaches the lungs is backwards from the aorta through the **ductus arteriosus**. That makes lung flow **ductal-dependent**, so the baby is stable while the duct is open and crashes when it closes on day 1 to 3. The rescue is exactly C1 concept 4: **PGE1 keeps the duct open**; indomethacin would slam it shut and be fatal.
- **The lesion list to hang on this branch: tetralogy of Fallot** (concept 2, the anchor), **tricuspid atresia** (concept 4), **pulmonary atresia** (concept 5), **critical pulmonary stenosis**, and **severe Ebstein anomaly**. All cyanotic, all with dark lung fields.
- **What Ebstein anomaly is (since it is on the list):** the **septal and posterior tricuspid leaflets are displaced downward (apically) into the right ventricle**, so the upper part of the RV becomes "atrialized" (thin, functionally part of the right atrium) and the true functional RV is small. That gives **severe tricuspid regurgitation plus a small, weak RV**, so forward flow to the lungs falls (decreased pulmonary flow) and blood shunts **right-to-left across an ASD / PFO** into the left heart, causing cyanosis. **Key associations: WPW / accessory pathways** (about a quarter of cases, so SVT and delta waves, ties straight back to EPS E2) and **maternal lithium** exposure. The CXR is a **massive "wall-to-wall" / box-shaped heart** (a huge right atrium); the ECG shows RBBB, tall P waves, and sometimes pre-excitation. Severe neonatal forms present with cyanosis (this bucket); milder forms present later with arrhythmias.
- **Example / exam anchor:** C1's **Q6** ("marked cyanotic defect with DECREASED pulmonary vascularity, treat first with PGE1") is describing a lesion from THIS branch. Any decreased-flow lesion that is duct-dependent gets PGE1 first.
- **Hook:** "Obstruct the road to the lungs, right-side pressure wins, blood shunts **right-to-left across the VSD into the aorta**: blue baby, dark lungs. If the obstruction is near-total, lung flow is **duct-dependent, so PGE1**."

2) Tetralogy of Fallot: one embryological error, four features (owns Q19)

- **Synonym:** TOF; the commonest cyanotic CHD beyond the newborn period.
- **The core fact first:** the "tetralogy" sounds like four separate defects but is **ONE developmental error, an anterosuperior deviation of the outflow (conal) septum**, which produces all four features at once. Memorise them with **PROVe**: **P**ulmonary (RVOT) stenosis, **R**ight ventricular hypertrophy, **O**verriding aorta, **V**SD.
- **The 4 components (click each one in the visual):** 1. **VSD (ventricular septal defect):** a large hole in the interventricular septum. It is the door through which the right-to-left shunt escapes into the aorta. 2. **Overriding aorta:** the aorta sits ASTRIDE the VSD, straddling both ventricles, so it receives blood from the RV as well as the LV. It literally collects the deoxygenated right-heart blood. 3. **Right-ventricular outflow tract obstruction (infundibular / pulmonary stenosis):** the narrowing below and at the pulmonary valve. This is the **key driver**: it determines how blue the baby is, and its muscular (infundibular)

part is what SPASMS in a tet spell. 4. **Right-ventricular hypertrophy (RVH):** the RV wall thickens because it is pumping against the outflow obstruction at systemic pressures. This is the CONSEQUENCE of feature 3, and it is what upturns the cardiac apex on the X-ray.

- **The mechanism (which feature does what):** the RVOT obstruction (3) raises RV pressure; the VSD (1) plus overriding aorta (2) give the blood a right-to-left escape into the aorta; RVH (4) is the downstream result. The severity of the cyanosis tracks the RVOT obstruction, not the VSD size.
- **The test signatures (all three appear in Q19):**
- **Boot-shaped heart (coeur en sabot)** on chest X-ray: the RVH upturns the apex and the small pulmonary artery leaves a **concave (empty) pulmonary bay** on the left heart border, so the silhouette looks like a boot. See the CXR schematic in the visual.
- **Decreased pulmonary vascularity** (dark lung fields): less blood reaching the lungs.
- **ECG: right axis deviation (RAD) + right-ventricular hypertrophy.** This is the discriminator that separates TOF from tricuspid atresia (concept 4, which has LEFT axis).
- **Why it matters (the whole of Q19):** a cyanotic neonate whose colour worsens with crying (spell physiology), with RAD and RV prominence on ECG and a boot-shaped heart with dark lungs on CXR, is TOF. The cyanosis-worse-with-crying detail is spell behaviour, which belongs to TOF, and the RIGHT axis is what excludes tricuspid atresia.
- **Example / exam anchor: Master Exam Q19 = A (tetralogy of Fallot).** Contrast the distractors: **TGA** gives a narrow "egg on a string" mediastinum with INCREASED flow (a C3 lesion); **tricuspid atresia** gives LEFT axis deviation (the key ECG discriminator); **pulmonary atresia with intact septum** is duct-dependent but does not fit the spell-with-crying + boot-shaped picture as cleanly; **TAPVR below the diaphragm** gives a "snowman" with INCREASED, congested flow. The RAD nails TOF over tricuspid atresia.
- **Hook:** "TOF = one error, four features (**PROVe**: Pulmonary stenosis, RVH, Overriding aorta, VSD). RVOT obstruction sets the cyanosis and spells; the X-ray is a **boot-shaped heart with dark lungs**; the ECG is **RIGHT axis + RVH**."

3) The tet spell (hypercyanotic spell) and its management (owns Q8)

- **Synonym:** hypercyanotic spell, blue spell, cyanotic spell of TOF.
- **The core idea first:** a tet spell is a sudden, dangerous deepening of cyanosis, and it happens because the TOF obstruction has a **DYNAMIC** part. Two things stack: 1. **The infundibular muscle SPASMS**, tightening the RVOT obstruction (triggered by crying, feeding, waking, agitation, fever, catecholamines). 2. **Systemic vascular resistance (SVR) FALLS** (crying, warmth, dehydration, vasodilatation). Both changes make the right-to-left shunt bigger, so more deoxygenated blood pours into the aorta and the baby goes acutely, deeply blue. It is a vicious cycle: hypoxia and acidosis drive hyperpnoea (deep fast breathing), which drops SVR further and worsens the spell.
- **Reads as (why the treatments are all one idea):** the fix is the exact mirror image of the two problems. **RAISE SVR** (so aortic pressure beats right-heart pressure and the shunt reverses back toward left-to-right) and **RELAX the infundibulum / break the cycle**. Every correct treatment does one or both:
- **Knee-chest position (infant) / squatting (toddler):** kinks the femoral arteries, **raises SVR**. First move, free, immediate.
- **Oxygen:** improves oxygenation and helps relax pulmonary vessels; supportive.
- **Morphine: calms the child, abolishes the hyperpnoea, and relaxes the infundibulum**, breaking the vicious cycle. A key spell drug.
- **Propranolol (beta-blocker): relaxes the dynamic infundibular muscle and slows the heart rate**, giving more filling time; used acutely (IV) and for prevention (oral).
- **Phenylephrine (alpha-agonist):** a pure vasoconstrictor that **raises SVR** directly.

- **IV fluids:** restore preload and SVR.
- **Sodium bicarbonate:** corrects the **metabolic acidosis** that is driving the hyperpnoea, breaking the cycle.
- **The trap (why digoxin is the EXCEPTION): digoxin is a positive inotrope**, it increases the FORCE of ventricular contraction. In a tet spell the problem is a **dynamic muscular (infundibular) obstruction**, so squeezing the RV harder makes that muscular narrowing WORSE and DEEPENS the right-to-left shunt. Digoxin is therefore **contraindicated** in a tet spell. This is exactly why beta-blockers (which RELAX the infundibulum) are the chronic prevention and inotropes are avoided. In Q8 every other option (oxygen, knee-chest, morphine, propranolol, sodium bicarbonate) helps, and digoxin is the impostor.
- **Why it matters:** the general reflex "sick heart plus cyanosis, give digoxin" is wrong here. The dynamic-obstruction physiology inverts the usual rule: you want LESS contractile force and MORE resistance, the opposite of standard heart-failure management.
- **Example / exam anchor: Master Exam Q8.** "A tet spell is treated with all of the following EXCEPT: oxygen, knee-chest position, morphine, digoxin, propranolol, sodium bicarbonate." Answer = **D, digoxin** (positive inotrope worsens the dynamic RVOT obstruction). Everything else raises SVR or relaxes the infundibulum or breaks the acidosis cycle.
- **Hook:** "A tet spell = **infundibular spasm + falling SVR**, so shunt goes MORE right-to-left. Fix = **raise SVR** (knee-chest, squat, phenylephrine, fluids) and **relax the infundibulum / break the cycle** (O2, morphine, propranolol, bicarbonate). **Digoxin is banned**: more contractility = worse dynamic obstruction."

4) Tricuspid atresia: obligatory mixing, and the LEFT-axis discriminator

- **Synonym:** absent tricuspid valve; the classic LEFT-axis-deviation cyanotic lesion.
- **The core idea first:** in tricuspid atresia there is **no tricuspid valve at all**, so blood cannot pass from the right atrium into the right ventricle the normal way. Two things follow: the **right ventricle is hypoplastic (small and underdeveloped)** because nothing fills it, and blood from the right atrium is FORCED **right-to-left across an ASD / patent foramen ovale** into the left atrium. To reach the lungs it then has to cross a **VSD** into the small RV and out the pulmonary artery, so pulmonary flow is usually **decreased** (a DECREASED-flow, cyanotic lesion) and depends on the VSD and often the duct.
- **Reads as (the obligatory circuit):** RA to LA across the ASD (obligatory right-to-left) to LV. The single big left ventricle then supplies both the body (aorta) and, via a VSD, the lungs. It is an obligatory-mixing lesion with a dominant LEFT ventricle.
- **The exam signature (the whole reason it is in this lesson):** because the RV is hypoplastic and the LV is dominant, the ECG shows **LEFT axis deviation (LAD)** with left-ventricular forces, which is UNUSUAL for a cyanotic newborn. This is the single best discriminator from tetralogy of Fallot, which has **RIGHT axis deviation + RVH**. So in a cyanotic decreased-flow baby: **LEFT axis to tricuspid atresia, RIGHT axis to TOF.**
- **Why it matters:** Q19 gives you a neonate with **right axis deviation (axis 120) and RV prominence**, which is TOF, not tricuspid atresia. If the same stem had said LEFT axis, the answer would flip to tricuspid atresia. The axis is the discriminator the question is testing.
- **Example / exam anchor:** tricuspid atresia is the cyanotic lesion that classically breaks the "cyanotic newborn = right-sided ECG" rule by showing **LEFT axis deviation**. Pair it in memory directly against TOF's right axis.
- **Hook:** "Tricuspid atresia = **no tricuspid valve, small RV, obligatory right-to-left across the ASD**, dominant LV. So the ECG is **LEFT axis** (the discriminator): **LAD = tricuspid atresia, RAD = TOF**."

5) Pulmonary atresia: the purest ductal-dependent-pulmonary lesion

- **Synonym:** atresia of the pulmonary valve / pulmonary outflow; pulmonary atresia with intact ventricular septum (PA-IVS) or with a VSD.

- **The core idea first:** in pulmonary atresia there is **no functional outlet from the right ventricle to the pulmonary artery at all**. Blood therefore cannot reach the lungs the front way, so pulmonary flow is **totally ductal-dependent**: the lungs are perfused only backwards from the aorta through the **ductus arteriosus**. This is the most extreme DECREASED-flow lesion.
- **Reads as (the causal chain):** no RV-to-PA outlet, so RA blood shunts **right-to-left across an ASD** into the left heart (obligatory mixing) and out the aorta (cyanosis), while the lungs get blood ONLY through the open duct. With an **intact ventricular septum** the RV has no way to decompress, so it stays **small (hypoplastic)**; with a VSD the RV can eject through the VSD and is better developed.
- **Why it matters (the C1 payoff):** because lung flow is entirely duct-dependent, the baby is stable while the duct is patent and develops **severe cyanosis in the first hours to days as the duct closes**. The emergency treatment is exactly C1 concept 4: **PGE1 to keep the duct open**, then surgery / catheter intervention. This is the cleanest example of the C1 Q6 logic (cyanotic + decreased flow to PGE1).
- **The imaging:** decreased pulmonary vascularity (dark lung fields); with an intact septum a small RV and often cardiomegaly from a dilated right atrium.
- **Example / exam anchor:** in Q19, "pulmonary atresia with intact ventricular septum" is a distractor. It is duct-dependent and cyanotic, but it does not fit the spell-with-crying, boot-shaped, RAD-with-a-formed-RV picture of TOF, so TOF is the better answer. Pulmonary atresia is the pure PGE1 lesion.
- **Hook:** "Pulmonary atresia = **no RV-to-lung outlet, lung flow 100% duct-dependent**. Duct closes to severe cyanosis in the first days. Treat with **PGE1** (C1 Q6 logic). Intact septum = small RV."

⚑ Treated differently (the exceptions to flag)

- **Digoxin is BANNED in a tet spell (the whole of Q8).** The normal reflex "cyanotic sick baby, give an inotrope" is wrong here. The obstruction is a DYNAMIC muscular one, so a positive inotrope (digoxin) increases contractility and TIGHTENS the RVOT obstruction, deepening the shunt. You want the opposite: relax the infundibulum (propranolol, morphine) and raise SVR (knee-chest, phenylephrine).
- **Squatting and knee-chest work by raising SVR, not by "resting."** Kinking the femoral arteries raises systemic vascular resistance so aortic pressure exceeds right-heart pressure and the shunt swings back toward left-to-right. It is haemodynamics, not fatigue. Same physiology as giving phenylephrine.
- **TOF vs tricuspid atresia is decided by the ECG AXIS, not the cyanosis.** Both are cyanotic decreased-flow lesions. **TOF = RIGHT axis + RVH; tricuspid atresia = LEFT axis.** A cyanotic newborn with LEFT axis deviation is the tell for tricuspid atresia, because the hypoplastic RV and dominant LV shift the axis left. This is the exact discriminator in Q19.
- **A tet spell is a DYNAMIC obstruction, the fixed anatomy has not changed.** The valve and VSD are the same second to second; what changes in a spell is infundibular muscle tone and SVR. That is why a spell can be broken in minutes by position and drugs without any surgery.
- **The cyanosis in TOF tracks the RVOT obstruction, not the VSD.** A big VSD alone would be a pink, high-flow lesion; it is the pulmonary/infundibular stenosis that makes TOF blue. So the "blueness" grades the obstruction severity.
- **Not every decreased-flow lesion needs PGE1, only the duct-dependent ones.** Pulmonary atresia and critical pulmonary stenosis are duct-dependent (PGE1 immediately). A well-balanced TOF with mild obstruction may be pink for weeks and is NOT initially duct-dependent. Match the PGE1 to the severity of the obstruction.
- **Tricuspid atresia and pulmonary atresia both force a right-to-left ATRIAL shunt.** With no tricuspid inflow (tricuspid atresia) or no RV outflow with an intact septum (pulmonary atresia), the right atrium can only empty across an ASD / patent foramen ovale, so an adequate atrial communication is life-sustaining in both.

Mnemonics to lock in

- **The whole lesson in one line: "Block the road to the lungs, blood shunts right-to-left across the VSD into the aorta: blue baby, dark lungs. TOF is one error, four features; a tet spell is dynamic obstruction plus low SVR (raise SVR, relax the infundibulum, never digoxin); the worst are duct-dependent, so PGE1."**
- **TOF's four features (PROVe): "P**ulmonary (RVOT) stenosis, **R**ight ventricular hypertrophy, **O**verriding aorta, **V**SD." (The RVOT stenosis is the driver; RVH is the consequence.)
- **TOF test triad: "Boot-shaped heart, dark lung fields, RIGHT axis + RVH."**
- **Tet spell fix (two jobs): "RAISE SVR (knee-chest / squat, phenylephrine, fluids) and RELAX the infundibulum / break the cycle (Oxygen, Morphine, Propranolol, Bicarbonate). NEVER digoxin (inotrope worsens the dynamic obstruction)."**
- **Why squatting helps: "Squat / knee-chest = kink the femorals = SVR UP = aorta beats the right heart = shunt swings back left-to-right."**
- **The axis discriminator: "TOF = RIGHT axis (RVH). Tricuspid atresia = LEFT axis (small RV, big LV). Cyanotic newborn with LEFT axis = tricuspid atresia."**
- **The duct-dependent ones: "Pulmonary atresia and critical pulmonary stenosis: lungs fed only by the duct, so PGE1 (C1 Q6). Intact septum = small RV."**
- **Digoxin trap: "Tet spell inotrope = WRONG. More force = tighter infundibulum = worse shunt. Digoxin is the EXCEPTION in Q8."**

🧪 Quiz yourself (commit your answer, key is sealed at the bottom)

Q8. A "tet spell" (hypercyanotic / blue spell) of tetralogy of Fallot is treated with all of the following EXCEPT: A. Oxygen B. Knee-chest position C. Morphine D. Digoxin E. Propranolol F. Sodium bicarbonate

Q19. A 6-hour-old neonate develops cyanosis that worsens with crying, born at 37 weeks. The ECG shows an axis of 120 degrees (right axis deviation) with right-ventricular prominence, and the chest radiograph shows a boot-shaped heart with decreased pulmonary vascularity. The most likely diagnosis is: A. Tetralogy of Fallot B. Transposition of the great vessels (D-TGA) C. Tricuspid atresia D. Pulmonary atresia with intact ventricular septum E. TAPVR below the diaphragm

Q-A (tet-spell mechanism). The acute deepening of cyanosis in a tet spell is caused mainly by: A. Fixed valvar pulmonary stenosis suddenly tightening B. A rise in systemic vascular resistance C. Dynamic infundibular (muscular) spasm together with a FALL in systemic vascular resistance, increasing the right-to-left shunt D. Closure of the VSD E. A left-to-right shunt across the VSD

Q-B (TOF components). Which of the four features of tetralogy of Fallot is the main determinant of how cyanosed the patient is, and the part that spasms in a tet spell? A. The ventricular septal defect B. The overriding aorta C. The right-ventricular outflow (infundibular) obstruction D. The right-ventricular hypertrophy E. The size of the aorta

Q-C (axis discriminator). A cyanotic newborn with decreased pulmonary vascularity has an ECG showing LEFT axis deviation. Compared with tetralogy of Fallot, this finding points to: A. Tetralogy of Fallot after all B. Tricuspid atresia C. Transposition of the great arteries D. TAPVR E. A normal variant

Q-D (ductal-dependent link to C1). A term neonate with pulmonary atresia and an intact ventricular septum becomes severely cyanosed at 24 hours as the duct closes. The immediate treatment is: A. Indomethacin B. Digoxin C. Prostaglandin E1 (alprostadil) to keep the duct open D. A beta-blocker E. Oxygen alone to close the duct faster

🔒 Answer key (only after you have committed)

- **Q8 = D (Digoxin).** A tet spell is a DYNAMIC infundibular (RVOT) spasm plus a fall in SVR, which deepens the right-to-left shunt. Treatment does the mirror image: **raise SVR** (knee-chest / squat, phenylephrine, IV fluids) and **relax the infundibulum or break the acidosis-hyperpnoea cycle** (oxygen, morphine, propranolol, sodium bicarbonate). **Digoxin is a positive inotrope**: increasing contractility TIGHTENS the dynamic muscular obstruction and WORSENS the shunt, so it is contraindicated. Every other listed option helps, which makes digoxin the exception.
- **Q19 = A (Tetralogy of Fallot).** The discriminators all point to TOF: **cyanosis worse with crying** is spell physiology (TOF), **right axis deviation (axis 120) + RV prominence** on ECG, and a **boot-shaped heart with decreased pulmonary vascularity** on CXR. Why not the others: **TGA (B)** classically has a narrow "egg on a string" mediastinum and INCREASED pulmonary flow; **tricuspid atresia (C)** classically has **LEFT axis deviation** (the key discriminator, here the axis is RIGHT); **pulmonary atresia with intact septum (D)** is duct-dependent but does not fit the spell-with-crying, boot-shaped, formed-RV picture; **TAPVR below the diaphragm (E)** gives a "snowman" with INCREASED, congested flow. The RIGHT axis is what nails TOF over tricuspid atresia.
- **Q-A = C (Dynamic infundibular spasm plus a fall in SVR).** The spell is not the fixed valve suddenly changing; it is the muscular infundibulum spasming AND systemic resistance dropping (crying, warmth, dehydration), both of which enlarge the right-to-left shunt. Hypoxia and acidosis then drive hyperpnoea, dropping SVR further, a vicious cycle. That is why the treatments raise SVR and relax the infundibulum.
- **Q-B = C (The RVOT / infundibular obstruction).** The degree of pulmonary/infundibular stenosis sets how much blood is diverted right-to-left, so it determines the cyanosis, and its muscular (infundibular) part is what spasms in a spell. The VSD (A) is just the escape door, the overriding aorta (B) collects the blood, and the RVH (D) is the consequence of the obstruction, not the cause of the cyanosis.
- **Q-C = B (Tricuspid atresia).** Tricuspid atresia has a hypoplastic RV and a dominant LV, so the ECG shows **LEFT axis deviation**, unusual for a cyanotic newborn. That is the single best discriminator from TOF, which shows RIGHT axis + RVH. LEFT axis in a cyanotic decreased-flow baby points to tricuspid atresia.
- **Q-D = C (Prostaglandin E1 to keep the duct open).** Pulmonary atresia has no RV-to-pulmonary-artery outlet, so lung flow is entirely ductal-dependent. When the duct closes, pulmonary blood flow vanishes and cyanosis is severe. **PGE1 (alprostadil) reopens/maintains the duct** (exactly C1 concept 4 / Q6). **Indomethacin (A)** would close the duct and be fatal; digoxin (B) and a beta-blocker (D) do nothing for the duct; oxygen to close the duct (E) is the opposite of what you want here.

Diagrams in the visual (hand-built inline SVG, nothing embedded from the web)

1. **DECREASED-flow physiology splitter (concept 1):** the obstruction-to-lungs chain shown as a mini schematic, with the right-to-left shunt across the VSD animating into the aorta and the C2 lesion list, tied back to the C1 DECREASED bucket and PGE1.
2. **Interactive tetralogy-of-Fallot heart (concept 2):** a hand-drawn SVG heart with the four TOF features. Clicking VSD, overriding aorta, RVOT/pulmonary stenosis, or RVH highlights that structure on the anatomy and explains it (which feature drives the cyanosis, which one spasms).
3. **Boot-shaped-heart CXR schematic (concept 2):** a hand-drawn chest-X-ray silhouette with the upturned apex (from RVH) and the concave, empty pulmonary bay, plus dark lung fields, labelled to explain why the silhouette looks like a boot.

4. **Tet-spell animation (concept 3):** an SVR meter and a right-to-left-shunt gauge with a baby whose colour deepens as SVR drops and the infundibulum spasms. Clicking a treatment (knee-chest, oxygen, morphine, propranolol) reverses the cycle inline (SVR rises, shunt shrinks, baby pinks up); a digoxin button shows it WORSENING the spell (the Q8 trap).
5. **TOF vs tricuspid-atresia axis card (concepts 4 and 5):** a side-by-side comparison of the ECG axis (RIGHT axis / RVH for TOF vs LEFT axis for tricuspid atresia), with the pulmonary-atresia duct-dependent / PGE1 note.

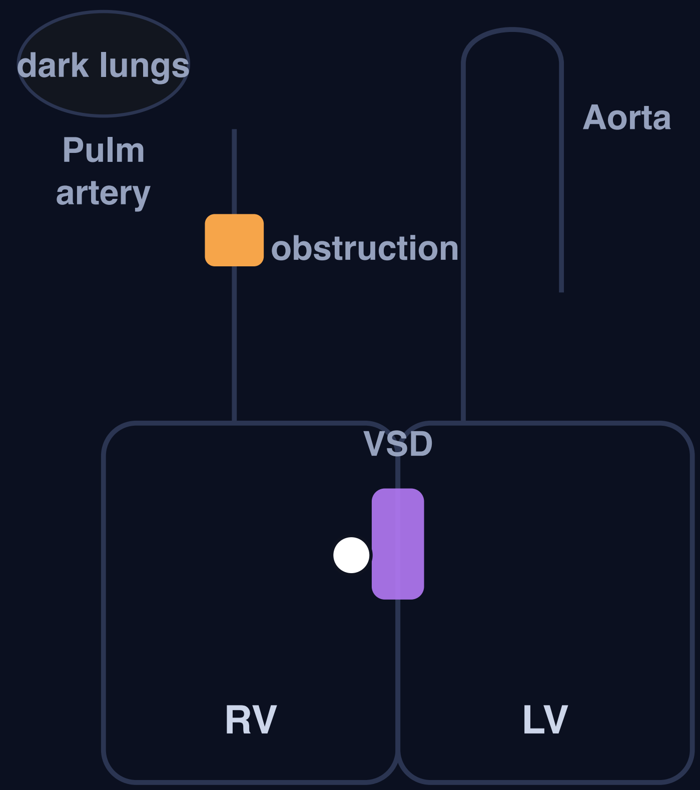
C2 - Interactive visual (rendered): diagrams & schematics

The live, toggleable version is the HTML file on the CHD hub. Below is a static render of its default view; open the hub to step through the other toggle states.

C2 - Cyanotic CHD with DECREASED pulmonary blood flow

The DECREASED-flow branch of C1. Right-heart outflow obstruction pushes blood right-to-left into the aorta (blue baby, dark lungs). Anchors: Q8 a tet spell is treated with everything EXCEPT digoxin (an inotrope worsens the dynamic RVOT obstruction); Q19 cyanosis worse with crying + right axis + boot-shaped heart = tetralogy of Fallot (right axis nails it over tricuspid atresia's left axis).

1. THE DECREASED-FLOW PHYSIOLOGY: OBSTRUCT THE LUNGS, SHUNT RIGHT-TO-LEFT (CLICK TO RUN THE SHUNT)



RV outflow is obstructed, so RV pressure beats LV pressure and deoxygenated blood shunts **right-to-left across the VSD** into the overriding aorta. Hand-drawn, nothing embedded.

Blue baby, dark lungs

Chain: RV outflow obstruction to RV pressure > LV pressure to right-to-left shunt across the VSD into the aorta to **cyanosis** + **oligaemic (dark) lung fields**.

The severest lesions (pulmonary atresia, critical PS, severe TOF) are **ductal-dependent**: lungs are perfused only backwards through the duct, so **PGE1** keeps it open (C1 Q6).

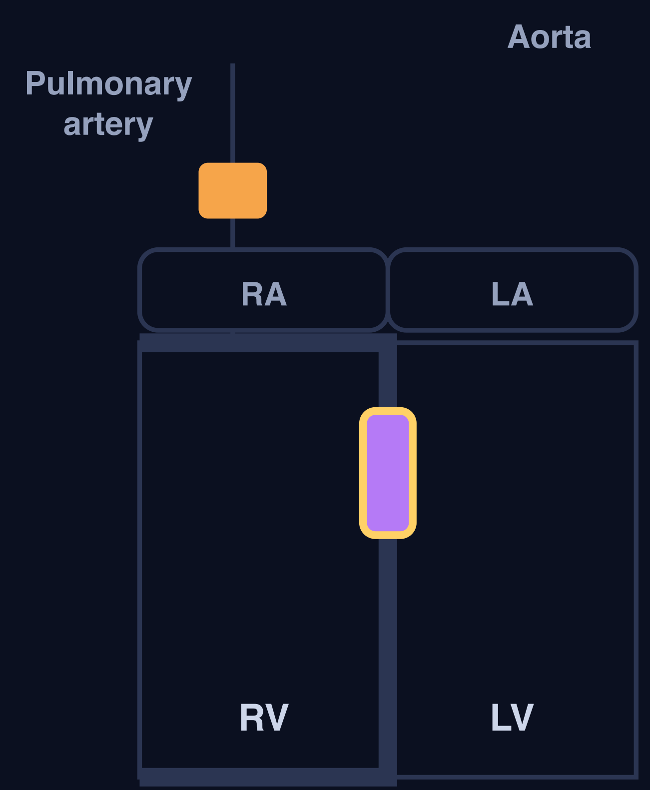
DECREASED (oligaemic, dark) = C2 lesions

- Tetralogy of Fallot (the anchor)
- Tricuspid atresia (LEFT axis)
- Pulmonary atresia (pure duct-dependent)
- Critical pulmonary stenosis
- Severe Ebstein anomaly

Ebstein anomaly: tricuspid leaflets displaced apically into the RV, so the upper RV is "atrialized" + severe TR + a small weak RV to poor forward flow to the lungs and right-to-left across an ASD to cyanosis. Associations: **WPW / accessory pathways** and **maternal lithium**. CXR = massive "wall-to-wall" heart (huge RA).

2. TETRALOGY OF FALLOT: ONE ERROR, FOUR FEATURES (CLICK EACH TO HIGHLIGHT IT ON THE HEART)

1. VSD (the escape door) | **2. Overriding aorta** | **3. RVOT / pulmonary stenosis** | **4. RV hypertrophy**



PROVe: Pulmonary stenosis, RVH, Overriding aorta, VSD. The RVOT obstruction is the driver; RVH is its consequence.

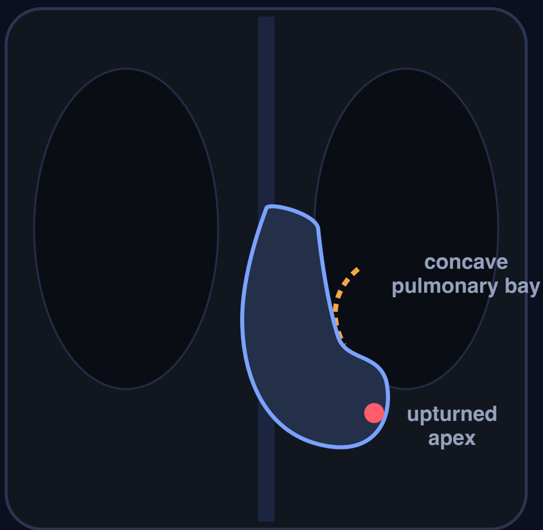
1. VSD (ventricular septal defect)

The escape door. A large hole in the interventricular septum. Because the RV outflow is obstructed, RV pressure exceeds LV pressure and deoxygenated blood shunts **right-to-left through this VSD** into the aorta.

Not the cause of the cyanosis by itself: the obstruction sets how much blood is diverted through it.

Q19: cyanosis worse with crying (spell), right axis deviation + RVH on ECG, and a boot-shaped heart with dark lungs on CXR = tetralogy of Fallot. The right axis is what separates it from tricuspid atresia (left axis).

3. THE BOOT-SHAPED HEART (COEUR EN SABOT): THE TOF CHEST X-RAY



Upturned apex: the right-ventricular hypertrophy lifts the cardiac apex up and out (off the diaphragm) instead of pointing down.

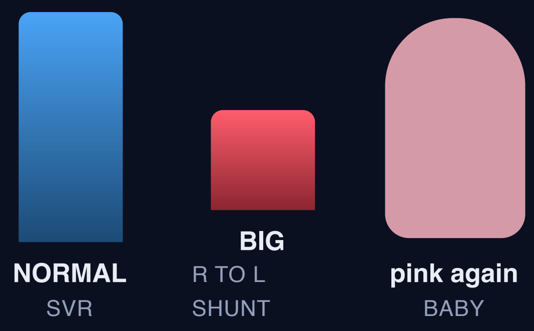
Concave pulmonary bay: the small, obstructed pulmonary artery leaves an empty, scooped-out segment on the left heart border where the normal bulge would be.

Dark lung fields: decreased pulmonary vascularity, because less blood is reaching the lungs.

Upturned apex + concave (empty) pulmonary bay = a silhouette shaped like a **boot**. Pair it with the ECG (right axis + RVH) and the spell history and you have TOF.

4. THE TET SPELL: SVR DROPS, THE SHUNT GROWS, THE BABY TURNS BLUE (CLICK A TREATMENT TO REVERSE IT)

Trigger a spell (crying: SVR drops, infundibulum spasms)	Knee-chest / squat (raise SVR)	Oxygen	Morphine
Propranolol		Digoxin	



Resting TOF

Baseline: fixed obstruction, a small right-to-left shunt, mild cyanosis. Trigger a spell, then break it with a treatment.

Q8: a tet spell is treated with all EXCEPT digoxin. It is a positive inotrope, so more contractility TIGHTENS the dynamic infundibular obstruction and DEEPENS the shunt. Correct moves RAISE SVR (knee-chest, squat, phenylephrine, fluids) or RELAX the infundibulum / break the cycle (O2, morphine, propranolol, bicarbonate).

5. THE AXIS DISCRIMINATOR: TOF (RIGHT AXIS) VS TRICUSPID ATRESIA (LEFT AXIS)

Tetralogy of Fallot

RIGHT axis deviation (~+120) + RVH.

RV pumps against the outflow obstruction, so it hypertrophies and the axis swings right. Boot-shaped heart, spells, formed RV.

Tricuspid atresia

LEFT axis deviation.

No tricuspid valve, so the RV is hypoplastic and the LV dominates: the axis swings LEFT. Obligatory right-to-left across the ASD.

The discriminator in Q19: both are cyanotic decreased-flow lesions, but a LEFT axis points to tricuspid atresia and a RIGHT axis (as in the stem, +120) points to TOF.

Pulmonary atresia (concept 5): no RV-to-lung outlet at all, so lung flow is **100% ductal-dependent**. Duct closes to severe cyanosis in the first days. Treat with **PGE1** (C1 Q6 logic). Intact ventricular septum = small RV.

C2 visual - Module 5 CHD. All diagrams hand-drawn inline SVG, fully self-contained, no external assets.

C3 - Cyanotic CHD with INCREASED pulmonary blood flow: the physiology of mixing and transposition, D-transposition of the great arteries (the anchor), TAPVR, truncus arteriosus, double-outlet right ventricle and single ventricle (full lesson)

> Visual: `study/visuals/C3-cyanotic-increased-flow.html` (the INCREASED-flow physiology splitter that carries over from C1, where the blood reaches the lungs fine but oxygenated and deoxygenated blood are not separated, so the baby is blue with PLETHORIC lungs, an interactive D-transposition diagram showing the two parallel circuits and how a mixing point at the ASD / VSD / PDA rescues the baby when you click it, a TAPVR card showing the pulmonary veins draining the wrong way into the systemic veins with the obstructed infracardiac emergency, a DORV card contrasting subaortic VSD (TOF / VSD-like) with subpulmonic VSD Taussig-Bing (TGA-like) and what pulmonary stenosis does to the flow, and a truncus arteriosus + 22q11 note). This is the INCREASED-pulmonary-flow branch of the master framework built in C1 (cyanotic CHD splits into DECREASED vs INCREASED lung blood flow), and the mirror image of C2. Directly tested (Master Exam): **Q5** cyanotic CHD with INCREASED pulmonary flow are all of the following EXCEPT (answer **B, D-TGA with VSD and severe pulmonary stenosis**, because the severe PS OBSTRUCTS flow to the lungs, so that lesion sits in the DECREASED bucket, not the increased one, which makes it the odd one out). **Q18** a cyanotic neonate with increased pulmonary markings, a normal heart size, loud S2, no murmur, and SIDE-BY-SIDE great arteries on echo (key answer **D, DORV with subaortic VSD and PS**, but flag the conflict: the stem's increased markings + no murmur + loud S2 + side-by-side arteries fit a Taussig-Bing subpulmonic-VSD, no-PS, TGA-like physiology, which argues for option **a**, so confirm with the course). This lesson also ties back to C1's PGE1: D-TGA is ductal-dependent for MIXING.

Step 0. The big theory: the blood reaches the lungs, but the two circuits never meet

In C1 you learned the one sorting question for a blue baby: what do the lung fields show? C2 was the whole DECREASED-flow (dark, oligaemic) bucket, where the road to the lungs is blocked. This lesson is the whole other half, the INCREASED-flow (PLETHORIC, congested, wet) bucket, and it is a completely different mechanism. Everything hangs off five linked ideas, in order:

- **The lungs get plenty of blood, but the oxygen never reaches the body.** In this bucket the pathway to the lungs is open (often too open, so the lung fields are PLETHORIC and congested), yet the baby is still blue. The reason is not obstruction, it is a FAILURE TO SEPARATE the blue blood from the red blood. Either the great arteries are transposed (the aorta carries the blue blood back to the body while the pulmonary artery recycles the red blood back to the lungs), or all the blood is forced to mix in one common chamber. So the baby is cyanotic WITH increased flow.
- **D-transposition of the great arteries is the anchor, and it is a plumbing swap.** The aorta comes off the RIGHT ventricle and the pulmonary artery comes off the LEFT ventricle. That creates two PARALLEL circuits instead of one circuit in series: blue blood goes body to RV to aorta to body forever, and red blood goes lungs to LV to pulmonary artery to lungs forever. The two loops never touch, which is incompatible

with life UNLESS there is a mixing point (an ASD, a VSD, or a PDA) letting some red blood cross into the systemic loop.
- **Mixing is the whole game in this bucket.** Because the problem is separation, the treatment and the survival both depend on how well the two bloods mix. In D-TGA you keep the duct open with PGE1 and, if that is not enough, you tear the atrial septum open with a balloon (Rashkind balloon atrial septostomy) to force mixing at the atrial level. The definitive repair is the arterial switch, which puts the great arteries back on the correct ventricles.
- **Pulmonary stenosis flips a lesion from the increased bucket to the decreased bucket.** This is the exam hinge in Q5 and Q18. Any of these mixing lesions normally has increased flow, but ADD significant pulmonary stenosis and you now obstruct the road to the lungs, so the flow becomes DECREASED and the lung fields go dark. So "D-TGA with severe PS" or "DORV with subaortic VSD and PS" behave like TOF (decreased flow), while the same lesions without PS behave like transposition (increased flow). PS is the switch.
- **Obstructed pulmonary VEINS are a different emergency (TAPVR).** In total anomalous pulmonary venous return the pulmonary veins drain the wrong way into the systemic veins, so an ASD is obligatory just to fill the left heart. The supracardiac form gives a "snowman / figure-of-8" heart, but the INFRACARDIAC (below-the-diaphragm) form is typically OBSTRUCTED as the draining vein is squeezed, so it is a neonatal emergency with pulmonary oedema, and here PGE1 does NOT help.

Hook: **"In the increased-flow bucket the lungs get plenty of blood but the red and blue never separate, so the baby is blue with WET (plethoric) lungs. D-TGA is a plumbing swap into two parallel circuits that only survive if blood MIXES (ASD / VSD / PDA), so keep the duct open with PGE1 and tear the atrial septum with a Rashkind balloon. Add pulmonary stenosis and any of these flips into the DECREASED bucket. Obstructed infracardiac TAPVR is a wet-lung emergency."**

Step 1. The map: the six ideas of C3

These numbers are the SAME as the concept numbers in Step 2 (map row 1 = concept 1, and so on).

#	Idea	The one-line mechanism	Exam anchor
1	**The INCREASED-flow physiology (mixing / transposition)**	The road to the lungs is OPEN (flow is normal or high, so lung fields are **plethoric / congested**), but oxygenated and deoxygenated blood are **not separated** (transposition or complete mixing). Result: **cyanosis + increased (plethoric) lung fields**. Add **pulmonary stenosis** and the lesion flips into the DECREASED bucket	This is the C1 INCREASED bucket. The lesions are the **5 Ts** (TGA, TAPVR, Truncus, Tricuspid atresia with big VSD/no PS, Single ventricle) + DORV. **Q5** is the "EXCEPT" that uses the PS-flips-the-bucket rule
2	**D-transposition of the great arteries (the anchor)**	Aorta off the **RV**, pulmonary artery off the **LV**: two **parallel** circuits that never meet, incompatible with life unless there is **mixing (ASD / VSD / PDA)**. Test signature: **egg on a string** (narrow mediastinum) + increased flow	Management is the point: **PGE1 to keep the duct open + Rashkind balloon atrial septostomy** to improve mixing; **arterial switch** is the definitive repair. Ties to **C1 Q6 (PGE1)**, this time duct-dependent for MIXING not for flow
3	**Total anomalous pulmonary venous return (TAPVR)**	The pulmonary veins drain into the **systemic venous** circulation (not the LA), so an **ASD is obligatory** to fill the left heart. Supracardiac = **snowman / figure-of-8**. Infracardiac (below diaphragm) is typically **OBSTRUCTED = neonatal emergency** with pulmonary oedema	The obstructed infracardiac form is the trap: severe cyanosis + a WET, congested chest that looks like lung disease, and **PGE1 does not help** (the block is in the veins). Emergency surgery
4	**Truncus arteriosus**	A **single common arterial trunk** overrides a **VSD** and gives off the aorta, pulmonaries and coronaries together, so systemic and pulmonary blood fully mix: cyanosis + **increased flow**	Strongly associated with **22q11 deletion (DiGeorge)**: think truncus + interrupted aortic arch + hypocalcaemia + absent thymus. One truncal valve, often regurgitant
5	**Double-outlet right ventricle (DORV, owns Q18)**	**Both** great arteries arise from the **RV**; the physiology depends on **where the VSD sits** and **whether there is PS**. **Subaortic VSD = TOF /**	**Q18 (key = D)**: side-by-side arteries, increased markings, loud S2, no murmur. Key says subaortic VSD + PS, but that PS would

#	Idea	The one-line mechanism	Exam anchor
		VSD-like; **subpulmonic VSD (Taussig-Bing) = TGA-like**. **PS decreases flow**	DECREASE flow and contradict the increased markings, so the clinical picture fits **subpulmonic (Taussig-Bing), no PS (option a)**. ⚠ confirm with the course
6	**Single ventricle**	One functional ventricle receives BOTH atria, so systemic and pulmonary venous blood **completely mix** in one chamber: cyanosis + usually **increased flow** (unless PS limits it)	The purest complete-mixing lesion, and a distractor in Q5 ("single ventricle with malposed great vessels" = increased flow, NOT the exception). Leads into the single-ventricle / Fontan pathway (C6)

The one rule that ties it together: in every INCREASED-flow lesion the road to the lungs is OPEN, so the lungs are plethoric, but the red and blue blood are not separated (transposed or mixed), so the baby is still blue. The survival hinges on MIXING (open the duct with PGE1, tear the atrial septum with a Rashkind balloon in D-TGA). The one modifier that changes everything is pulmonary stenosis: add PS and the lesion falls out of this bucket into the DECREASED (dark-lung) bucket. That single rule answers both Q5 and Q18.

Step 2. The concepts you must know (numbered to match the map in Step 1: concept N = map row N)

1) The INCREASED-flow physiology: open road to the lungs, but the two bloods never separate (owns the Q5 rule)

- **Synonym:** cyanotic CHD with increased (plethoric / congested) pulmonary blood flow; the mixing / transposition group; the "5 Ts".
- **The core idea first:** these lesions all share one mechanism, and it is NOT obstruction (that was C2). Here the pathway to the lungs is open, so the lungs get plenty of blood (often too much, so the chest X-ray shows **plethoric, congested lung fields**), yet the baby is still cyanotic. The reason is a **failure to separate oxygenated from deoxygenated blood**. Either the great arteries are transposed (so the aorta recycles blue blood back to the body while the pulmonary artery recycles red blood back to the lungs), or all the returning blood is forced to mix in one common chamber. Two consequences fall out: the aorta carries poorly oxygenated (mixed or transposed) blood so the baby is **cyanosed**, and the open pulmonary pathway means **increased pulmonary blood flow (plethora)**.
- **Reads as (the causal chain):** 1. The road to the lungs is **open** (no significant right-heart outflow obstruction). 2. But the circulation **fails to separate red from blue**: transposition (parallel circuits) OR complete mixing in a common chamber / trunk. 3. The aorta therefore carries **desaturated (mixed or transposed) blood**, so the baby is **cyanosed**. 4. Because the pulmonary pathway is open, **pulmonary blood flow is increased**, so the lung fields are **plethoric / congested** and the baby is at risk of pulmonary over-circulation and heart failure.
- **The lesion list to hang on this branch (the 5 Ts + DORV): Transposition (D-TGA)** (concept 2), **TAPVR** (concept 3), **Truncus arteriosus** (concept 4), **Tricuspid atresia with a large VSD and no PS** (an increased-flow variant), **Single ventricle** without PS (concept 6), and **DORV** (concept 5). All cyanotic, all with plethoric lungs (when there is no PS).
- **Why it matters (the PS switch, which is the whole of Q5):** add significant **pulmonary stenosis** to any of these and you now obstruct the road to the lungs, so the flow becomes **DECREASED** and the lung fields go dark. That is why **D-TGA with VSD and severe PS is NOT an increased-flow lesion** (the severe PS drops it into the C2 / decreased bucket) and is the EXCEPTION in Q5. The same rule makes tricuspid atresia, DORV and single ventricle land in either bucket depending on whether PS is present. PS is the switch.
- **Example / exam anchor: Master Exam Q5** = "cyanotic CHD with INCREASED pulmonary flow, all EXCEPT: (A) tricuspid atresia with ASD and VSD, (B) D-TGA with VSD and severe pulmonary stenosis,

(C) single ventricle with malposed great vessels, (D) truncus arteriosus type 1." Answer = **B**: the severe PS obstructs pulmonary flow, so that lesion has DECREASED (not increased) flow, the odd one out. A, C and D are all mixing / increased-flow lesions.
- **Hook:** "Increased-flow = the road to the lungs is OPEN (plethoric, wet lungs) but the red and blue never separate (transposed or mixed), so the baby is still blue. Add **pulmonary stenosis** and the lesion flips into the DECREASED (dark-lung) bucket: that flip is the whole of Q5."

2) D-transposition of the great arteries: two parallel circuits that only survive by mixing (the anchor)

- **Synonym:** D-TGA; transposition of the great arteries; the classic "egg on a string" cyanotic newborn.
- **The core fact first:** in D-TGA the two great arteries are connected to the WRONG ventricles, a **ventriculo-arterial discordance**. The **aorta arises from the right ventricle** and the **pulmonary artery arises from the left ventricle**. This turns the normal single loop (in series) into **two separate PARALLEL loops**: deoxygenated blood goes body to RA to RV to aorta to body forever, and oxygenated blood goes lungs to LA to LV to pulmonary artery to lungs forever. The oxygen never reaches the body. This is **incompatible with life unless the two loops are connected by a mixing point**.
- **The mixing points (click each in the visual):** 1. **ASD / patent foramen ovale (atrial level):** the most important and most reliable mixing site; oxygenated blood crosses from LA to RA and into the systemic loop. This is why the atrial septum is deliberately torn open in the Rashkind procedure. 2. **VSD (ventricular level):** allows mixing between the ventricles; a large VSD gives better saturation but also over-circulates the lungs (heart failure). 3. **PDA (ductal level):** the duct lets blood cross between aorta and pulmonary artery; keeping it open with PGE1 buys time and improves mixing, especially with an intact septum.
- **The mechanism (why it is an emergency on day 1):** with an intact ventricular septum, the only mixing is a small foramen ovale and the closing duct. As the duct closes over the first hours to days, mixing collapses and the baby becomes **profoundly, refractorily cyanotic** with a loud single S2 and often NO murmur (nothing is obstructed, blood is just in the wrong loop). The cyanosis does not improve with oxygen because the problem is circulation, not oxygenation.
- **The test signatures:**
- **Egg on a string (egg-shaped heart on a narrow mediastinum)** on chest X-ray: the great arteries lie antero-posteriorly (one in front of the other) instead of side by side, so the mediastinum looks narrow, like a string holding an egg.
- **Increased pulmonary vascularity** (plethora).
- **Severe cyanosis in the first day of life**, single loud S2, often no murmur.
- **Why it matters (the management is the point):** the whole treatment logic is MIXING. **PGE1 (alprostadil)** keeps the ductus arteriosus open to improve mixing (C1 concept 4 logic, but here duct-dependent for MIXING, not for pulmonary flow). If saturations are still too low, **balloon atrial septostomy (the Rashkind procedure)** tears the atrial septum open with a balloon catheter to force atrial-level mixing. The **definitive repair is the arterial switch operation (Jatene)**, which detaches and re-implants the great arteries onto the correct ventricles (with coronary transfer), usually in the first weeks of life.
- **Example / exam anchor:** D-TGA is the anchor of this bucket and the reason PS matters: plain **D-TGA (with or without a VSD) has INCREASED flow**, but **D-TGA with VSD and severe PS has DECREASED flow** and is therefore the EXCEPTION in **Q5 = B**. Adult D-TGA options (Rastelli for TGA + VSD + PS, and the atrial-switch late complications) belong to C7 (Q17).
- **Hook:** "D-TGA = aorta off the RV, PA off the LV, so two **parallel** loops that never meet. Survival = **MIXING** (ASD / VSD / PDA). **Egg on a string**, increased flow, loud single S2, no murmur. Treat: **PGE1 + Rashkind balloon septostomy**, then the **arterial switch**. Add severe PS and it flips to the decreased bucket (Q5)."

3) Total anomalous pulmonary venous return (TAPVR): the pulmonary veins drain the wrong way

- **Synonym:** TAPVR / TAPVD (total anomalous pulmonary venous drainage / connection); anomalous pulmonary venous return.
- **The core idea first:** in TAPVR the **pulmonary veins do not connect to the left atrium** at all. Instead they drain the oxygenated lung blood the WRONG way, into the **systemic venous circulation** (via a vertical vein to the innominate / SVC in the supracardiac type, to the coronary sinus / right atrium in the cardiac type, or down through the diaphragm to the portal / hepatic veins in the infracardiac type). So ALL the blood, systemic and pulmonary, arrives back at the right atrium and mixes there. An **ASD (or patent foramen ovale) is OBLIGATORY**: it is the only way any blood reaches the left heart to be pumped to the body. It is therefore a complete-mixing, increased-flow, cyanotic lesion.
- **Reads as (the obligatory circuit):** pulmonary veins to systemic veins to RA (total mixing) to across the ASD to LA to LV to aorta. Without the ASD, the left heart has nothing to pump, so the ASD is life-sustaining.
- **The two faces (the exam split):**
- **Supracardiac / non-obstructed TAPVR:** the classic **"snowman" or "figure-of-8"** heart on chest X-ray (the dilated vertical vein + SVC form the snowman's head over the enlarged heart body). Increased flow, cyanosis, but not an immediate emergency.
- **Infracardiac (below-the-diaphragm) TAPVR:** the descending vein passes through the diaphragm and is usually **OBSTRUCTED** (squeezed as it passes the diaphragm / through the liver). This backs blood up into the lungs, so it presents as a **neonatal EMERGENCY** with severe cyanosis and **pulmonary oedema**, a wet, congested chest X-ray that looks like lung disease or respiratory distress. It needs emergency surgery.
- **Why it matters (the trap):** obstructed infracardiac TAPVR is one of the few cyanotic lesions where **PGE1 does not help and can even worsen it** (the block is in the pulmonary veins, not the duct), and where the CXR mimics primary lung disease. Recognise it as a surgical emergency, not a medical one.
- **Example / exam anchor:** the Master-Exam Q20 variant is a TAPVD suprasternal-view case (vertical vein). In this lesson, TAPVR is the increased-flow lesion whose **obstructed infracardiac form is a wet-lung neonatal emergency**, and whose supracardiac form gives the **snowman** sign.
- **Hook:** "TAPVR = pulmonary veins drain the WRONG way into the systemic veins, everything mixes in the RA, so an **ASD is obligatory**. Supracardiac = **snowman**. Infracardiac (below diaphragm) = **OBSTRUCTED = wet-lung emergency**, and **PGE1 does not help**."

4) Truncus arteriosus: one common trunk over a VSD, plus 22q11

- **Synonym:** truncus arteriosus; common arterial trunk; persistent truncus.
- **The core idea first:** in truncus arteriosus the embryonic common arterial trunk **never divides** into a separate aorta and pulmonary artery. So a **single great vessel arises over a VSD** and gives off the coronaries, the aorta and the pulmonary arteries all together, guarded by **one truncal valve**. Because both ventricles empty into that common trunk through the VSD, systemic and pulmonary blood **completely mix**: the baby is **cyanotic** (mixed blood to the body) and has **increased pulmonary flow** (the low-resistance lungs steal a lot of the trunk's output), so it over-circulates the lungs and goes into heart failure early. The single truncal valve is often abnormal (stenotic or regurgitant).
- **Reads as (the causal chain):** common trunk over a VSD to both ventricles eject into one vessel to full mixing to cyanosis + plethoric lungs to early pulmonary over-circulation and congestive failure.
- **Why it matters (the association is the exam hook):** truncus is **strongly associated with 22q11.2 deletion (DiGeorge syndrome)**. Think of the DiGeorge cluster: conotruncal defects (**truncus** and interrupted aortic arch), **hypocalcaemia** (absent parathyroids), **absent / hypoplastic thymus** (immune

deficiency), and the facial features. If the exam pairs a cyanotic increased-flow newborn with hypocalcaemia or an immune problem, think truncus + 22q11.

- **Example / exam anchor:** truncus arteriosus **type 1** is a distractor in **Q5** (it is a genuine increased-flow / mixing lesion, so it is NOT the exception; the exception is D-TGA + severe PS). Remember truncus as one of the "5 Ts" and the classic **22q11 / DiGeorge** lesion.
- **Hook:** "Truncus = **one common trunk over a VSD**, both ventricles empty into it, so full mixing = cyanosis + plethoric lungs + early failure. The buzzword association is **22q11 / DiGeorge** (truncus, interrupted arch, hypocalcaemia, absent thymus)."

5) Double-outlet right ventricle (DORV): the VSD position decides the physiology (owns Q18)

- **Synonym:** DORV; both great arteries from the RV; the "chameleon" lesion.
- **The core idea first:** in DORV **both great arteries arise from the right ventricle**, so the ONLY way the left ventricle can eject is through a **VSD**. The physiology of DORV is not fixed, it is a chameleon that mimics another lesion depending on **two things**: (1) **where the VSD sits** relative to the great arteries, and (2) **whether there is pulmonary stenosis**. This is why DORV is the hardest of the bucket and why Q18 is subtle.
- **The subtypes (click each in the visual):** 1. **Subaortic VSD (VSD under the aorta):** LV blood streams preferentially to the aorta, and the RV blood goes to the pulmonary artery. This behaves like a **large VSD (acyanotic, increased flow)**, and if there is added PS it behaves exactly like **tetralogy of Fallot (cyanotic, DECREASED flow)**. So subaortic VSD = **VSD / TOF-like**. 2. **Subpulmonic VSD (VSD under the pulmonary artery) = the Taussig-Bing anomaly:** LV blood streams preferentially to the PULMONARY artery, while the RV (systemic, blue) blood goes to the aorta. That is functionally **like transposition (TGA-like)**: the poorly oxygenated blood reaches the body, so the baby is **cyanotic with INCREASED pulmonary flow**. Often the great arteries are **side by side**.
- **The PS modifier (the switch again): pulmonary stenosis DECREASES pulmonary flow** in DORV exactly as it does everywhere in this bucket. So "DORV with subaortic VSD **and PS**" behaves like TOF (dark lungs), while "DORV with subpulmonic VSD, **no PS**" (Taussig-Bing) behaves like TGA (plethoric lungs). Match the lung fields to the PS.
- **Why it matters (the whole of Q18, with the conflict flagged):** Q18 describes a neonate with cyanosis, **no increase in heart size, INCREASED pulmonary vascular markings, a loud S2 and no significant murmur, and echo showing SIDE-BY-SIDE great arteries.** Reading the physiology: increased markings mean flow is HIGH (so NO significant PS), side-by-side arteries + TGA-like physiology point to a **subpulmonic VSD (Taussig-Bing)**, and the loud S2 with no murmur fits unobstructed transposition-type streaming. That picture fits **option a (DORV, subpulmonic VSD, no PS)**. The provided **key says D (DORV, subaortic VSD, with PS)**, but PS would DECREASE the markings and contradict the stem, so the key and the clinical picture conflict.
- **Example / exam anchor: Master Exam Q18, key = D**, but flag it: the increased markings + no murmur + loud S2 + side-by-side arteries argue for **a (Taussig-Bing, subpulmonic VSD, no PS, TGA-like increased flow)**. Teach the physiology cleanly and **confirm the intended answer with the course**, exactly the way C2 / other lessons flag key-vs-clinical conflicts. Do NOT just assert D.
- **Hook:** "DORV = both arteries off the RV, so the LV escapes through a VSD, and the VSD POSITION decides everything. **Subaortic VSD = VSD / TOF-like**; **subpulmonic VSD = Taussig-Bing = TGA-like** (cyanotic, increased flow, side-by-side arteries). **PS decreases flow.** Q18 stem (increased markings, no murmur, side-by-side) fits Taussig-Bing / option a, but the key says D: confirm with the course."

6) Single ventricle: one chamber, complete mixing

- **Synonym:** single / common / univentricular heart; functionally single ventricle.

- **The core idea first:** in a single ventricle there is **one functional pumping chamber** that receives blood from **both atria**, so systemic (blue) and pulmonary (red) venous blood **completely mix** in that one ventricle before being pumped out. The result is a **cyanotic** baby (the aorta carries mixed blood) with **usually increased pulmonary flow** (plethora), unless there is associated **pulmonary stenosis** to limit it (the PS switch again).
- **Reads as (the causal chain):** both atria empty into one ventricle to complete mixing to cyanosis + (without PS) increased pulmonary flow; add PS and the flow drops and the baby is more cyanotic but with darker lungs.
- **Why it matters (the pathway ahead):** a single functional ventricle cannot be given a two-ventricle repair, so it is committed to the **single-ventricle (Glenn to Fontan) staged palliation pathway** covered in C6. The balance of pulmonary vs systemic flow (and whether a band or a shunt is needed) is set by how much PS there is.
- **Example / exam anchor:** in **Q5**, "single ventricle with malposed great vessels" is a genuine **increased-flow / complete-mixing** lesion, so it is a DISTRACTOR, NOT the exception (the exception is D-TGA + severe PS). Single ventricle is the purest complete-mixing lesion in the bucket.
- **Hook:** "Single ventricle = one chamber, both atria empty in, **complete mixing** = cyanosis + (no PS) increased flow. It is a distractor in Q5 (still increased flow) and the entry to the **Fontan pathway** (C6). PS is what would drop its flow."

⚑ Treated differently (the exceptions to flag)

- **Pulmonary stenosis flips the bucket (the whole of Q5, and half of Q18).** Every lesion here is an increased-flow lesion by default, but add significant PS and it moves into the DECREASED (dark-lung) bucket. That is why **D-TGA with severe PS** is the EXCEPTION in Q5, and why "DORV subaortic + PS" behaves like TOF. Read the lung fields: plethoric = no significant PS; dark = PS is present.
- **D-TGA is duct-dependent for MIXING, not for pulmonary flow.** In C2 the duct was kept open to feed the lungs (pulmonary blood flow). In D-TGA the lungs are already over-perfused; the duct (and the torn atrial septum) is kept open so the two parallel circuits can MIX. Same drug (PGE1), different job.
- **The Rashkind balloon atrial septostomy is a mixing procedure, not a repair.** It tears the atrial septum open to improve atrial-level mixing and buy time; the definitive fix is the **arterial switch**. Do not confuse the emergency palliation with the corrective surgery.
- **Obstructed infracardiac TAPVR is a SURGICAL emergency, and PGE1 does NOT help.** Unlike the duct-dependent decreased-flow lesions, the block here is in the pulmonary VEINS. The chest X-ray mimics primary lung disease (wet, congested), the baby is profoundly cyanotic, and the only treatment is emergency surgery. Giving PGE1 does nothing for a venous obstruction and may worsen pulmonary over-circulation.
- **A cyanotic newborn with NO murmur and a loud single S2 is a red flag for transposition-type physiology.** Nothing is obstructed, so there is no murmur; the anteriorly placed aorta gives a loud single S2. That points to D-TGA (or Taussig-Bing DORV), not to an obstructive lesion. This is the exact clue in Q18 that argues for the TGA-like (Taussig-Bing / option a) reading.
- **DORV is a chameleon: name the VSD position and the PS before you name the physiology.** Do not memorise DORV as one disease. Subaortic VSD to VSD / TOF-like; subpulmonic VSD (Taussig-Bing) to TGA-like; then let PS decide the lung fields. This is the discipline Q18 is testing.
- **Truncus is the 22q11 / DiGeorge lesion.** If a cyanotic increased-flow newborn also has hypocalcaemia, seizures, or an immune / thymic problem, think truncus arteriosus (and interrupted aortic arch) with a 22q11.2 deletion.

Mnemonics to lock in

- **The whole lesson in one line: "Open road to the lungs but the red and blue never separate: blue baby, WET (plethoric) lungs. D-TGA = two parallel circuits that survive only by MIXING (PGE1 + Rashkind balloon, then arterial switch). Add pulmonary stenosis and any of them flips into the DECREASED (dark-lung) bucket."**
- **The increased-flow lesions (the 5 Ts + DORV): "Transposition, TAPVR, Truncus, Tricuspid atresia (big VSD, no PS), plus Single ventricle and DORV. All mix, all plethoric, until PS makes them dark."**
- **D-TGA: "Aorta off the RIGHT ventricle, PA off the LEFT: two parallel loops, egg on a string, loud single S2, no murmur. Rescue = MIXING (ASD / VSD / PDA), so PGE1 + Rashkind balloon, then the arterial switch."**
- **The PS switch (answers Q5): "Increased-flow lesion + significant pulmonary stenosis = DECREASED flow. D-TGA + severe PS is the EXCEPTION in Q5 because the PS drops it into the dark-lung bucket."**
- **TAPVR: "Pulmonary veins drain the WRONG way, so an ASD is obligatory. Supracardiac = SNOWMAN. Infracardiac = OBSTRUCTED = wet-lung emergency, and PGE1 does NOT help."**
- **Truncus: "One trunk over a VSD, full mixing, plethoric lungs, early failure. Buzzword: 22q11 / DiGeorge (truncus, interrupted arch, hypocalcaemia, absent thymus)."**
- **DORV (answers Q18): "Both arteries off the RV, LV escapes through the VSD. Subaortic VSD = TOF / VSD-like. Subpulmonic VSD = Taussig-Bing = TGA-like (cyanotic, plethoric, side-by-side arteries). PS decreases flow. Q18 stem fits Taussig-Bing / option a, key says D: confirm with course."**
- **Single ventricle: "One chamber, both atria in, complete mixing, plethoric (no PS). Distractor in Q5, entry to the Fontan pathway (C6)."**

🧪 Quiz yourself (commit your answer, key is sealed at the bottom)

Q5. Congenital cyanotic heart diseases with INCREASED pulmonary blood flow include all of the following EXCEPT: A. Tricuspid atresia with ASD and VSD B. D-transposition of the great arteries (D-TGA) with VSD and severe pulmonary stenosis C. Single ventricle with malposed great vessels D. Truncus arteriosus type 1

Q18. A neonate has cyanosis, is mildly tachypnoeic with retractions, and has an O2 saturation of 65%. The chest X-ray shows no increase in cardiothoracic ratio with INCREASED pulmonary vascular markings. There is a loud S2 and no significant murmur. Echocardiography shows the great arteries lying SIDE BY SIDE. The most likely diagnosis is: a. DORV with subpulmonic VSD and no PS b. DORV with subpulmonic VSD and PS c. DORV with subaortic VSD and no PS d. DORV with subaortic VSD and PS e. DORV with subaortic VSD and suprasystemic pulmonary hypertension

Q-A (D-TGA parallel circuits). In D-transposition of the great arteries, why is the lesion incompatible with life unless there is an ASD, VSD or PDA? A. Because the pulmonary artery is severely stenosed B. Because the aorta arises from the RV and the pulmonary artery from the LV, creating two PARALLEL circuits that never meet, so oxygenated blood can reach the body only through a mixing point C. Because the ductus arteriosus is the only source of pulmonary blood flow D. Because the pulmonary veins drain the wrong way E. Because both great arteries arise from the right ventricle

Q-B (obstructed infracardiac TAPVR). A newborn is deeply cyanotic and in respiratory distress with a wet, congested ("white-out") chest X-ray. Echo shows the pulmonary veins draining below the diaphragm into an obstructed descending vein. The correct immediate management is: A. Start prostaglandin E1 to keep the duct open B. Give indomethacin to close the duct C. Emergency surgical repair (PGE1 does not relieve a pulmonary VENOUS obstruction) D. Digoxin and diuretics only E. Reassure, this is transient tachypnoea of the newborn

Q-C (truncus association). A cyanotic neonate with increased pulmonary markings and early heart failure is found to have hypocalcaemia and an absent thymus. The most likely cardiac lesion and its genetic association are: A. D-TGA with a 21 trisomy B. TAPVR with Turner syndrome C. Truncus arteriosus with a 22q11.2 deletion (DiGeorge) D. Tricuspid atresia with Noonan syndrome E. Single ventricle with Marfan syndrome

Q-D (the PS switch / bucket discriminator). Adding significant pulmonary stenosis to a D-TGA (or to a DORV) does what to the pulmonary blood flow, and why does that matter for Q5? A. Increases flow, so it stays in the increased-flow bucket B. Has no effect on flow C. DECREASES flow, moving the lesion into the decreased-flow (dark-lung) bucket, which is exactly why "D-TGA with severe PS" is the EXCEPTION in Q5 D. Causes left-to-right shunting E. Closes the ductus arteriosus

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🔒 Answer key (only after you have committed)

- **Q5 = B (D-TGA with VSD and severe pulmonary stenosis).** This bucket is defined by INCREASED pulmonary flow, but the **severe pulmonary stenosis obstructs the road to the lungs**, so this particular lesion has **DECREASED** flow (it behaves like the C2 / dark-lung group), which makes it the odd one out. The other three are genuine increased-flow / mixing lesions: **tricuspid atresia with a large VSD and no PS** over-circulates the lungs, **single ventricle with malposed great vessels** is complete mixing with plethora, and **truncus arteriosus type 1** is a common trunk over a VSD with full mixing and increased flow. The PS is the switch that drops (B) out of the bucket.
- **Q18: provided key = D (DORV, subaortic VSD, with PS). ⚠ CONFLICT with the clinical picture.** Read the stem physiologically: **INCREASED pulmonary vascular markings** mean flow is HIGH, so there is **no significant pulmonary stenosis** (PS would make the markings DECREASED / dark); **side-by-side great arteries** with a **loud single S2 and no murmur** and TGA-like cyanosis point to a **subpulmonic VSD (the Taussig-Bing anomaly)**, where the systemic (blue) RV blood streams to the aorta. That fits **option a (DORV, subpulmonic VSD, no PS)**, a TGA-like increased-flow lesion. Option D (subaortic VSD + PS) would behave like TOF with DECREASED markings, contradicting the stem. So the defensible clinical answer is **a**, while the course key says **D**: **confirm the intended answer with the course** (flagged, exactly like the C2 / other key-vs-clinical conflicts). What you MUST take from Q18 either way: name the **VSD position** (subaortic = TOF / VSD-like; subpulmonic = Taussig-Bing = TGA-like) and the **PS status** (PS decreases flow) before you name the physiology.
- **Q-A = B.** D-TGA is a ventriculo-arterial discordance: the **aorta arises from the RV and the pulmonary artery from the LV**, so deoxygenated blood recirculates through the body and oxygenated blood recirculates through the lungs in **two parallel circuits that never meet**. Survival depends entirely on a **mixing point (ASD, VSD or PDA)** letting oxygenated blood cross into the systemic loop, which is why PGE1 (keep the duct open) and the Rashkind balloon atrial septostomy (tear the atrial septum) are the emergency moves, and the arterial switch is the definitive repair. (A) and (C) describe decreased-flow / TOF-type physiology, (D) is TAPVR, and (E) is DORV.
- **Q-B = C (emergency surgical repair).** In obstructed infracardiac TAPVR the block is in the **pulmonary VEINS** (the descending vein is squeezed as it passes the diaphragm), so blood dams back into the lungs, giving a wet, congested chest X-ray and profound cyanosis. **PGE1 does not relieve a venous obstruction** (it opens the duct, which is irrelevant here) and diuretics/digoxin do not fix the anatomy: it is a **surgical emergency**. (A) and (D) delay definitive care, (B) is harmful, and (E) misreads a life-threatening lesion as benign.

- **Q-C = C (truncus arteriosus with a 22q11.2 deletion / DiGeorge).** Truncus is a **conotruncal** defect, and the DiGeorge (22q11.2 deletion) cluster is conotruncal heart disease + **hypocalcaemia** (absent parathyroids) + **absent / hypoplastic thymus** (immune deficiency) + facial features. A cyanotic increased-flow newborn with hypocalcaemia and an absent thymus is the classic truncus + 22q11 pairing. The other options mismatch the lesion with the wrong syndrome.
- **Q-D = C.** Pulmonary stenosis **obstructs the road to the lungs, so it DECREASES pulmonary blood flow** and moves the lesion out of the increased-flow bucket into the decreased-flow (dark-lung) bucket. That is precisely why **"D-TGA with VSD and severe PS" is the EXCEPTION in Q5**: the severe PS makes its flow decreased, not increased. The same rule explains why "DORV subaortic + PS" behaves like TOF while "DORV subpulmonic, no PS" (Taussig-Bing) behaves like TGA.

Diagrams in the visual (hand-built inline SVG, nothing embedded from the web)

1. **INCREASED-flow physiology splitter (concept 1):** the open-road-to-the-lungs / failure-to-separate chain shown as a mini schematic with plethoric (congested) lungs, an animated mixing example, the C3 lesion list (the 5 Ts + DORV), and the PS-flips-the-bucket rule tied back to the C1 INCREASED bucket.
2. **Interactive D-transposition diagram (concept 2):** a hand-drawn SVG showing the aorta off the RV and the pulmonary artery off the LV as two PARALLEL circuits (blue loop and red loop that never meet). Clicking ASD, VSD or PDA opens that mixing point and animates oxygenated blood crossing into the systemic loop (the baby pinks up), showing why mixing is life-sustaining and why PGE1 + Rashkind are used.
3. **TAPVR card (concept 3):** a hand-drawn schematic of the pulmonary veins draining the WRONG way into the systemic veins with the obligatory ASD, plus a toggle between the supracardiac "snowman" sign and the OBSTRUCTED infracardiac (below-diaphragm) emergency with wet lungs.
4. **DORV card (concept 4):** a side-by-side comparison of subaortic VSD (TOF / VSD-like) vs subpulmonic VSD Taussig-Bing (TGA-like), with the PS-decreases-flow modifier and the Q18 key-vs-clinical conflict called out.
5. **Truncus + 22q11 note (concept 4 area):** a hand-drawn single common trunk over a VSD giving off the aorta, pulmonaries and coronaries, with the DiGeorge (22q11.2) association box.

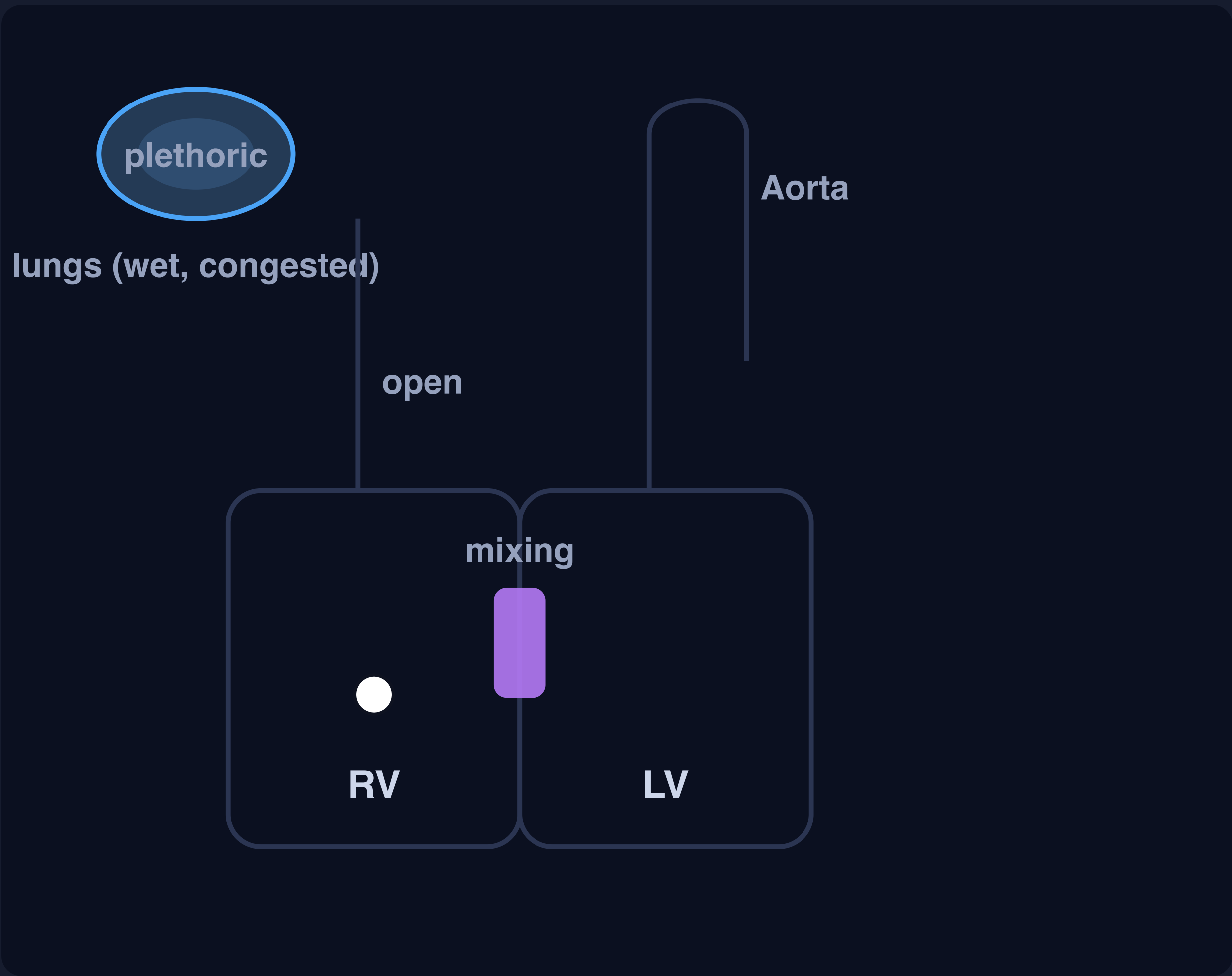
C3 - Interactive visual (rendered): diagrams & schematics

The live, toggleable version is the HTML file on the CHD hub. Below is a static render of its default view; open the hub to step through the other toggle states.

C3 - Cyanotic CHD with INCREASED pulmonary blood flow

The INCREASED-flow branch of C1 and the mirror image of C2. The road to the lungs is OPEN (so lung fields are plethoric / wet), but oxygenated and deoxygenated blood are not separated (transposition or complete mixing), so the baby is still blue. Anchors: Q5 all of these have increased flow EXCEPT D-TGA with VSD and severe pulmonary stenosis (the PS obstructs flow, dropping it into the decreased bucket); Q18 a cyanotic neonate with increased markings, loud S2, no murmur and side-by-side arteries (key says D, but the physiology fits a Taussig-Bing subpulmonic VSD with no PS = option a: confirm with the course).

1. THE INCREASED-FLOW PHYSIOLOGY: OPEN ROAD TO THE LUNGS, BUT THE TWO BLOODS NEVER SEPARATE (CLICK TO RUN THE MIXING)



Flow to the lungs is OPEN (wide pulmonary artery, plethoric lungs), but red and blue are not separated, so blood **mixes** and the aorta carries desaturated blood. Hand-drawn, nothing embedded.

Blue baby, WET lungs

Chain: road to the lungs is OPEN (plethoric lungs) but the circulation fails to separate red from blue (transposed or completely mixed) to the aorta carries desaturated blood to **cyanosis + increased (plethoric) lung fields**.

The switch: add significant **pulmonary stenosis** and the road to the lungs is now obstructed, so the flow becomes DECREASED (dark lungs). That flip is the whole of Q5.

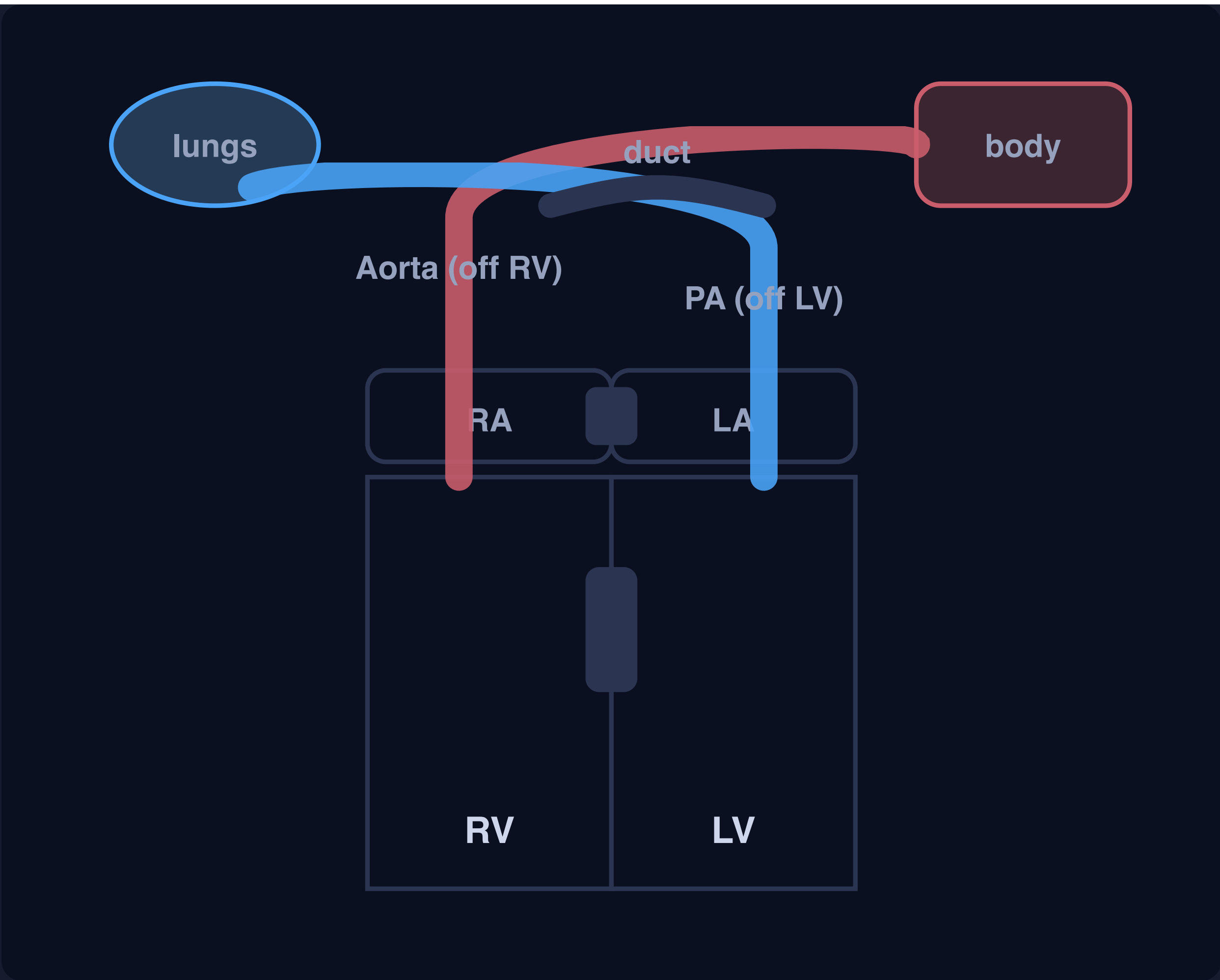
INCREASED (plethoric, wet) = C3 lesions (the 5 Ts + DORV)

- Transposition (D-TGA), the anchor
- TAPVR (obstructed infracardiac = emergency)
- Truncus arteriosus (22q11 / DiGeorge)
- Tricuspid atresia with big VSD, no PS
- Single ventricle (complete mixing)
- DORV (chameleon: VSD position + PS decide)

Q5: all of these are INCREASED-flow EXCEPT D-TGA with VSD and severe pulmonary stenosis. The severe PS obstructs the road to the lungs, so that lesion has DECREASED flow (it behaves like the C2 / dark-lung group). PS is the switch that flips the bucket.

2. D-TRANSPOSITION: TWO PARALLEL CIRCUITS THAT SURVIVE ONLY BY MIXING (CLICK A MIXING POINT TO RESCUE THE BABY)

No mixing (two parallel loops) | ASD (atrial mixing / Rashkind) | VSD (ventricular mixing) | PDA (ductal mixing / PGE1)



Aorta off the RIGHT ventricle, PA off the LEFT: the blue loop (body to RV to aorta to body) and the red loop (lungs to LV to PA to lungs) never meet. Only a MIXING point lets oxygen reach the body.

No mixing: two parallel circuits

Aorta off the RV, PA off the LV. The blue loop (body to RV to aorta to body) and the red loop (lungs to LV to PA to lungs) run in **parallel and never meet**. Oxygen never reaches the body. Incompatible with life as it stands.

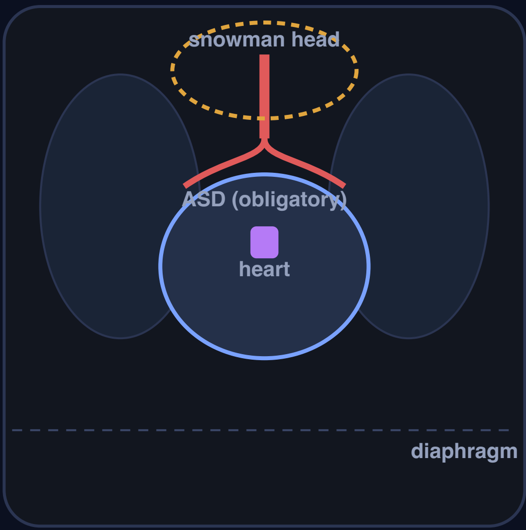
baby: deep blue (no mixing)

Test signature: egg on a string (narrow mediastinum, great arteries front-to-back) + increased flow + loud single S2, no murmur. Rescue = MIXING: PGE1 keeps the duct open and the Rashkind balloon atrial septostomy tears the atrial septum; the arterial switch is the definitive repair. Here the duct is kept open for MIXING, not for pulmonary flow.

3. TAPVR: THE PULMONARY VEINS DRAIN THE WRONG WAY (TOGGLE THE TWO FORMS)

Supracardiac (snowman, not obstructed)

Infracardiac (below diaphragm, OBSTRUCTED)



Supracardiac TAPVR (snowman)

The pulmonary veins do not reach the LA. They drain the oxygenated lung blood the WRONG way into the systemic veins, so everything mixes in the right atrium. An **ASD is obligatory**: it is the only way blood reaches the left heart and the body.

Supracardiac / not obstructed: the dilated vertical vein + SVC form the classic **snowman (figure-of-8)** heart on CXR. Increased flow, cyanosis, not an immediate emergency.

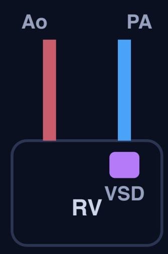
The trap: infracardiac (below-diaphragm) TAPVR is usually OBSTRUCTED (the descending vein is squeezed at the diaphragm), so it is a neonatal emergency with pulmonary oedema, a wet chest X-ray that mimics lung disease. PGE1 does NOT help (the block is in the veins, not the duct): it needs emergency surgery.

4. DORV: THE VSD POSITION DECIDES THE PHYSIOLOGY (SUBAORTIC = TOF-LIKE, SUBPULMONIC TAUSSIG-BING = TGA-LIKE)

Subaortic VSD

Behaves like a VSD / TOF. LV blood streams to the aorta. Add PS and it is exactly TOF (cyanotic, DECREASED flow, dark lungs).

VSD / TOF-like



Subpulmonic VSD (Taussig-Bing)

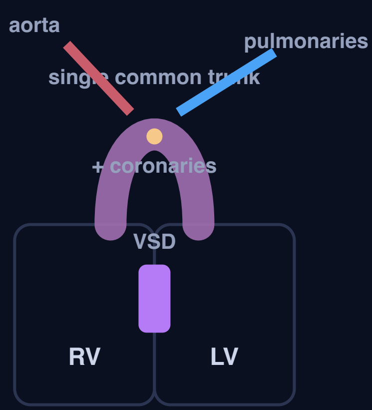
Behaves like transposition (TGA-like). LV blood streams to the PA; the systemic (blue) RV blood goes to the aorta, so the baby is cyanotic with INCREASED flow. Great arteries often **side by side**.

TGA-like (increased flow)

The PS switch again: in DORV, pulmonary stenosis DECREASES pulmonary flow. So "subaortic VSD + PS" behaves like TOF (dark lungs) and "subpulmonic VSD, no PS" (Taussig-Bing) behaves like TGA (plethoric lungs). Match the lung fields to the PS.

Q18 (key = D, DORV subaortic VSD + PS). ⚠ CONFLICT with the stem: increased markings mean flow is HIGH, so NO significant PS; side-by-side arteries + loud S2 + no murmur = a subpulmonic VSD (Taussig-Bing), no PS = option a (TGA-like). Option D's PS would make the markings DECREASED, contradicting the stem. Teach the physiology and confirm the intended answer with the course. Do not just assert D.

5. TRUNCUS ARTERIOSUS: ONE COMMON TRUNK OVER A VSD, PLUS 22Q11 (DIGEORGE)



One vessel does everything. The embryonic trunk never divides, so a single common arterial trunk overrides a VSD and gives off the aorta, the pulmonaries and the coronaries through one (often abnormal) truncal valve.

Both ventricles empty into it, so systemic and pulmonary blood **fully mix**: cyanosis + increased flow + early pulmonary over-circulation and heart failure.

22q11.2 deletion (DiGeorge)

- Conotruncal defects: **truncus** + interrupted aortic arch
- **Hypocalcaemia** (absent parathyroids)
- **Absent / hypoplastic thymus** (immune deficiency)
- Characteristic facies

In Q5, **truncus arteriosus type 1** is a genuine increased-flow / mixing lesion, so it is a distractor, NOT the exception. The exception is D-TGA + severe PS.

C4 - Acyanotic (left-to-right shunt) lesions: the physiology of over-circulation, atrial septal defect (types), ventricular septal defect, patent ductus arteriosus, atrioventricular septal defect (Down), and Eisenmenger as the endpoint (full lesson)

> Visual: `study/visuals/C4-acyanotic-shunts.html` (the acyanotic left-to-right shunt splitter that carries over from C1, where the baby is PINK not blue because oxygenated left-heart blood crosses back to the right side and over-circulates the lungs; an interactive shunt heart where clicking ASD (atrial level), VSD (ventricular level) or PDA (aorta to pulmonary artery) opens that shunt and animates oxygenated blood crossing from the LEFT heart to the RIGHT heart and up to the lungs; an ASD card contrasting secundum (RIGHT axis, RSR' in V1, wide FIXED split S2) vs primum (LEFT axis, AVSD spectrum); a VSD closure-indications card with the Q9 trap (a high pressure gradient across the VSD means it is a small restrictive VSD, so you do NOT close it); a PDA card (continuous machinery murmur, the Q4 silent-duct trap); an AVSD + Down syndrome + LEFT axis card; and an Eisenmenger endpoint mini-panel showing the shunt reversing to right-to-left, the patient turning cyanotic, and the defect no longer being closeable). This is the ACYANOTIC (left-to-right shunt) branch of the master framework built in C1: the baby is PINK, the shunt is oxygenated blood going the wrong way (left heart back to right heart / lungs), and the driver is the fall in pulmonary vascular resistance over the first weeks (C1). Directly tested (Master Exam / Congenital mock): **Q3** a 52 year old woman with SOB, RAD + RVH, a dilated RV and high-normal PA pressure (answer **B, secundum ASD** presenting in adulthood with RV volume overload and early pulmonary hypertension; a primum ASD would give LEFT axis). **Q4** PDA in adults, all true EXCEPT (answer **C**: you do NOT close truly silent / tiny ducts; A, B and D are true). **Q7** a 3 month old girl with trisomy 21, poor weight gain, tachypnoea, a low LLSB murmur, cardiomegaly and LEFT axis deviation (answer **B, complete AVSD**: Down + left axis + failure is the giveaway). **Q9** indications for VSD closure, all EXCEPT (answer **C**: a 100 mmHg gradient across the VSD means it is a small restrictive VSD, which is NOT an indication to close; the real indications are LV volume overload / dilatation, aortic regurgitation from valve prolapse, and failure to thrive).

Step 0. The big theory: a left-to-right shunt is oxygenated blood taking a wrong turn back to the lungs

C2 and C3 were the whole CYANOTIC (blue baby) half of C1: either the road to the lungs was blocked (C2, dark lungs) or the red and blue never separated (C3, wet lungs). This lesson is the ACYANOTIC half, and the mechanism is completely different. Here nothing is cyanotic to start with, because the defect lets **oxygenated (red) left-heart blood cross back into the right heart or the pulmonary artery** and recirculate through the lungs a second time. The baby is PINK. Everything in this bucket hangs off five linked ideas, in order:

- **The shunt goes LEFT to RIGHT because the pressures make it.** After birth the left heart is the high-pressure pump (it drives the whole body) and the right heart / lungs are the low-pressure side. So whenever there is a hole (atrial, ventricular) or a persistent connection (the duct), blood follows the pressure gradient and flows from the high-pressure LEFT side to the low-pressure RIGHT side. That oxygenated blood is already saturated, so there is no cyanosis, just wasted, recirculated flow.

- **The size of the shunt is set by the fall in pulmonary vascular resistance (this is the C1 link).** At birth the PVR is still high, so the left-to-right shunt is small and the baby often looks well for the first days to weeks. As the **PVR falls over the first weeks** (exactly the C1 first-breath transition), the right side and lungs become an even lower-pressure sink, so more and more blood shunts left to right. That is why these babies typically present at a few weeks to a few months, not on day one, with failure to thrive and tachypnoea as the shunt grows.
- **A big shunt over-circulates the lungs and floods the heart, so it causes heart failure.** The extra shunted blood goes round the lungs again, comes back to the left heart, and volume-overloads it. The result is **pulmonary plethora + volume-overload heart failure**: tachypnoea, poor feeding, sweating, failure to thrive in an infant, or exertional dyspnoea in an adult. The lung fields are plethoric on the chest X-ray, like C3, but the baby is PINK, not blue.
- **Restrictive (small) vs non-restrictive (large) is the whole clinical spectrum.** A small hole is RESTRICTIVE: it offers high resistance, so only a little blood crosses (small shunt) but at a big pressure drop, which makes a LOUD murmur, and it often causes no failure and may close on its own. A large hole is NON-RESTRICTIVE: the two sides communicate freely, pressures equalise, a huge volume shunts across, and the baby goes into failure and is at risk of pulmonary hypertension. Loud murmur + well baby usually means small; quiet or soft murmur + a sick, breathless baby can mean a big, non-restrictive defect.
- **The dreaded endpoint of an unrepaired big shunt is Eisenmenger: the shunt REVERSES and the patient turns blue.** If a large left-to-right shunt is left alone for years, the chronically over-circulated pulmonary arteries thicken and scar, so the PVR climbs until it is fixed and suprasystemic. Once the right-side pressure exceeds the left, the **shunt reverses to right-to-left**, deoxygenated blood now crosses into the systemic circulation, and the patient becomes **cyanotic (Eisenmenger)**. At that point the defect can no longer be closed (closing it would leave the right heart pumping against a fixed, unbeatable resistance). This is the acyanotic-turns-cyanotic bridge into adult CHD (C7).

Hook: **"Acyanotic = a left-to-right shunt: oxygenated left-heart blood is pushed by pressure back into the low-pressure right heart / lungs, so the baby is PINK but the lungs are over-circulated. The shunt GROWS as the PVR falls over the first weeks (C1), so these present at weeks to months with failure, not on day one. Small = restrictive = loud murmur, often closes; large = non-restrictive = failure + pulmonary hypertension. Leave a big one for years and it flips: fixed pulmonary hypertension REVERSES the shunt to right-to-left, the patient turns blue (Eisenmenger), and it can no longer be closed."**

Step 1. The map: the six ideas of C4

These numbers are the SAME as the concept numbers in Step 2 (map row 1 = concept 1, and so on).

#	Idea	The one-line mechanism	Exam anchor
1	**The acyanotic left-to-right shunt physiology (over-circulation)**	The LEFT heart is the high-pressure pump, so a hole (ASD, VSD) or a persistent duct (PDA) lets **oxygenated blood shunt LEFT to RIGHT** back to the lungs. No cyanosis (the blood is already red), but the lungs are **over-circulated (plethoric)**. The shunt **grows as PVR falls** over the first weeks (C1), so a big one causes failure and, long-term, pulmonary hypertension	This is the C1 ACYANOTIC bucket. **Restrictive (small) = loud murmur, small shunt, often closes; non-restrictive (large) = big shunt, failure, pulmonary hypertension.** The spectrum drives Q9 (VSD) and Q4 (PDA)
2	**Atrial septal defect (ASD): secundum vs primum (owns Q3)**	A hole in the atrial septum shunts LA to RA, so the RIGHT heart carries the extra volume: **RV volume overload**. **Secundum** (commonest) gives **RIGHT axis + RSR' in V1** (incomplete RBBB); **primum** (AVSD spectrum) gives **LEFT axis**. Classic sign: a **wide, FIXED split S2**. Often silent in childhood, can present in ADULTHOOD	**Q3 (answer B)**: a 52 year old with SOB, RAD + RVH, a dilated RV and high-normal PA pressure = a **secundum ASD** presenting late with RV volume overload and early pulmonary hypertension. A primum ASD would give LEFT axis, so it is not the answer

#	Idea	The one-line mechanism	Exam anchor
3	**Ventricular septal defect (VSD): the commonest defect, and the closure rules (owns Q9)**	A hole in the ventricular septum shunts LV to RV, a **harsh pansystolic murmur at the left sternal border**. **Small (restrictive) = LOUD murmur, small shunt, often closes spontaneously; large (non-restrictive) = over-circulates the lungs, failure, pulmonary hypertension**	**Q9 (answer C)**: indications to CLOSE a VSD are LV volume overload / dilatation, aortic regurgitation from valve prolapse, and failure to thrive. A **peak gradient of 100 mmHg across the VSD is NOT an indication**, because a big gradient means it is a **small restrictive VSD** (high resistance, small shunt)
4	**Patent ductus arteriosus (PDA): machinery murmur, and the silent-duct trap (owns Q4)**	The ductus fails to close, so blood shunts from the **aorta to the pulmonary artery** throughout the cardiac cycle: a **continuous "machinery" murmur** and a **wide pulse pressure**. A **restrictive PDA implies a normal PA pressure**	**Q4 (answer C, the EXCEPT)**: a restrictive PDA implies normal PA pressure (true), infective **endarteritis** is a dreaded complication (true), and small PDAs can be closed percutaneously (true), but **NOT every silent duct must be closed**: a truly silent, tiny PDA is generally left alone
5	**Atrioventricular septal defect (AVSD): Down syndrome + LEFT axis (owns Q7)**	The endocardial cushions fail to fuse, so there is a **common AV valve + a primum ASD + an inlet VSD**: a big central left-to-right shunt with early failure. ECG shows **LEFT axis deviation**	**Q7 (answer B)**: a 3 month old with **trisomy 21 (Down)**, poor weight gain, tachypnoea, a low LLSB murmur, cardiomegaly and **LEFT axis deviation** = a **complete AVSD**. Down + left axis + failure is the giveaway (VSD or secundum ASD would not give left axis)
6	**Eisenmenger: the endpoint where the shunt reverses and the patient turns blue**	Years of over-circulation thicken and scar the pulmonary arteries, so the **PVR becomes fixed and suprasystemic**, the **shunt REVERSES to right-to-left**, and the acyanotic patient becomes **CYANOTIC**. Once fixed, the defect can **no longer be closed**	The bridge from acyanotic to cyanotic and into adult CHD (**C7**). An unrepaired big ASD / VSD / PDA / AVSD is what gets you here. Recognise it: an adult with a known shunt who is now cyanotic and clubbed = do NOT close the defect

The one rule that ties it together: every lesion here is a LEFT-to-RIGHT shunt of oxygenated blood driven by the higher left-heart pressure, so the baby is PINK but the lungs are over-circulated, and the shunt GROWS as the PVR falls over the first weeks. Small (restrictive) defects make a loud murmur, shunt little, and often close; large (non-restrictive) defects over-circulate the lungs, cause failure, and drive pulmonary hypertension. If a big one is left for years, the pulmonary resistance becomes fixed, the shunt reverses, and the patient turns blue (Eisenmenger) and can no longer be closed. That single spine answers Q3, Q4, Q7 and Q9.

Step 2. The concepts you must know (numbered to match the map in Step 1: concept N = map row N)

1) The acyanotic left-to-right shunt physiology: oxygenated blood pushed back to the lungs (owns the restrictive-vs-non-restrictive spine)

- **Synonym:** acyanotic congenital heart disease; left-to-right shunt lesions; the over-circulation / pulmonary-plethora group (pink, not blue).
- **The core idea first:** in C2 and C3 the baby was BLUE. Here the baby is PINK, because the defect does not desaturate the systemic blood, it just lets already-oxygenated **left-heart blood cross back into the low-pressure right heart or pulmonary artery**. After birth the left ventricle is the high-pressure systemic pump and the right heart / lungs are the low-pressure circuit, so blood follows the gradient and shunts **LEFT to RIGHT**. The consequences are all about volume: the shunted blood goes round the lungs a second time (pulmonary over-circulation, so the lung fields are **plethoric**), comes back to the left heart, and volume-overloads it. There is no cyanosis, but there is wasted flow, a murmur, and, if the shunt is big, heart failure.
- **Reads as (the causal chain):** 1. After birth the LEFT heart is high-pressure and the RIGHT heart / lungs are low-pressure. 2. A defect (ASD, VSD) or a persistent duct (PDA) lets **oxygenated blood shunt from**

the high-pressure LEFT side to the low-pressure RIGHT side. 3. That blood **over-circulates the lungs** (plethora) and volume-overloads the receiving chamber, so the murmur and, if the shunt is big, **heart failure** appear. 4. The shunt **grows as the PVR falls over the first weeks** (the C1 transition), which is why these babies present at weeks to months, not on day one. 5. Left alone for years, chronic over-circulation drives **pulmonary hypertension**, and eventually the shunt can **reverse (Eisenmenger, concept 6)**.

- **The restrictive vs non-restrictive spectrum (this is the exam spine for Q9 and Q4):** a **RESTRICTIVE (small)** defect offers high resistance, so only a small volume crosses but at a big pressure drop. That gives a **LOUD murmur, a small shunt, no failure**, and it often **closes spontaneously**, and its PA pressure stays normal. A **NON-RESTRICTIVE (large)** defect lets the two sides communicate freely, so pressures equalise, a **huge volume shunts across**, and the baby goes into **failure** and is at risk of **pulmonary hypertension**. Counterintuitively, a loud murmur usually means a SMALL (restrictive) defect, and a soft murmur in a sick breathless baby can mean a big non-restrictive one.
- **Why it matters (the whole framework):** this is the C1 acyanotic bucket, the fourth and last box of the master sort (decreased-flow cyanotic = C2, increased-flow cyanotic = C3, acyanotic left-to-right shunt = C4, and obstructive = C5). Everything downstream (which lesion, when to close, when it is too late) is read off this physiology.
- **Example / exam anchor:** the restrictive-vs-non-restrictive rule is what makes **Q9 (a 100 mmHg gradient across a VSD is a small restrictive VSD, do NOT close)** and **Q4 (a restrictive PDA implies a normal PA pressure)** fall out. Both hinge on: a big pressure drop across the defect = restrictive = small shunt.
- **Hook:** "Acyanotic = a LEFT-to-RIGHT shunt: the high-pressure left heart pushes oxygenated blood back into the low-pressure right heart / lungs, so the baby is PINK but the lungs are over-circulated. The shunt GROWS as PVR falls (C1). Small = restrictive = loud murmur, often closes; large = non-restrictive = failure + pulmonary hypertension."

2) Atrial septal defect (ASD): RV volume overload, and secundum vs primum (owns Q3)

- **Synonym:** ASD; atrial-level left-to-right shunt. Types: **ostium secundum** (commonest), **ostium primum** (AVSD spectrum), and **sinus venosus** (upper septum, near the SVC, often with anomalous pulmonary venous drainage).
- **The core fact first:** a hole in the atrial septum lets oxygenated blood shunt from the **left atrium to the right atrium**, so the extra volume loads the RIGHT heart: **RV volume overload** with a dilated RA and RV. In childhood an ASD is usually **asymptomatic** (the atrial-level shunt is low-pressure and well tolerated), which is exactly why it commonly slips through to **ADULTHOOD**, when the patient presents with **exertional dyspnoea, a dilated RA and RV, atrial arrhythmias (flutter / fibrillation), and eventually pulmonary hypertension**.
- **The exam signatures (memorise these):**
- A **wide, FIXED split S2**: the extra right-heart volume delays pulmonary valve closure, and because the atria are in free communication the split does not vary with respiration (it is FIXED). This is the classic ASD auscultation sign.
- **RV volume overload on the ECG: right axis deviation (RAD)** and an **RSR' pattern in V1 (incomplete right bundle branch block)** for a **secundum** ASD.
- A soft **pulmonary flow (ejection systolic) murmur** at the upper left sternal border, from the increased flow across the pulmonary valve (not from the ASD itself, which is silent).
- **Secundum vs primum (the axis is the discriminator, and it is a favourite):**
- **Ostium SECUNDUM (commonest):** mid-septum, **RIGHT axis deviation** + RSR' in V1. This is the plain ASD.

- **Ostium PRIMUM:** low septum, part of the **AVSD (endocardial cushion) spectrum**, so it behaves like the AVSD group and gives **LEFT axis deviation**. Think of primum as "an ASD that has slid down into the AVSD family".
- One line to lock it: **secundum = RIGHT axis; primum = LEFT axis.**
- **Why it matters (the whole of Q3):** Q3 is a **52 year old woman with shortness of breath**, an ECG showing **RAD + RVH**, and echo showing a **dilated RV with high-normal PA pressure**. Read it: RV volume overload (RAD, RVH, dilated RV) presenting in **adulthood** with beginning **pulmonary hypertension** = a **secundum ASD** that was silent through childhood. The RIGHT axis is the tell: a **primum** ASD would give LEFT axis, so option A (primum) is wrong. TOF (C) and truncus (D) are cyanotic neonatal lesions, not a breathless 52 year old. Answer = **B, secundum ASD**.
- **Example / exam anchor: Q3 = B (secundum ASD).** The pairing to keep: adult + RV volume overload (RAD + RVH + dilated RV) + early pulmonary hypertension + fixed split S2 = secundum ASD. Change the axis to LEFT and it becomes a primum ASD (AVSD spectrum).
- **Hook:** "ASD = LA to RA shunt = **RV volume overload**, silent in childhood, often presents in ADULTHOOD (dyspnoea, dilated RA/RV, atrial arrhythmias, pulmonary hypertension). Sign = **wide FIXED split S2**. **Secundum = RIGHT axis + RSR' in V1**; **primum = LEFT axis (AVSD spectrum)**. Q3's adult with RAD + RVH + dilated RV = secundum ASD."

3) Ventricular septal defect (VSD): the commonest defect, and when to close it (owns Q9)

- **Synonym:** VSD; ventricular-level left-to-right shunt. The single **commonest congenital cardiac defect**.
- **The core fact first:** a hole in the ventricular septum lets oxygenated blood shunt from the high-pressure **left ventricle to the right ventricle** in systole, giving a **harsh PANSYSTOLIC (holosystolic) murmur at the left sternal border**. Because this is a ventricular-level shunt driven by the big LV-to-RV systolic pressure difference, the murmur is prominent, and the physiology is dominated by the restrictive-vs-non-restrictive spectrum from concept 1.
- **Small (restrictive) vs large (non-restrictive), the whole clinical split:**
- **Small / restrictive VSD:** high resistance across a tiny hole, so a **small shunt but a big pressure drop = a LOUD murmur**, a well child, no failure, and a strong chance of **spontaneous closure**. The loud murmur is reassuring, not alarming: it means the defect is small and the pressures are still very different across it.
- **Large / non-restrictive VSD:** the ventricles communicate freely, pressures equalise, a **big volume over-circulates the lungs**, and the infant develops **heart failure and failure to thrive**, and is at risk of **pulmonary hypertension** and, eventually, **Eisenmenger** if not repaired.
- **The indications to CLOSE a VSD (memorise the list, then the trap):** you close a VSD when the shunt is doing harm, namely a **significant shunt (a high Qp:Qs)** with **LV volume overload / dilatation**, **failure to thrive**, or **aortic regurgitation from prolapse of an aortic cusp into the defect** (a doming cusp can prolapse and become incompetent). A recurrent important VSD with pulmonary hypertension is also closed while still operable.
- **The Q9 TRAP (this is the whole question):** a **peak pressure gradient of 100 mmHg ACROSS the VSD is NOT an indication to close it**. A large gradient across the defect means the LV pressure is much higher than the RV pressure at the hole, which is the signature of a **small, RESTRICTIVE VSD (high resistance, small shunt, protected lungs)**. That is the good kind, the kind that often closes on its own, so a big gradient is a reason to LEAVE it, not to close it. The genuine indications are the volume-overload and complication ones (LV dilatation, AR from prolapse, failure to thrive), not the pressure gradient.
- **Why it matters (the whole of Q9):** Q9 asks for the indication that is the EXCEPT. A **dilated left ventricle (A)**, **aortic valve prolapse with aortic regurgitation (B)** and **failure to thrive (D)** are all real indications to close. A **100 mmHg gradient across the VSD (C)** is not: it flags a small restrictive VSD, so it is the odd one out. Answer = **C**.

- **Example / exam anchor: Q9 = C.** The one line: a big gradient across a VSD = a small restrictive VSD = do NOT close. Real reasons to close = LV volume overload / dilatation, AR from cusp prolapse, failure to thrive (and a significant Qp:Qs).
- **Hook:** "VSD = the commonest defect, LV to RV shunt, **harsh pansystolic murmur at the LSB**. **Small restrictive = LOUD murmur, small shunt, often closes; large non-restrictive = failure + pulmonary hypertension.** Close for LV dilatation, AR from cusp prolapse, or failure to thrive. Q9 trap: a **100 mmHg gradient across the VSD = a small restrictive VSD = do NOT close**."

4) Patent ductus arteriosus (PDA): the machinery murmur, and the silent-duct trap (owns Q4)

- **Synonym:** PDA; persistent ductus arteriosus; aorto-pulmonary shunt at ductal level.
- **The core fact first:** the ductus arteriosus is the fetal connection between the aorta and the pulmonary artery that should close after birth (C1). If it stays open, aortic pressure exceeds pulmonary pressure throughout the cardiac cycle, so blood shunts from the **aorta to the pulmonary artery in BOTH systole and diastole**. That continuous shunt gives the two signatures: a **continuous "machinery" murmur** (loudest at the left infraclavicular / upper left sternal border, rolling through S2) and a **wide pulse pressure with bounding pulses** (the diastolic run-off into the low-pressure pulmonary artery drops the diastolic pressure).
- **The size spectrum (same restrictive rule as concept 1):** a **restrictive (small) PDA** offers high resistance, so the shunt is small and the **pulmonary artery pressure stays normal** (this is Q4 option A, and it is TRUE). A **large PDA** over-circulates the lungs, causes failure, and drives pulmonary hypertension, and if untreated can reach Eisenmenger with **differential cyanosis** (blue lower body, pink upper body, once the shunt reverses).
- **The complications and the management nuance:** a feared complication of a PDA is **infective endarteritis** (endocarditis of the ductus / pulmonary side), so a haemodynamically significant or audible PDA is closed, and **small ones can be closed percutaneously** (device / coil). The nuance the exam tests: a **truly silent, tiny PDA (found incidentally on echo, no murmur, no volume load) is generally LEFT ALONE**, because the procedural risk outweighs the negligible benefit. "Close every duct even the silent ones" is the wrong, over-aggressive statement.
- **Why it matters (the whole of Q4, an EXCEPT):** Q4 asks which statement about adult PDA is FALSE. A restrictive PDA implies a normal PA pressure (**A, true**), infective endarteritis is a dreaded complication (**B, true**), and small PDAs should be closed percutaneously (**D, true**). The false one is "**all PDAs should be closed even silent ducts (C)**": you do NOT chase truly silent, tiny ducts. Answer = **C**.
- **Example / exam anchor: Q4 = C.** The one line: continuous machinery murmur + wide pulse pressure = PDA; a restrictive PDA = normal PA pressure; endarteritis is the dread complication; but a **silent tiny duct is left alone**, so "close even silent ducts" is the false statement.
- **Hook:** "PDA = aorta-to-PA shunt in systole AND diastole = **continuous machinery murmur + wide pulse pressure / bounding pulses**. **Restrictive PDA = normal PA pressure.** Feared complication = **infective endarteritis**. Close significant or audible ducts (small ones percutaneously), but **LEAVE truly silent tiny ducts**. Q4's false statement = 'close even silent ducts'."

5) Atrioventricular septal defect (AVSD): Down syndrome + LEFT axis (owns Q7)

- **Synonym:** AVSD; atrioventricular canal defect; endocardial cushion defect; AV canal. The complete form = a common (single) AV valve.
- **The core fact first:** the endocardial cushions are the tissue that normally divides the central heart into separate atrial and ventricular septa and two separate AV valves. In AVSD they **fail to fuse**, leaving a defect at the CENTRE of the heart: a **primum ASD + an inlet VSD + a common (single) atrioventricular valve** all together. This is a big central left-to-right shunt at both atrial and ventricular levels, plus AV-valve

regurgitation, so it over-circulates the lungs heavily and causes **early heart failure and failure to thrive** in infancy.

- **The two buzzword associations (both are exam tells):**
- **Strongly associated with Down syndrome (trisomy 21).** A complete AVSD is the classic cardiac lesion of trisomy 21. If the stem says "Down / trisomy 21" plus a breathless, failing infant, think AVSD first.
- **LEFT axis deviation on the ECG.** Because it is an endocardial-cushion / primum defect, the conduction axis is shifted, giving **LEFT axis deviation** (superior axis). This is the same reason a **primum** ASD (concept 2) gives left axis: primum and AVSD are the same family.
- **The presentation:** an infant (weeks to a few months, as the PVR falls) with **poor weight gain, tachypnoea, sweating with feeds, cardiomegaly on the chest X-ray, a low-pitched systolic murmur at the lower left sternal border** (from the VSD / AV-valve regurgitation), and **LEFT axis deviation**. Repair is surgical (patch the septa, reconstruct the AV valves).
- **Why it matters (the whole of Q7):** Q7 is a **3 month old girl with trisomy 21**, poor weight gain, tachypnoea, a **low-pitched grade 2 systolic murmur at the lower left sternal border**, **cardiomegaly** on CXR, and **LEFT axis deviation** on ECG. That is the exact AVSD picture: Down + failure + LEFT axis. A plain VSD (D) or a secundum ASD (E) would NOT give left axis; coarctation (A) and PDA (C) do not fit the Down + left-axis + LLSB-murmur combination. Answer = **B, complete AVSD**.
- **Example / exam anchor: Q7 = B (complete AVSD).** The one line: **Down syndrome + LEFT axis deviation + infant heart failure = complete AVSD.** The LEFT axis is what separates it from a plain VSD or secundum ASD.
- **Hook:** "AVSD = endocardial cushions fail to fuse = **common AV valve + primum ASD + inlet VSD** = big central shunt + AV-valve regurgitation = early failure. Two tells: **Down syndrome (trisomy 21)** and **LEFT axis deviation** (same family as the primum ASD). Q7's Down infant with failure + left axis = complete AVSD."

6) Eisenmenger: the endpoint where the shunt reverses and the patient turns blue

- **Synonym:** Eisenmenger syndrome / reaction; the reversed shunt; pulmonary vascular obstructive disease from a long-standing shunt.
- **The core idea first:** Eisenmenger is not a separate defect, it is the **end-stage of any unrepaired large left-to-right shunt** (ASD, VSD, PDA or AVSD). Years of pulmonary over-circulation batter the pulmonary arteries: they undergo medial hypertrophy, intimal proliferation and eventually irreversible obliteration, so the **PVR climbs until it is fixed and suprasystemic**. Once the right-side pressure exceeds the left-side pressure, the pressure gradient across the defect **reverses**, so the **shunt turns RIGHT to LEFT**: now deoxygenated blood crosses into the systemic circulation and the previously pink, acyanotic patient becomes **CYANOTIC** and clubbed.
- **Reads as (the causal chain):** big left-to-right shunt for years to chronic pulmonary over-circulation to pulmonary arterial remodelling to **fixed, suprasystemic PVR** to the gradient reverses to **right-to-left shunt** to **cyanosis (Eisenmenger)**.
- **Why it matters (the point of no return):** once the PVR is fixed, the defect can **no longer be closed**. Closing it would remove the "pop-off" and leave the right heart pumping against an unbeatable resistance with nowhere to decompress, which is fatal. So the whole clinical urgency of the earlier concepts (close the significant shunt before the lungs are ruined) is defined by Eisenmenger being the endpoint you are racing to prevent. In an adult with a known shunt who is now cyanotic and clubbed, the message is: do NOT close it, this is Eisenmenger.
- **Example / exam anchor:** Eisenmenger is the acyanotic-to-cyanotic bridge and belongs in full to adult CHD (**C7**). Here it is the endpoint that explains WHY you close big shunts early: an unrepaired large ASD / VSD / PDA / AVSD is exactly what reaches it. It also underlies the PDA "differential cyanosis" picture once a big duct reverses.

- **Hook:** "Eisenmenger = the endpoint of an unrepaired big shunt: years of over-circulation to **fixed suprasystemic pulmonary hypertension** to the **shunt REVERSES to right-to-left** to the patient turns **CYANOTIC**. Once fixed, the defect can **no longer be closed**. It is WHY you close big shunts early (full detail in C7)."

⚑ Treated differently (the exceptions to flag)

- **A LOUD murmur usually means a SMALL (restrictive) defect, not a dangerous one.** This is the counterintuitive core of the bucket. A small hole makes a big pressure drop and a loud noise but shunts little and often closes; a big, dangerous, non-restrictive defect can be quiet because the pressures have equalised. Loud + well baby = small; soft + sick breathless baby = worry about a big non-restrictive shunt.
- **A high pressure gradient ACROSS a VSD is a reason NOT to close it (the Q9 trap).** A 100 mmHg gradient means the LV and RV pressures are still very different at the hole = a small restrictive VSD = protected lungs = often self-closing. Do not read a big gradient as severity. The real closure indications are LV volume overload / dilatation, AR from aortic-cusp prolapse, and failure to thrive.
- **A restrictive (small) PDA implies a NORMAL PA pressure, and a truly silent duct is left alone (the Q4 trap).** Same restrictive logic as the VSD: a small, high-resistance duct protects the lungs, so the PA pressure is normal. And you do NOT close every duct: a silent, tiny, incidental PDA is generally left alone. "Close even silent ducts" is the wrong statement.
- **Secundum ASD = RIGHT axis; primum ASD = LEFT axis (Q3 vs the AVSD family).** The axis is the fastest discriminator. Secundum (the plain, commonest ASD) gives right axis deviation + RSR' in V1; primum sits in the AVSD / endocardial-cushion spectrum and gives left axis. Q3's adult with RAD + RVH is a secundum ASD; flip to left axis and it is a primum.
- **Down syndrome + LEFT axis + infant failure = complete AVSD (Q7).** The LEFT axis is what separates an AVSD (or a primum ASD) from a plain VSD or secundum ASD. If you see trisomy 21 and left axis together in a failing infant, it is an AVSD until proven otherwise.
- **An ASD is the acyanotic shunt that classically presents in ADULTHOOD.** Because the atrial-level shunt is low-pressure and well tolerated, it is often missed in childhood and turns up in an adult with dyspnoea, a dilated RA/RV, atrial arrhythmias, or early pulmonary hypertension (exactly Q3). VSD, PDA and AVSD usually declare themselves in infancy as the PVR falls.
- **Once a big shunt has reached Eisenmenger, you do NOT close the defect.** The reversed, cyanotic patient with fixed suprasystemic pulmonary hypertension needs the defect kept open as a pop-off. Closing it is fatal. This is the one situation where an "obvious" hole must be left alone.

Mnemonics to lock in

- **The whole lesson in one line: "Acyanotic = a LEFT-to-RIGHT shunt of oxygenated blood, so the baby is PINK but the lungs are over-circulated. The shunt GROWS as PVR falls (C1). Small = restrictive = loud murmur, often closes; large = non-restrictive = failure + pulmonary hypertension. Leave a big one for years and it REVERSES: Eisenmenger, blue, no longer closeable."**
- **The four holes and their level: "ASD (atrial), VSD (ventricular), PDA (aorta-to-PA duct), AVSD (the whole centre: primum ASD + inlet VSD + common AV valve). All left-to-right, all plethoric, all pink."**
- **ASD: "LA to RA = RV volume overload. Wide FIXED split S2. Silent in childhood, presents in ADULTHOOD. Secundum = RIGHT axis + RSR' in V1; primum = LEFT axis (AVSD family). Q3 adult with RAD + RVH = secundum ASD."**
- **VSD (answers Q9): "Commonest defect, harsh pansystolic LSB murmur. Small restrictive = LOUD, often closes; large = failure + pulmonary hypertension. Close for LV dilatation, AR from cusp prolapse, failure to thrive. A 100 mmHg gradient across it = small restrictive VSD = do NOT close."**

- **PDA (answers Q4): "Continuous machinery murmur + wide pulse pressure. Restrictive PDA = normal PA pressure. Feared complication = infective endarteritis. Close significant ducts (small ones percutaneously), but LEAVE silent tiny ducts. 'Close even silent ducts' is FALSE."**
- **AVSD (answers Q7): "Endocardial cushions fail = common AV valve + primum ASD + inlet VSD = early failure. Buzzwords: Down syndrome (trisomy 21) + LEFT axis deviation. Down + left axis + infant failure = complete AVSD."**
- **Eisenmenger: "Unrepaired big shunt to fixed suprasystemic pulmonary hypertension to the shunt REVERSES to right-to-left to CYANOTIC and clubbed. Once fixed, do NOT close the defect. It is WHY you close big shunts early (C7)."**

🧪 Quiz yourself (commit your answer, key is sealed at the bottom)

Q3. A 52 year old woman presents with shortness of breath. Her ECG shows right axis deviation and right ventricular hypertrophy, and echocardiography shows a dilated right ventricle with a high-normal pulmonary artery pressure. The most probable diagnosis is: A. A primum atrial septal defect B. A secundum atrial septal defect C. Tetralogy of Fallot D. Truncus arteriosus

Q4. Regarding patent ductus arteriosus in adults, all of the following are true EXCEPT: A. A restrictive PDA indicates a normal pulmonary artery pressure B. Infective endarteritis is a dreadful complication C. All PDAs should be closed, even silent ducts D. Small PDAs should be closed percutaneously

Q7. A 3 month old girl with trisomy 21 has poor weight gain, tachypnoea, a low-pitched grade 2 systolic murmur at the lower left sternal border, cardiomegaly on the chest X-ray, and left axis deviation on the ECG. The most likely diagnosis is: A. Coarctation of the aorta B. Complete atrioventricular septal defect C. Patent ductus arteriosus D. Ventricular septal defect E. Secundum atrial septal defect

Q9. All of the following are indications for closure of a ventricular septal defect EXCEPT: A. A dilated left ventricle B. Aortic valve prolapse with aortic regurgitation C. A peak pressure gradient of 100 mmHg across the VSD D. Failure to thrive

Q-A (the ASD auscultation + ECG signature). A young adult is found to have a wide, FIXED split of the second heart sound, and the ECG shows an RSR' pattern in V1 with right axis deviation. The single most likely lesion, and its specific type, is: A. A ventricular septal defect B. A secundum atrial septal defect C. A patent ductus arteriosus D. A primum atrial septal defect E. Mitral stenosis

Q-B (secundum vs primum axis). An atrial septal defect on the ECG shows LEFT axis deviation. Compared with the commonest ASD, this tells you the defect is: A. A secundum ASD (right axis is expected) B. A primum ASD, which sits in the atrioventricular septal / endocardial-cushion spectrum C. A sinus venosus ASD with no clinical significance D. Not an ASD at all E. A restrictive VSD

Q-C (the PDA murmur and the Eisenmenger endpoint). A child has a continuous "machinery" murmur and bounding pulses with a wide pulse pressure. If this lesion is large and left unrepaired for many years, the feared endpoint is: A. Spontaneous closure of the duct B. Progressive aortic stenosis C. Fixed pulmonary hypertension with reversal of the shunt to right-to-left (Eisenmenger), turning the patient cyanotic and making the duct no longer closeable D. Complete heart block E. Rheumatic mitral stenosis

Q-D (AVSD and Down). Which single combination is the strongest giveaway of a complete atrioventricular septal defect in a failing infant? A. Trisomy 21 (Down syndrome) plus LEFT axis deviation B. Trisomy 21 plus right axis deviation C. A restrictive VSD with a 100 mmHg gradient D. A wide fixed split S2 with right axis deviation E. Differential cyanosis with a machinery murmur

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🔒 Answer key (only after you have committed)

- **Q3 = B (secundum atrial septal defect).** Read the physiology: **right axis deviation + RVH + a dilated RV** is **RV volume overload**, and presenting at **52 with dyspnoea and a high-normal PA pressure** is the classic **secundum ASD that was silent through childhood** and has now, in adulthood, produced RV dilatation and beginning pulmonary hypertension. The **RIGHT axis is the tell**: a **primum ASD (A) would give LEFT axis** (it sits in the AVSD spectrum), so A is wrong. TOF (C) and truncus (D) are cyanotic neonatal lesions, not a breathless 52 year old. An ASD is precisely the acyanotic shunt that classically presents in adulthood.
- **Q4 = C (all PDAs should be closed, even silent ducts) is the FALSE statement.** A **restrictive (small) PDA does imply a normal PA pressure (A, true)** by the same restrictive logic as a small VSD (high resistance across the duct protects the lungs). **Infective endarteritis is a dreaded complication (B, true). Small PDAs are closed percutaneously with a device or coil (D, true).** But you do **NOT** chase truly **silent, tiny, incidental ducts (C)**: with no murmur and no volume load, the procedural risk outweighs the negligible benefit, so they are generally left alone. "Close even silent ducts" is the over-aggressive, false statement.
- **Q7 = B (complete atrioventricular septal defect).** The combination is the giveaway: **trisomy 21 (Down)** + an **infant in heart failure** (poor weight gain, tachypnoea, cardiomegaly) + a **low LLSB murmur** + **LEFT axis deviation**. A complete **AVSD** is the classic cardiac lesion of Down syndrome, and being an endocardial-cushion / primum defect it gives **left axis deviation**. That LEFT axis is exactly what rules out a plain **VSD (D)** and a **secundum ASD (E)** (both give a normal or right axis), and coarctation **(A)** and PDA **(C)** do not fit the Down + left-axis + LLSB picture. Answer = **B**.
- **Q9 = C (a peak pressure gradient of 100 mmHg across the VSD) is NOT an indication.** The genuine indications to close a VSD are the ones that show the shunt is doing harm: **a dilated left ventricle (A, LV volume overload)**, **aortic valve prolapse with aortic regurgitation (B, a cusp prolapsing into the defect)**, and **failure to thrive (D)**, all with a significant Qp:Qs. A **100 mmHg gradient ACROSS the VSD (C)** means the LV pressure is still far higher than the RV pressure at the hole, which is the signature of a **small, RESTRICTIVE VSD** (high resistance, small shunt, protected lungs, often self-closing). A big gradient is a reason to LEAVE the VSD, not to close it, so it is the odd one out.
- **Q-A = B (a secundum atrial septal defect).** A **wide, FIXED split S2** is the classic ASD sign (the extra right-heart volume delays pulmonary closure, and free atrial communication fixes the split against respiration), and **RSR' in V1 with right axis deviation** is RV volume overload of the **secundum** type specifically. A VSD (A) gives a pansystolic murmur, a PDA (C) a continuous machinery murmur, a primum ASD (D) would give LEFT axis, and mitral stenosis (E) is a different, acquired lesion.
- **Q-B = B (a primum ASD, in the atrioventricular septal / endocardial-cushion spectrum).** The commonest ASD (**secundum**) gives **RIGHT axis**; **LEFT axis deviation** on an ASD tells you it is a **primum ASD**, which sits in the same **endocardial-cushion / AVSD family** (the same reason a complete AVSD gives left axis). So left axis = primum, not secundum (A), not an insignificant sinus venosus defect (C), and it is still a genuine ASD (D wrong).
- **Q-C = C (fixed pulmonary hypertension with reversal of the shunt to right-to-left = Eisenmenger).** A **continuous machinery murmur with a wide pulse pressure and bounding pulses** is a PDA. A large, unrepaired duct over-circulates the lungs for years, driving **irreversible pulmonary vascular disease** until the **PVR is fixed and suprasystemic**; the **shunt then reverses to right-to-left**, the patient turns

cyanotic (with differential cyanosis in a PDA), and the duct can **no longer be closed**. That is Eisenmenger, the endpoint you close big shunts early to prevent.

- **Q-D = A (trisomy 21 / Down syndrome plus LEFT axis deviation).** A complete **AVSD** is the classic lesion of Down syndrome, and being an endocardial-cushion defect it gives **LEFT axis deviation**. That specific pairing (Down + LEFT axis in a failing infant) is the strongest single giveaway. Down + RIGHT axis (B) is wrong (AVSD gives left axis), a restrictive VSD with a big gradient (C) is a different, self-limiting lesion, a wide fixed split S2 with right axis (D) points to a secundum ASD, and differential cyanosis with a machinery murmur (E) is a reversed PDA.

Diagrams in the visual (hand-built inline SVG, nothing embedded from the web)

1. **Interactive left-to-right shunt heart (concept 1 + the three levels):** a hand-drawn anterior-view heart (RA upper-left, LA upper-right, RV lower-left, LV lower-right; SVC and IVC into the RA on the screen-left; aorta arising from the LV on the right, pulmonary artery from the RV on the left). Clicking **ASD (atrial level)**, **VSD (ventricular level)** or **PDA (aorta-to-PA duct)** opens that shunt and animates an oxygenated (red) dot crossing from the LEFT heart to the RIGHT heart and up the pulmonary artery to the lungs, which turn plethoric. Every shunt originates from the correct left-heart structure and flows the correct direction (screen right-to-left for the atria / ventricles, since LA/LV are on the screen-right).
2. **ASD card (concept 2):** secundum (RIGHT axis, RSR' in V1, wide FIXED split S2, commonest) vs primum (LEFT axis, AVSD spectrum), with the Q3 adult-presentation note.
3. **VSD closure-indications card (concept 3):** the real indications (LV dilatation, AR from cusp prolapse, failure to thrive) against the Q9 trap (a high gradient across the VSD = a small restrictive VSD = do NOT close).
4. **PDA card (concept 4):** the aorta-to-PA duct, the continuous machinery murmur and wide pulse pressure, and the Q4 silent-duct trap (leave truly silent tiny ducts; endarteritis is the dread complication; a restrictive PDA = normal PA pressure).
5. **AVSD card (concept 5):** the common AV valve + primum ASD + inlet VSD at the centre of the heart, with the Down syndrome and LEFT axis buzzwords (Q7).
6. **Eisenmenger endpoint mini-panel (concept 6):** a toggle from the left-to-right shunt (pink) to the reversed right-to-left shunt (blue), showing fixed suprasystemic pulmonary hypertension, cyanosis, and "no longer closeable".

C4 - Interactive visual (rendered): diagrams & schematics

The live, toggleable version is the HTML file on the CHD hub. Below is a static render of its default view; open the hub to step through the other toggle states.

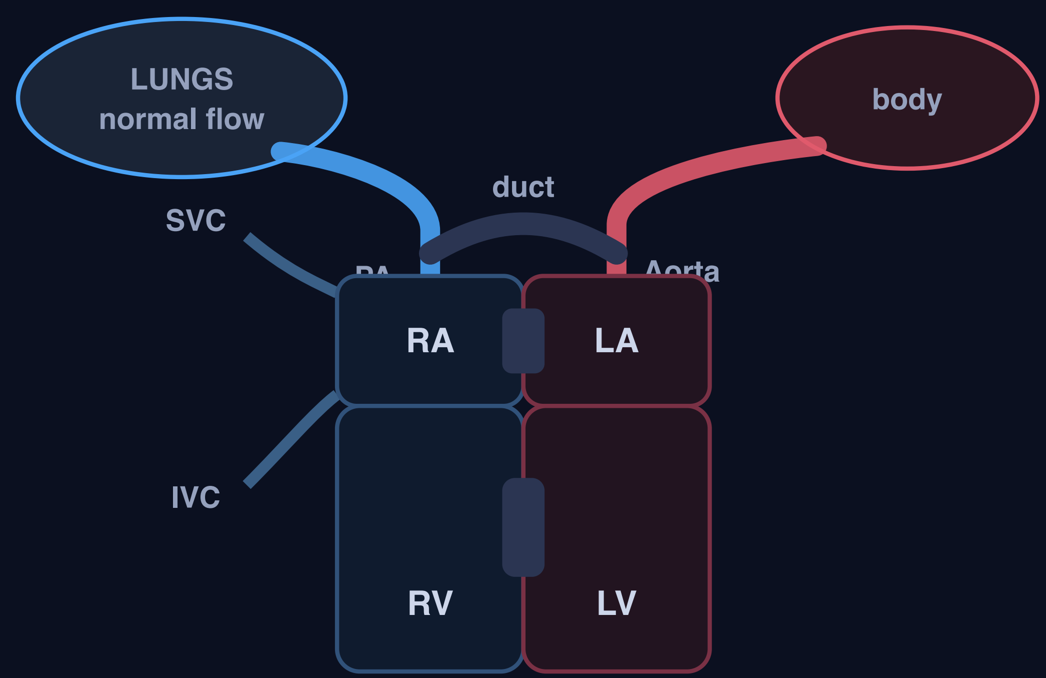
C4 - Acyanotic (left-to-right shunt) lesions: ASD, VSD, PDA, AVSD

The ACYANOTIC branch of C1: the baby is PINK, not blue, because the higher-pressure LEFT heart pushes oxygenated blood back into the low-pressure RIGHT heart / lungs, so the lungs are over-circulated. The shunt GROWS as the pulmonary vascular resistance falls over the first weeks (C1), so a big one causes failure and, long-term, pulmonary hypertension. Anchors: Q3 an adult with RAD + RVH + a dilated RV = secundum ASD; Q4 you do NOT close silent tiny ducts; Q7 Down + LEFT axis + infant failure = complete AVSD; Q9 a 100 mmHg gradient across a VSD = a small restrictive VSD = do NOT close.

1. THE LEFT-TO-RIGHT SHUNT HEART: CLICK A LEVEL TO OPEN THAT SHUNT (OXYGENATED BLOOD CROSSES LEFT HEART TO RIGHT HEART TO LUNGS)

No shunt (normal separation) | ASD (atrial level) | VSD (ventricular level) | PDA (aorta to PA duct)

right heart / deoxygenated (blue) · left heart / oxygenated (red) · the defect



Standard anterior view: RA upper-left, LA upper-right, RV lower-left, LV lower-right; SVC and IVC into the RA on the left; aorta off the LV (right), PA off the RV (left). Every shunt sends oxygenated (red) blood from the LEFT heart to the RIGHT heart and up to the lungs. Hand-drawn, nothing embedded.

Normal separation (no shunt)

The septa are intact and the duct is closed. The LEFT heart (red, oxygenated) and the RIGHT heart (blue, deoxygenated) stay separate, so nothing recirculates. Click a level to open a shunt.

baby: pink, lungs normal

The spectrum that runs the whole bucket: a SMALL (restrictive) defect = high resistance = small shunt but a big pressure drop = a LOUD murmur, often closes on its own, normal PA pressure. A LARGE (non-restrictive) defect = free communication = big shunt = heart failure + pulmonary hypertension. Loud murmur + well baby usually means small; a soft murmur in a sick, breathless baby can mean a big one.

The shunt is oxygenated blood taking a wrong turn, so the baby is PINK (acyanotic), but the lungs are over-circulated (plethoric). The shunt GROWS as the PVR falls over the first weeks (C1), so these present at weeks to months, not on day one. Left for years, a big shunt drives pulmonary hypertension to Eisenmenger (card 6).

2. ATRIAL SEPTAL DEFECT: SECUNDUM (RIGHT AXIS) VS PRIMUM (LEFT AXIS) - OWNS Q3

LA to RA shunt = **RV volume overload** (dilated RA + RV). Classic sign: a **wide, FIXED split S2**. Often silent in childhood, so it commonly presents in **ADULTHOOD** (dyspnoea, dilated RA/RV, atrial arrhythmias, pulmonary hypertension). The axis tells you the type.

Ostium secundum (commonest)

Mid-septum. **RIGHT axis deviation** + **RSR' in V1** (incomplete RBBB). The plain ASD.

RIGHT axis = secundum

Ostium primum

Low septum, part of the **AVSD / endocardial-cushion spectrum**, so it behaves like the AVSD family: **LEFT axis deviation**.

LEFT axis = primum

Q3 = B (secundum ASD). A 52 year old with SOB, RAD + RVH and a dilated RV with high-normal PA pressure = a secundum ASD presenting late with RV volume overload and early pulmonary hypertension. The RIGHT axis is the tell: a primum ASD (A) would give LEFT axis; TOF (C) and truncus (D) are cyanotic neonatal lesions. Secundum = RIGHT axis; primum = LEFT axis.

3. VENTRICULAR SEPTAL DEFECT: WHEN TO CLOSE IT - OWNS Q9

The commonest congenital defect. LV to RV shunt = a **harsh pansystolic murmur at the left sternal border**. **Small (restrictive) = LOUD murmur, small shunt, often closes; large (non-restrictive) = over-circulation, failure, pulmonary hypertension.**

CLOSE the VSD when (real indications)

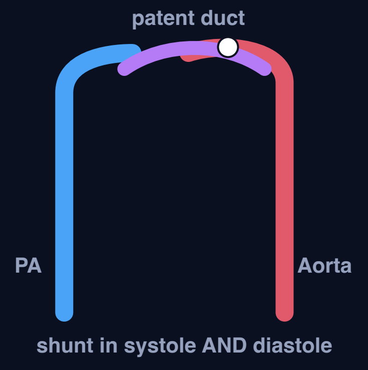
- A **significant shunt (high Qp:Qs)** with **LV volume overload / dilatation**
- **Aortic regurgitation** from an aortic cusp prolapsing into the defect
- **Failure to thrive** (a big non-restrictive shunt)
- Pulmonary hypertension while still operable

Do NOT close (the Q9 trap)

- A **peak gradient of 100 mmHg ACROSS the VSD**
- Why: a big gradient means LV pressure is still far above RV pressure at the hole
- = a **small, RESTRICTIVE VSD** (high resistance, small shunt, protected lungs)
- That is the good kind that often self-closes: **leave it**

Q9 = C. Real indications are LV dilatation (A), AR from cusp prolapse (B) and failure to thrive (D). A 100 mmHg gradient across the VSD (C) is NOT an indication: it flags a small restrictive VSD, so it is the odd one out. Do not read a big gradient as severity.

4. PATENT DUCTUS ARTERIOSUS: MACHINERY MURMUR AND THE SILENT-DUCT TRAP - OWNS Q4



The duct fails to close (C1), so aortic pressure exceeds pulmonary pressure all through the cycle and blood shunts **aorta to PA in systole AND diastole**.

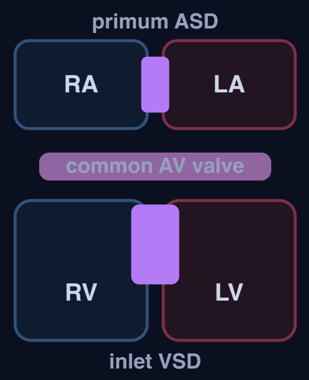
Signatures: a **continuous "machinery" murmur** and a **wide pulse pressure with bounding pulses** (diastolic run-off into the PA).

Q4 facts (find the FALSE one)

- A **restrictive PDA = a NORMAL PA pressure** (true)
- **Infective endarteritis** is a dreaded complication (true)
- Small PDAs are closed **percutaneously** (true)
- "Close every duct, even SILENT ones" = **FALSE** (leave silent tiny ducts)

Q4 = C (the EXCEPT). You do NOT chase a truly silent, tiny, incidental duct (no murmur, no volume load): the procedural risk outweighs the negligible benefit. A, B and D are all true, so "close even silent ducts" is the false statement.

5. ATRIOVENTRICULAR SEPTAL DEFECT: DOWN SYNDROME + LEFT AXIS - OWNS Q7



The endocardial cushions fail to fuse, so the CENTRE of the heart is open: a **common (single) AV valve + a primum ASD + an inlet VSD**. A big central left-to-right shunt plus AV-valve regurgitation = heavy over-circulation and **early heart failure**.

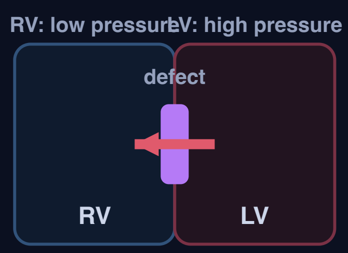
The two buzzword tells

- **Down syndrome (trisomy 21)** - the classic cardiac lesion
- **LEFT axis deviation** (endocardial-cushion / primum family)
- Infant with failure to thrive, tachypnoea, low LLSB murmur, cardiomegaly

Q7 = B (complete AVSD). A 3 month old with trisomy 21, poor weight gain, tachypnoea, a low LLSB murmur, cardiomegaly and LEFT axis deviation = complete AVSD. The LEFT axis rules out a plain VSD (D) and a secundum ASD (E); coarctation (A) and PDA (C) do not fit. Down + LEFT axis + infant failure = AVSD.

6. EISENMENGER: THE ENDPOINT WHERE THE SHUNT REVERSES AND THE PATIENT TURNS BLUE

Early: LEFT to RIGHT shunt (pink, closeable) | **Late: shunt REVERSED right to left (blue)**



Early: left-to-right shunt

The high-pressure LV drives blood into the low-pressure RV through the defect. The patient is **PINK (acyanotic)**, the lungs are over-circulated, and the defect is still **closeable**.

Years of over-circulation thicken and scar the pulmonary arteries, so the PVR becomes fixed and suprasystemic. Once the right-side pressure exceeds the left, the shunt REVERSES to right-to-left, deoxygenated blood crosses to the body, and the patient turns CYANOTIC (Eisenmenger). Now the defect can no longer be closed (closing it leaves the right heart pumping against an unbeatable resistance). This is WHY you close big shunts early. Full adult detail in C7.

C4 visual - Module 5 CHD. Hand-drawn inline SVG, self-contained.

C5 - Obstructive lesions and the "other" group: coarctation of the aorta (the anchor), valvar pulmonary stenosis, Ebstein anomaly, congenitally corrected TGA, vascular ring and heterotaxy / atrial isomerism (full lesson)

> Visual: `study/visuals/C5-obstructive-other.html` (a coarctation diagram drawn per the anatomy rule showing the arch narrowing just distal to the left subclavian, upper-limb hypertension vs weak / delayed femoral pulses, and the collateral intercostal arteries that cause rib notching; a valvar pulmonary-stenosis + Noonan card with the Q13 trap; an Ebstein card showing the apically displaced tricuspid leaflets, the atrialized RV, the WPW / associations and the wall-to-wall heart, with the Q16 conflict flagged; a ccTGA "double discordance" flow card; a vascular-ring card showing the ring around the trachea and oesophagus with the barium + CT/MRI work-up and the echo-is-limited Q11 trap; and an isomerism card contrasting right isomerism = asplenia vs left isomerism = polysplenia, with the Q15 conflict flagged). This is the OBSTRUCTIVE-and-other branch of the CHD module: lesions that do not fit the cyanotic decreased-flow (C2) or increased-flow (C3) buckets or the simple acyanotic shunts (C4). Some are pure obstructions (coarctation, pulmonary stenosis), some are valve / connection anomalies (Ebstein, ccTGA), and some are extracardiac or laterality problems (vascular ring, heterotaxy). C5 owns four Master-Exam questions. **Q11** vascular-ring diagnosis, all true EXCEPT (answer **D**: echo is LIMITED for a vascular ring, the tools are the barium swallow + CT/MRI). **Q13** CHD associations, all true EXCEPT (answer **E**: Noonan is associated with **pulmonary stenosis**, not mitral stenosis). **Q15** a 6-year-old girl with RIGHT atrial isomerism, expected feature (the course key says **B, polysplenia**, but standard teaching is right isomerism = **asplenia**, so teach it as **A** and confirm with the course). **Q16** Ebstein associations, all EXCEPT (the course key says **A, ASD**, but ASD is STRONGLY associated with Ebstein, so the defensible odd-one-out is **B, paradoxical splitting of S2** because Ebstein gives a WIDELY split S2 from RBBB; teach the real associations and confirm with the course).

Step 0. The big theory: obstruction on the left, obstruction on the right, and a set of "shapes" that do not fit the buckets

C1 to C4 sorted congenital heart disease by shunt and by lung blood flow: blue with dark lungs (C2), blue with wet lungs (C3), pink with a left-to-right shunt (C4). This lesson is the leftover pile, and it splits into three clean sub-ideas that you can hold in order:

- **Left-heart outflow obstruction = coarctation of the aorta.** A narrowing at or just distal to the ductus / left subclavian raises the pressure UPSTREAM (the head and arms) and starves everything DOWNSTREAM (the abdomen and legs). That single mechanical fact generates the whole classic picture: upper-limb hypertension, weak and delayed femoral pulses (radio-femoral delay), a murmur over the back, and, in an older child, rib notching from the collateral arteries that grow to bypass the block. The associations you must own are the bicuspid aortic valve and Turner syndrome, and a critical neonatal coarctation is duct-dependent for the SYSTEMIC circulation (PGE1, straight back to C1).
- **Right-heart outflow obstruction = valvar pulmonary stenosis.** The mirror image on the right side: the RV has to push through a narrowed pulmonary valve, so you hear a systolic ejection murmur with a click at the upper LEFT sternal border and the RV hypertrophies. The fix is balloon valvuloplasty. The syndrome to

marry to it is Noonan (a dysplastic pulmonary valve), and that pairing is the whole of Q13: Noonan is pulmonary stenosis, NOT mitral stenosis.

- **The "other" shapes that do not obey the buckets.** Ebstein anomaly (the tricuspid valve is pushed down into the RV, so part of the RV is "atrialized," giving severe tricuspid regurgitation, a tiny working RV, and a set of buzzword associations: an ASD, WPW / pre-excitation, a widely split S1, maternal lithium, and a huge "wall-to-wall" heart). Congenitally corrected TGA (a DOUBLE discordance that cancels itself out, so the patient is pink, but the morphologic right ventricle is doing the systemic work and eventually fails, with a high rate of complete heart block). A vascular ring (an aortic-arch anomaly that literally encircles the trachea and oesophagus, so an infant has stridor and feeding difficulty, and the work-up is a barium swallow + CT/MRI, NOT echo). And heterotaxy / atrial isomerism (the body loses its left-right asymmetry: right isomerism = two right sides = ASPLENIA and severe cyanotic heart disease; left isomerism = two left sides = POLYSPLENIA, an interrupted IVC and heart block).

Hook: **"Obstruct the left outflow and you get coarctation (high pressure and bounding pulses ABOVE, weak delayed femorals BELOW, rib notching, bicuspid valve / Turner). Obstruct the right outflow and you get pulmonary stenosis (ejection click at the upper left sternal border, RVH, balloon it, think Noonan). Then the odd shapes: Ebstein (tricuspid pushed down, atrialized RV, WPW, wall-to-wall heart), ccTGA (double discordance so pink now but the RV fails and heart-blocks later), a vascular ring (stridor + feeding trouble, barium + CT/MRI not echo), and isomerism (right = asplenia, left = polysplenia)."**

Step 1. The map: the six ideas of C5

These numbers are the SAME as the concept numbers in Step 2 (map row 1 = concept 1, and so on).

#	Idea	The one-line mechanism	Exam anchor
1	**Coarctation of the aorta (the anchor)**	A narrowing **at / just distal to the ductus and left subclavian** obstructs the aorta: pressure is HIGH upstream (head + arms) and LOW downstream, giving **upper-limb hypertension with weak, delayed femoral pulses (radio-femoral delay)**, a murmur over the back, and **rib notching** from collateral intercostals in older children	Associations: **bicuspid aortic valve** and **Turner syndrome**. Critical neonatal coarctation is **duct-dependent-systemic to PGE1** (ties to C1). The radio-femoral-delay / upper-vs-lower pressure split is the classic sign
2	**Valvar pulmonary stenosis**	RV outflow obstruction at the pulmonary valve: a **systolic ejection murmur + click at the upper LEFT sternal border**, **RV hypertrophy**, fixed by **balloon valvuloplasty**	The syndrome to own is **Noonan (dysplastic pulmonary valve)**. **Q13**: Noonan is associated with **pulmonary stenosis, NOT mitral stenosis** (answer E)
3	**Ebstein anomaly**	**Apical (downward) displacement of the septal and posterior tricuspid leaflets to an "atrialized" RV + severe tricuspid regurgitation + a small functional RV.** A **massive "wall-to-wall" / box-shaped heart** on CXR	Associations you must own: **an ASD / PFO, ventricular pre-excitation (WPW, accessory pathways, atrial flutter / SVT), a widely split S1**; maternal **lithium**. **Q16** (key = A / ASD, but ASD is strongly associated; the defensible EXCEPT is **B, paradoxical S2 splitting**, because Ebstein gives a WIDELY split S2 from RBBB; confirm with the course)
4	**Congenitally corrected TGA (ccTGA / L-TGA)**	**Double discordance**: BOTH the atrio-ventricular AND the ventriculo-arterial connections are swapped, so the two wrongs cancel and the circulation is physiologically "corrected", so the patient is **acyanotic**. But the **morphologic RV is the systemic ventricle** (it fails long-term)	High rate of **complete heart block** (the conduction axis is abnormally placed). Contrast with C3's D-TGA (a SINGLE discordance, cyanotic). A distractor / exam-sign lesion
5	**Vascular ring**	An **aortic-arch anomaly (e.g. a double aortic arch)** that **encircles the trachea and oesophagus**, compressing both, so an infant	Work-up = **barium oesophagogram (the most useful noninvasive tool) + CT / MRI (very accurate)**; **echo is LIMITED** for defining a vascular

#	Idea	The one-line mechanism	Exam anchor
		has **stridor + breathing and feeding / swallowing difficulty**	ring. **Q11**: all true EXCEPT **D** (echo can give detailed anatomy = false)
6	**Heterotaxy / atrial isomerism**	The body loses its normal left-right asymmetry, so both sides become mirror copies. **RIGHT atrial isomerism (bilateral right-sidedness) = ASPLENIA**; **LEFT atrial isomerism (bilateral left-sidedness) = POLYSPLENIA**	Right = asplenia + bilateral tri-lobed lungs + severe cyanotic CHD + overwhelming sepsis (Howell-Jolly bodies). Left = polysplenia + bilateral bi-lobed lungs + interrupted IVC + heart block. **Q15** (key = B / polysplenia, but standard teaching is right isomerism = **asplenia = A**; confirm with the course)

The one rule that ties it together: obstruction chooses a side (coarctation on the left outflow, pulmonary stenosis on the right outflow), and each obstruction advertises itself with a specific pulse or murmur (radio-femoral delay for coarctation, an ejection click at the upper left sternal border for PS). The "other" shapes are recognised by their buzzword associations, not by lung blood flow: Ebstein by WPW + a wall-to-wall heart + lithium, ccTGA by "pink now, heart block and RV failure later," a vascular ring by stridor + feeding trouble (worked up with barium + CT/MRI, never echo), and isomerism by the spleen (right = asplenia, left = polysplenia). Four exam questions live in this pile, and two of them (Q15, Q16) have a key that fights the standard teaching, so you must know the real answer and flag the conflict.

Step 2. The concepts you must know (numbered to match the map in Step 1: concept N = map row N)

1) Coarctation of the aorta: a narrowing that splits the body into a high-pressure top and a low-pressure bottom (the anchor)

- **Synonym:** coarctation of the aorta (CoA); aortic coarctation; a juxtaductal / "adult-type" aortic narrowing.
- **The core fact first:** coarctation is a **discrete narrowing of the aorta, classically at or just distal to the origin of the left subclavian artery, at the site of the ductus arteriosus (juxtaductal)**. Mechanically it is a dam across the aorta. Everything ARISING BEFORE the dam (the coronaries, the head and neck vessels, and the arms via the subclavians) sees a HIGH pressure, and everything BEYOND the dam (the abdominal aorta, the kidneys and the legs) sees a LOW pressure and a weak, delayed pulse. That one pressure split generates the entire clinical picture.
- **Reads as (the causal chain):** 1. A discrete narrowing sits **at / just distal to the left subclavian and the duct**. 2. Pressure is **HIGH proximal** to it (head + both arms) to **upper-limb hypertension and bounding brachial / radial pulses**. 3. Pressure is **LOW distal** to it (abdomen + legs) to **weak, delayed femoral pulses = radio-femoral delay** (the femoral pulse arrives late and feeble compared with the radial). 4. Over years, blood finds a way around the block through **collateral arteries** (internal mammary to intercostals), which enlarge and pulsate and **erode the undersurface of the ribs to rib notching** on the chest X-ray in older children and adults. 5. A **systolic murmur is heard over the back** (between the scapulae), from the turbulent flow across the coarctation and the collaterals.
- **The two associations you MUST own:** a **bicuspid aortic valve** (present in a large proportion of coarctations, and itself a source of aortic stenosis / regurgitation later) and **Turner syndrome (45,X)** (a short female with a webbed neck, in whom coarctation is the classic left-heart lesion). If the exam gives you a girl with Turner features and weak leg pulses, that is coarctation.
- **The neonatal emergency version (ties straight to C1):** a **critical / severe coarctation presents in the neonate when the duct closes**, because in utero the lower body was fed partly by the duct. As the duct shuts, the lower body suddenly loses its blood supply and the baby collapses with shock, acidosis and absent femoral pulses. This is a **duct-dependent-SYSTEMIC lesion**, so the emergency treatment is **PGE1 to reopen the duct** and restore lower-body perfusion (exactly the C1 logic, but here the duct feeds the systemic circulation distal to the coarctation). Do not confuse it with the duct-dependent-PULMONARY lesions of C2.

- **Example / exam anchor:** coarctation is the model left-outflow obstruction and the source of the **radio-femoral delay** sign and the **rib-notching** X-ray. It is the classic **Turner** and **bicuspid-valve** lesion. It underlies the "match the sign to the lesion" style of question (upper-limb hypertension + weak femorals = coarctation).
- **Hook:** "Coarctation = a dam just past the left subclavian: **high pressure and bounding pulses ABOVE (upper-limb hypertension), weak and DELAYED femoral pulses BELOW (radio-femoral delay), a murmur over the back, rib notching from collaterals.** Marry it to the **bicuspid aortic valve** and **Turner syndrome**. A critical neonatal coarctation is **duct-dependent-systemic to PGE1** (C1)."

2) Valvar pulmonary stenosis: the right-sided mirror of aortic outflow obstruction, and the Noonan lesion (owns the Q13 trap)

- **Synonym:** valvar pulmonary stenosis (PS); pulmonary valve stenosis; RV outflow tract obstruction at valve level.
- **The core fact first:** in valvar pulmonary stenosis the **pulmonary valve is narrowed**, so the **right ventricle must generate a high pressure to eject through it**. Turbulent flow across the narrowed valve produces a **systolic ejection murmur, loudest at the upper LEFT sternal border**, preceded by an **ejection click** (a domed, mobile valve popping open), and the pressure load makes the **right ventricle hypertrophy (RVH)**. Unlike the cyanotic lesions of C2 and C3, isolated PS is usually **acyanotic** (there is no shunt to desaturate the blood, the RV simply has to work harder); severe PS can drop the output and, with a patent foramen, cause a right-to-left shunt and cyanosis.
- **Reads as (the causal chain):** narrowed pulmonary valve to the RV must push harder to systolic ejection murmur + click at the upper LEFT sternal border to RV hypertrophies to (if severe) RV failure or a right-to-left shunt at the atrial level.
- **Why it matters (the treatment + the syndrome):** the definitive treatment for a typical domed valvar PS is **balloon pulmonary valvuloplasty** (a catheter balloon splits the fused commissures), which is one of the great successes of interventional paediatric cardiology. The syndrome to marry to PS is **Noonan syndrome** (a "male Turner-like" phenotype: short stature, webbed neck, pectus, in either sex, with a normal karyotype), in which the pulmonary valve is characteristically **dysplastic (thick, immobile, poorly responsive to a balloon)**. Noonan is also linked to hypertrophic cardiomyopathy, but the headline CHD is pulmonary stenosis.
- **Example / exam anchor (the whole of Q13): Q13** asks for the CHD association that is FALSE. The true pairs are Holt-Oram (ASD + a triphalangeal / absent thumb), maternal lithium and Ebstein, maternal rubella and PDA, and maternal antiepileptics and an interrupted aortic arch. The FALSE pair is **"Noonan syndrome and MITRAL stenosis"** (answer **E**), because **Noonan is associated with pulmonary stenosis (a dysplastic valve), not mitral stenosis**. That is the trap: they swap the correct valve.
- **Hook:** "Valvar pulmonary stenosis = the RIGHT-sided outflow obstruction: **ejection click + systolic murmur at the upper LEFT sternal border, RVH, treated by balloon valvuloplasty.** The syndrome is **Noonan (dysplastic pulmonary valve)**. Q13 = Noonan is **pulmonary** stenosis, NOT mitral stenosis (answer E)."

3) Ebstein anomaly: the tricuspid valve is pushed down into the right ventricle (owns the Q16 associations)

- **Synonym:** Ebstein anomaly / malformation of the tricuspid valve; apical displacement of the tricuspid valve.
- **The core fact first:** in Ebstein anomaly the **septal and posterior leaflets of the tricuspid valve are displaced DOWNWARD (apically) into the body of the right ventricle**, instead of sitting at the true atrio-ventricular junction. Two consequences fall out of that single fact: (1) the part of the right ventricle

above the displaced valve is "**atrialized**" (it is anatomically ventricle but functions as part of the right atrium), so the **effective right atrium is huge** and the **functional pumping RV is small**; and (2) the malformed valve **leaks badly, giving severe tricuspid regurgitation**. The result is a big, floppy right heart with a small working ventricle, which on the chest X-ray produces a **massive "wall-to-wall" / box-shaped cardiomegaly**.

- **Reads as (the causal chain):** septal + posterior tricuspid leaflets displaced apically to atrialized RV (huge functional RA) + a small functional RV + severe TR to a giant, "wall-to-wall" heart, right-heart failure, and (through the associated ASD) a right-to-left shunt with cyanosis.
- **The associations you MUST own (this is Q16):**
- **An ASD or patent foramen ovale** (very commonly present; it lets the high-pressure regurgitant right atrium shunt right-to-left, causing cyanosis and a risk of paradoxical embolism). **ASD is STRONGLY associated with Ebstein.**
- **Ventricular pre-excitation / WPW** (accessory pathways are common, so Ebstein patients get **pre-excitation, atrial flutter and SVT**; arrhythmia is a leading cause of trouble). Right-sided accessory pathways in particular.
- **A widely split first heart sound (S1)** (the large, sail-like anterior tricuspid leaflet closes late and loudly, the "sail sound").
- The auscultation is generally that of tricuspid regurgitation plus the widely split S1, and because of the associated right bundle branch block the **second heart sound (S2) is WIDELY split, not paradoxically split**.
- Cause / association: **maternal lithium** exposure in the first trimester (the classic teratogen link, and the "infant of a psychiatric mother" pairing in Q13b).
- **Why it matters (the Q16 conflict, spelled out): Q16** asks which is NOT associated with Ebstein, listing ASD, paradoxical splitting of S2, ventricular pre-excitation, a widely split S1, and atrial flutter. The provided **course key says A (ASD)**, but that is hard to defend, because **ASD / PFO is one of the strongest, most classic associations of Ebstein**. The genuinely odd item is **B, paradoxical splitting of S2**: Ebstein (with its right bundle branch block) produces a **WIDELY split S2**, whereas paradoxical (reversed) splitting is a LEFT-sided phenomenon (aortic stenosis, LBBB). So the defensible EXCEPT is **B (paradoxical S2 splitting)**. Teach the real associations (ASD, WPW / pre-excitation, widely split S1, atrial flutter / SVT, maternal lithium) and **flag that the key says A but ASD is strongly associated, so the defensible odd-one-out is B; confirm the intended answer with the course**. Do NOT just assert A.
- **Example / exam anchor:** Ebstein was introduced briefly in C2 as a right-heart lesion; here it is the full version. Exam signatures: **apically displaced tricuspid valve, atrialized RV, severe TR, a wall-to-wall heart, WPW / arrhythmias, maternal lithium.** It owns Q16 (associations) and is the "b" pair (lithium) in Q13.
- **Hook:** "Ebstein = the tricuspid valve **falls down into the RV**: an **atrialized RV**, a small working RV, **severe TR**, and a **wall-to-wall (box-shaped) heart**. Associations = an **ASD, WPW / pre-excitation (flutter / SVT), a widely split S1**, and **maternal lithium**. Q16: the key says A (ASD), but ASD is strongly associated, so the defensible EXCEPT is **B, paradoxical S2 splitting** (Ebstein gives a WIDELY split S2 from RBBB): confirm with the course."

4) Congenitally corrected TGA (ccTGA / L-TGA): a double discordance that cancels itself out

- **Synonym:** congenitally corrected transposition (ccTGA); L-transposition (L-TGA); atrioventricular AND ventriculo-arterial discordance; "physiologically corrected" transposition.
- **The core fact first:** ccTGA is a **DOUBLE discordance**. There are two connections in the heart, the atrio-ventricular connection (atria to ventricles) and the ventriculo-arterial connection (ventricles to great arteries), and in ccTGA **BOTH are swapped**. The right atrium connects to the morphologic LEFT ventricle, which connects to the pulmonary artery; and the left atrium connects to the morphologic RIGHT ventricle,

which connects to the aorta. Because there are TWO errors in series, they **cancel out**: blue systemic blood still ends up going to the lungs and red pulmonary blood still ends up going to the body. So unlike C3's D-TGA (a SINGLE discordance, in which the circulations run in parallel and the baby is deeply cyanotic), ccTGA is **physiologically "corrected" and the patient is ACYANOTIC**, often picked up incidentally or in adulthood.

- **Reads as (the causal chain):** RA to morphologic LV to pulmonary artery to lungs (blue path is normal), and LA to morphologic RV to aorta to body (red path is normal), so the flow is correct to acyanotic. BUT the morphologic RV is now doing the systemic (high-pressure) work.
- **Why it matters (the two long-term problems):** 1. **The systemic ventricle is a morphologic RIGHT ventricle.** The RV was never built to pump against systemic pressure for a lifetime, so it eventually **dilates and fails**, and its tricuspid valve (now the systemic AV valve) leaks. Systemic-ventricle failure is the main late problem. 2. **A very high rate of complete (third-degree) heart block.** The conduction axis is abnormally placed and fragile in ccTGA, so patients develop AV block at a rate of roughly 2% per year, and many end up needing a pacemaker.
- **Example / exam anchor:** ccTGA is the "double discordance = physiologically corrected = acyanotic, but RV fails and heart-blocks" lesson. Contrast it cleanly with D-TGA (C3): **D-TGA = single discordance = cyanotic newborn emergency; ccTGA = double discordance = pink, presents late with RV failure and complete heart block.** It appears as an exam-sign / matching distractor.
- **Hook:** "ccTGA = a **DOUBLE discordance** (atria-to-ventricles AND ventricles-to-arteries both swapped), so the two errors cancel and the patient is **PINK (acyanotic)**. The catch: the **morphologic RV is the systemic ventricle**, so it **fails long-term**, and there is a **high rate of complete heart block**. Opposite of D-TGA (single discordance, cyanotic)."

5) Vascular ring: an aortic-arch anomaly that strangles the trachea and oesophagus (owns Q11)

- **Synonym:** vascular ring; double aortic arch (the commonest complete ring); a right aortic arch with an aberrant left subclavian + ligamentum (another ring).
- **The core fact first:** a vascular ring is an **anomaly of the aortic arch and its branches that forms a complete ring of vessels ENCIRCLING the trachea and the oesophagus**. The classic example is a **double aortic arch**, where the aorta splits into two limbs that pass on either side of the trachea and oesophagus and rejoin, trapping both tubes inside a vascular noose. Because the ring squeezes the airway and the food-pipe, the presentation is not cardiac at all, it is **respiratory and feeding**: an **infant with stridor (noisy breathing), respiratory distress that is worse with feeding, recurrent chest infections, and difficulty swallowing / feeding**. The child may extend the neck to relieve the airway.
- **Reads as (the causal chain):** an arch anomaly forms a complete vascular ring to the ring compresses the trachea (to stridor, noisy / obstructed breathing) AND the oesophagus (to feeding and swallowing difficulty) to an infant with breathing + feeding trouble.
- **Why it matters (the WHOLE of Q11, the diagnostic tools):** the point of the exam question is **how you diagnose it**, and the trick is that the best tools are NOT echo:
- **Barium oesophagogram (barium swallow) is the most useful noninvasive tool**: the ring makes a characteristic **posterior / bilateral indentation on the barium-filled oesophagus**, which is highly suggestive.
- **CT and MRI are very accurate** and define the exact anatomy of the arch and the ring (they are the definitive imaging).
- **Echocardiography is LIMITED for a vascular ring**: echo is superb for intracardiac anatomy, but it cannot reliably trace the arch branches and the ligamentum through the mediastinum, so it does NOT give detailed anatomic information about the vascular anomaly.

- **Example / exam anchor (Q11): Q11** = "regarding the diagnosis of a vascular ring, all are true EXCEPT," with options: (A) suspected when a child has breathing and feeding difficulties [TRUE], (B) barium oesophagogram is the most useful noninvasive tool [TRUE], (C) CT and MRI are very accurate [TRUE], (D) echocardiography can provide detailed anatomic information about the vascular anomaly [FALSE]. Answer = **D**, because **echo is limited for defining a vascular ring**; the diagnostic tools are the barium swallow plus CT / MRI.
- **Hook:** "Vascular ring (classically a **double aortic arch**) = a ring of vessels **strangling the trachea and oesophagus**, so an infant has **stridor + breathing and FEEDING difficulty**. Diagnose with a **barium swallow (most useful noninvasive) + CT / MRI (very accurate)**; **ECHO is LIMITED** for a ring. Q11 = the false statement is that echo gives detailed anatomy (answer D)."

6) Heterotaxy / atrial isomerism: the body loses its left-right sidedness (owns Q15)

- **Synonym:** heterotaxy syndrome; atrial isomerism; the "splenic" syndromes; right / left isomerism.
- **The core fact first:** normally the body is asymmetric, one right side and one left side (the right lung has three lobes, the left has two; there is one spleen on the left; the IVC and the liver are right-sided). In **heterotaxy** that left-right patterning breaks, and the body becomes **isomeric**, that is, it has **two copies of the same side**. Which side is duplicated splits into two syndromes, and the **spleen is the fastest discriminator**:
- **RIGHT atrial isomerism = bilateral right-sidedness = ASPLENIA (Ivemark syndrome).** Both sides look like a right side: **bilateral tri-lobed lungs with eparterial bronchi**, and, because the spleen is normally a LEFT-sided organ, there is **NO spleen (asplenia)**. Right isomerism carries **severe, complex cyanotic congenital heart disease** (often a common atrium, AVSD, TGA, anomalous pulmonary venous drainage) and, because of the absent spleen, a **risk of overwhelming bacterial sepsis** (the blood film shows **Howell-Jolly bodies**, the marker of a non-functioning / absent spleen).
- **LEFT atrial isomerism = bilateral left-sidedness = POLYSPLENIA.** Both sides look like a left side: **bilateral bi-lobed lungs with hyparterial bronchi**, and **multiple small spleens (polysplenia)**. It classically has an **interrupted inferior vena cava (with azygos continuation)** and **conduction disease / heart block** (absent or abnormal sinus node). The cardiac disease is generally less severe than in right isomerism.
- **Reads as (the mnemonic logic): RIGHT isomerism to two RIGHT sides to no left-sided organ to no spleen = ASPLENIA (and severe cyanotic CHD + sepsis risk + Howell-Jolly bodies). LEFT isomerism to two LEFT sides to extra left-sided organ to many spleens = POLYSPLENIA (and interrupted IVC + heart block).**
- **Why it matters (the Q15 conflict, spelled out): Q15** describes a 6-year-old girl with **RIGHT atrial isomerism** and asks the expected feature, options including asplenia, polysplenia, two functional left lungs, T-cell deficiency, trisomy 21. By the standard teaching, right isomerism = **asplenia**, so the answer is **A (asplenia)**. The provided **course key says B (polysplenia)**, which matches LEFT isomerism, not right, and so contradicts the standard teaching. Teach it correctly as **A (asplenia)** and **flag that the course key says B (polysplenia); confirm the intended answer with the course**. Do NOT just assert B. (Note the other distractors: "two functional left lungs" belongs to LEFT isomerism, T-cell deficiency is DiGeorge / 22q11, and trisomy 21 is AVSD, none of which is the right-isomerism feature.)
- **Example / exam anchor:** the whole point of Q15 is the spleen split. Lock it as **"right isomerism = a-splenia, left isomerism = poly-splenia"** and know that the given key inverts it.
- **Hook:** "Heterotaxy = the body loses its left-right sidedness and becomes **isomeric (two copies of one side)**. **RIGHT isomerism = two right sides = ASPLENIA** (bilateral tri-lobed lungs, severe cyanotic CHD, sepsis risk, Howell-Jolly bodies). **LEFT isomerism = two left sides = POLYSPLENIA** (bilateral bi-lobed lungs, interrupted IVC, heart block). Q15: the girl has RIGHT isomerism so the answer is **asplenia (A)**, but the key says B (polysplenia): confirm with the course."

⚑ Treated differently (the exceptions to flag)

- **Coarctation is duct-dependent-SYSTEMIC, the mirror of C2's duct-dependent-pulmonary lesions.** In C2 the duct was kept open to feed the LUNGS. In critical neonatal coarctation the duct is kept open (PGE1) to feed the lower BODY (the systemic circulation distal to the narrowing). Same drug, opposite territory. When the duct closes, a critical coarctation presents as shock with absent femoral pulses, not as cyanosis.
- **Radio-femoral delay + upper-limb hypertension = coarctation, not "essential" hypertension.** In a young person with high arm blood pressure, always feel the femoral pulses: a weak, delayed femoral pulse (and an arm-leg pressure gradient) is coarctation until proven otherwise. This is the sign the "match the exam finding to the lesion" questions are built on.
- **Noonan is PULMONARY stenosis, not mitral stenosis (the whole of Q13).** The trap swaps the valve. Noonan = a dysplastic PULMONARY valve (and HCM). If the exam pairs Noonan with mitral stenosis, that pair is FALSE.
- **Ebstein's S2 is WIDELY split, not paradoxically split (the whole of Q16 conflict).** ASD is strongly associated with Ebstein, so "ASD" is a poor EXCEPT. The genuinely wrong item is paradoxical (reversed) S2 splitting, which is a LEFT-sided sign (aortic stenosis, LBBB); Ebstein's right bundle branch block gives a WIDELY split S2. Teach the true associations (ASD, WPW, widely split S1, atrial flutter / SVT, lithium) and flag the key.
- **ccTGA is "corrected," D-TGA is not.** D-TGA (C3) = a SINGLE discordance = two parallel circuits = deeply cyanotic newborn emergency needing mixing / PGE1 / arterial switch. ccTGA = a DOUBLE discordance = the errors cancel = ACYANOTIC, presenting late with systemic (morphologic RV) failure and complete heart block. Do not let the shared word "transposition" merge them.
- **A vascular ring is worked up with a barium swallow + CT / MRI, NEVER echo (the whole of Q11).** Echo, which is the workhorse for almost every OTHER lesson in this module, is the WRONG tool for a vascular ring, because it cannot follow the arch branches through the mediastinum. The false statement in Q11 is the one that credits echo with detailed ring anatomy (D).
- **Isomerism: the spleen is inverted between the two forms, and the given key inverts it further (the whole of Q15).** RIGHT isomerism = ASplenia (two right sides, no left-sided spleen); LEFT isomerism = POLYsplenia. The course key labels right isomerism as polysplenia, which is the standard teaching for LEFT isomerism, so teach asplenia and flag the conflict.

Mnemonics to lock in

- **The whole lesson in one line: "Obstruct the LEFT outflow to coarctation (high pulses ABOVE, weak delayed femorals BELOW, rib notching, bicuspid valve / Turner, PGE1 if critical). Obstruct the RIGHT outflow to pulmonary stenosis (ejection click at the upper LEFT sternal border, RVH, balloon it, Noonan). Then the odd shapes: Ebstein, ccTGA, vascular ring, isomerism."**
- **Coarctation: "A dam past the left subclavian: HIGH and bounding above, WEAK and DELAYED femorals below (radio-femoral delay), a murmur over the back, rib notching from collaterals. Bicuspid aortic valve + Turner. Critical neonate = duct-dependent-systemic to PGE1."**
- **Pulmonary stenosis + Q13: "RIGHT outflow obstruction: ejection CLICK + systolic murmur at the upper LEFT sternal border, RVH, balloon valvuloplasty. Noonan = a DYSPLASTIC PULMONARY valve. Q13: Noonan is PULMONARY stenosis, NOT mitral stenosis (answer E)."**
- **Ebstein + Q16: "The tricuspid valve FALLS into the RV: atrialized RV, small working RV, severe TR, a WALL-TO-WALL heart. Associations = ASD, WPW / pre-excitation (flutter / SVT), a widely split S1, maternal LITHIUM. Q16: key says A (ASD) but ASD is strongly associated, so the defensible EXCEPT is B (paradoxical S2 splitting, because Ebstein gives a WIDELY split S2): confirm with the course."**

- **ccTGA: "DOUBLE discordance (atria-ventricles AND ventricles-arteries both swapped) = the errors cancel = PINK (acyanotic). But the morphologic RV is the systemic ventricle, so it FAILS, and there is a high rate of COMPLETE HEART BLOCK. Opposite of D-TGA (single discordance, cyanotic)."**
- **Vascular ring + Q11: "A double aortic arch STRANGLES the trachea + oesophagus: stridor + breathing and FEEDING trouble. Diagnose with a BARIUM swallow (most useful noninvasive) + CT / MRI (very accurate). ECHO is LIMITED. Q11: the false statement is that echo gives detailed anatomy (answer D)."**
- **Isomerism + Q15: "RIGHT isomerism = two RIGHT sides = A-SPLENIA (tri-lobed lungs both sides, severe cyanotic CHD, sepsis, Howell-Jolly). LEFT isomerism = two LEFT sides = POLY-SPLENIA (bi-lobed lungs both sides, interrupted IVC, heart block). Q15: the girl has RIGHT isomerism to asplenia (A), but the key says B (polysplenia): confirm with the course."**

🧪 Quiz yourself (commit your answer, key is sealed at the bottom)

Q11. Regarding the diagnosis of a vascular ring, all of the following are true EXCEPT: A. It is suspected when a child has breathing and feeding difficulties B. A barium oesophagogram is the most useful noninvasive diagnostic tool C. CT and MRI are very accurate for defining the anatomy D. Echocardiography can provide detailed anatomic information about the vascular anomaly

Q13. All of the following congenital-heart-disease associations are true EXCEPT: a. ASD and a triphalangeal thumb (Holt-Oram) b. An infant of a mother on lithium and Ebstein anomaly c. Maternal rubella and infant PDA d. An epileptic mother on antiepileptics and interrupted aortic arch e. Noonan syndrome and mitral stenosis

Q15. A 6-year-old girl is found to have RIGHT atrial isomerism. Which feature would you expect? A. Asplenia B. Polysplenia C. Two functional left lungs D. T-cell deficiency E. Trisomy 21

Q16. Each of the following is associated with Ebstein anomaly EXCEPT: A. Atrial septal defect B. Paradoxical splitting of the second heart sound C. Ventricular pre-excitation D. A widely split first heart sound E. Atrial flutter

Q-A (coarctation sign). A 14-year-old boy has a blood pressure of 160/95 in both arms but weak, delayed pulses in both legs and a systolic murmur audible over the back. Which is the single best explanation? A. Essential hypertension B. Coarctation of the aorta just distal to the left subclavian (radio-femoral delay) C. Valvar pulmonary stenosis D. Ebstein anomaly E. A vascular ring

Q-B (coarctation associations). A short girl with a webbed neck and widely spaced nipples is found to have upper-limb hypertension and weak femoral pulses. The most likely syndrome and the valve lesion you would look for are: A. Noonan syndrome and a dysplastic pulmonary valve B. Turner syndrome and a bicuspid aortic valve C. Down syndrome and an AVSD D. DiGeorge syndrome and truncus arteriosus E. Holt-Oram syndrome and an ASD

Q-C (Noonan lesion). A boy with short stature, a webbed neck, pectus and a normal karyotype has an ejection systolic murmur and click at the upper left sternal border. The lesion is: A. Mitral stenosis B. Aortic coarctation C. Valvar pulmonary stenosis (a dysplastic valve) D. A secundum ASD E. Mitral regurgitation

Q-D (Ebstein arrhythmia). A cyanotic newborn with a huge "wall-to-wall" heart on chest X-ray and a widely split S1 is later found to have episodes of a fast, regular tachycardia. The associated conduction abnormality is: A. Complete heart block B. Ventricular pre-excitation (WPW / accessory pathway) C. Mobitz I block D. Left bundle branch block E. Sick sinus syndrome

Q-E (isomerism split). A neonate with complex cyanotic heart disease has a blood film showing Howell-Jolly bodies and bilateral tri-lobed lungs. Which laterality syndrome does this indicate, and what is the splenic status? A. Left atrial isomerism, polysplenia B. Right atrial isomerism, asplenia C. Left atrial isomerism, asplenia D. Right atrial isomerism, polysplenia E. Normal situs, single spleen

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🔒 Answer key (only after you have committed)

- **Q11 = D (echocardiography can provide detailed anatomic information about the vascular anomaly).** This is the FALSE statement, so it is the EXCEPT answer. **Echo is limited for a vascular ring**: it images the heart beautifully but cannot reliably trace the arch branches and the ligamentum through the mediastinum, so it does NOT define ring anatomy. The tools that DO are the **barium oesophagogram (the most useful noninvasive test, showing the characteristic oesophageal indentation) and CT / MRI (very accurate)**. Options A (breathing + feeding difficulty), B (barium is the most useful noninvasive tool) and C (CT / MRI are very accurate) are all TRUE.
- **Q13 = E (Noonan syndrome and mitral stenosis).** This pairing is FALSE, so it is the EXCEPT answer. **Noonan syndrome is associated with valvar PULMONARY stenosis (a dysplastic pulmonary valve)** and with hypertrophic cardiomyopathy, NOT mitral stenosis, so the exam has swapped the valve. The other four pairs are all TRUE associations: (a) Holt-Oram = an ASD (or VSD) with an upper-limb / triphalangeal-thumb defect; (b) maternal lithium = Ebstein anomaly ("an infant of a psychiatric mother"); (c) maternal rubella = PDA (and peripheral pulmonary stenosis); (d) maternal antiepileptics = an interrupted aortic arch (a conotruncal association).
- **Q15: teach A (asplenia); the provided course key = B (polysplenia).** ⚠ **CONFLICT.** By the standard teaching, **RIGHT atrial isomerism = bilateral right-sidedness = ASPLENIA (Ivemark)**: two morphologically right sides, bilateral tri-lobed lungs with eparterial bronchi, severe complex cyanotic CHD, and (because the spleen is a left-sided organ that fails to form) NO spleen, with a sepsis risk and Howell-Jolly bodies on the film. **LEFT** atrial isomerism is the one that gives **polysplenia** (plus bilateral bi-lobed lungs, an interrupted IVC and heart block). So for a girl with RIGHT isomerism the expected feature is **asplenia (A)**. The course key says **B (polysplenia)**, which is the LEFT-isomerism feature and contradicts the standard teaching, so **teach asplenia and confirm the intended answer with the course**. (Distractors: C "two functional left lungs" = LEFT isomerism, D "T-cell deficiency" = DiGeorge / 22q11, E "trisomy 21" = AVSD.) Do NOT just assert B.
- **Q16: the defensible EXCEPT is B (paradoxical splitting of S2); the provided course key = A (ASD).** ⚠ **CONFLICT.** All of ASD, ventricular pre-excitation, a widely split S1 and atrial flutter ARE genuine Ebstein associations, and in particular **an ASD / PFO is one of the strongest, most classic associations of Ebstein** (it is what allows the right-to-left shunt and cyanosis), so calling ASD the "EXCEPT" is hard to defend. The genuinely non-associated item is **B, paradoxical (reversed) splitting of S2**: Ebstein carries a **right bundle branch block, which WIDENS the split of S2**, whereas paradoxical splitting is a LEFT-sided phenomenon (severe aortic stenosis, LBBB). So the defensible odd-one-out is **B**. Teach the real associations, ASD, WPW / pre-excitation (with the atrial flutter / SVT that follows), a widely split S1 (the sail sound), and maternal lithium, and **flag that the key says A but ASD is strongly associated, so B is the defensible answer; confirm with the course**. Do NOT just assert A.
- **Q-A = B (coarctation of the aorta).** Upper-limb hypertension with weak, delayed lower-limb pulses (**radio-femoral delay**) and a murmur over the back is the textbook coarctation picture: the narrowing just distal to the left subclavian raises pressure above it (both arms) and lowers it below it (both legs). Essential hypertension (A) would not give a radio-femoral delay or a back murmur; PS (C) and Ebstein (D) are right-

heart lesions without an arm-leg gradient; a vascular ring (E) presents with stridor and feeding trouble, not hypertension.

- **Q-B = B (Turner syndrome and a bicuspid aortic valve).** A short girl with a webbed neck and widely spaced nipples is **Turner syndrome (45,X)**, whose classic left-heart lesions are **coarctation of the aorta and a bicuspid aortic valve** (look for the bicuspid valve). Noonan (A) is a different phenotype (either sex, normal karyotype) linked to pulmonary stenosis; Down (C) to AVSD; DiGeorge (D) to truncus / interrupted arch; Holt-Oram (E) to an ASD with a limb defect.
- **Q-C = C (valvar pulmonary stenosis, a dysplastic valve).** The phenotype (short stature, webbed neck, pectus, normal karyotype) is **Noonan syndrome**, and its headline congenital lesion is **valvar pulmonary stenosis from a dysplastic (thick, poorly mobile) pulmonary valve**, giving an ejection systolic murmur and click at the upper LEFT sternal border. Noonan is NOT associated with mitral stenosis (A) (that is the Q13 trap).
- **Q-D = B (ventricular pre-excitation / WPW).** Ebstein anomaly is strongly associated with **accessory pathways and WPW**, so these patients get **pre-excitation with atrial flutter / SVT**; the huge "wall-to-wall" heart and the widely split S1 (sail sound) complete the picture. Complete heart block (A) and sick sinus (E) are the ccTGA / left-isomerism conduction problems, not Ebstein.
- **Q-E = B (right atrial isomerism, asplenia). Howell-Jolly bodies** signal a non-functioning or absent spleen, and **bilateral tri-lobed lungs** are the bilateral RIGHT-sided (eparterial-bronchus) pattern, so this is **RIGHT atrial isomerism with ASPLENIA** (Ivemark), which carries severe complex cyanotic CHD and a risk of overwhelming sepsis. Left atrial isomerism (A) is the polysplenia / interrupted-IVC / heart-block form. This is the same asplenia-vs-polysplenia split that the Q15 key inverts.

Diagrams in the visual (hand-built inline SVG, nothing embedded from the web)

1. **Coarctation of the aorta (concept 1):** a hand-drawn heart in the standard anterior view (RA upper-left, LA upper-right, RV lower-left, LV lower-right) with the aorta arising from the LV and the arch sweeping across the top, the head and arm vessels coming off the arch, a discrete narrowing just distal to the left subclavian, HIGH-pressure bounding pulses in the head and arms vs WEAK / delayed femoral pulses in the legs, and the collateral intercostal arteries with a rib-notching note. Animation: flow slows at the narrowing and lights up the collaterals.
2. **Valvar pulmonary stenosis + Noonan (concept 2):** the RV (lower-left) ejecting through a narrowed pulmonary valve into the PA (screen-left), an ejection-click / murmur label at the upper left sternal border, an RVH note, and a Noonan / balloon-valvuloplasty box with the Q13 "pulmonary not mitral" trap.
3. **Ebstein anomaly (concept 3):** the tricuspid valve displaced apically into the RV, an "atrialized" RV, a giant right atrium, a severe TR jet, the wall-to-wall heart outline, and an associations box (ASD, WPW / pre-excitation, widely split S1, lithium) with the Q16 conflict flagged.
4. **ccTGA double discordance (concept 4):** a schematic showing both connections swapped (RA to morphologic LV to PA, and LA to morphologic RV to aorta), the flow physiologically corrected (acyanotic), the morphologic RV labelled as the failing systemic ventricle, and a complete-heart-block note, contrasted with D-TGA.
5. **Vascular ring (concept 5):** a double aortic arch forming a complete ring around the central trachea and oesophagus, compression arrows, a stridor + feeding-difficulty label, and a work-up box (barium swallow + CT / MRI = yes; echo = limited, the Q11 trap).
6. **Heterotaxy / atrial isomerism (concept 6):** a two-column comparison, RIGHT isomerism (bilateral tri-lobed lungs, ASPLENIA, severe cyanotic CHD, Howell-Jolly) vs LEFT isomerism (bilateral bi-lobed lungs, POLYSPLENIA, interrupted IVC, heart block), with the Q15 conflict flagged.

C5 - Interactive visual (rendered): diagrams & schematics

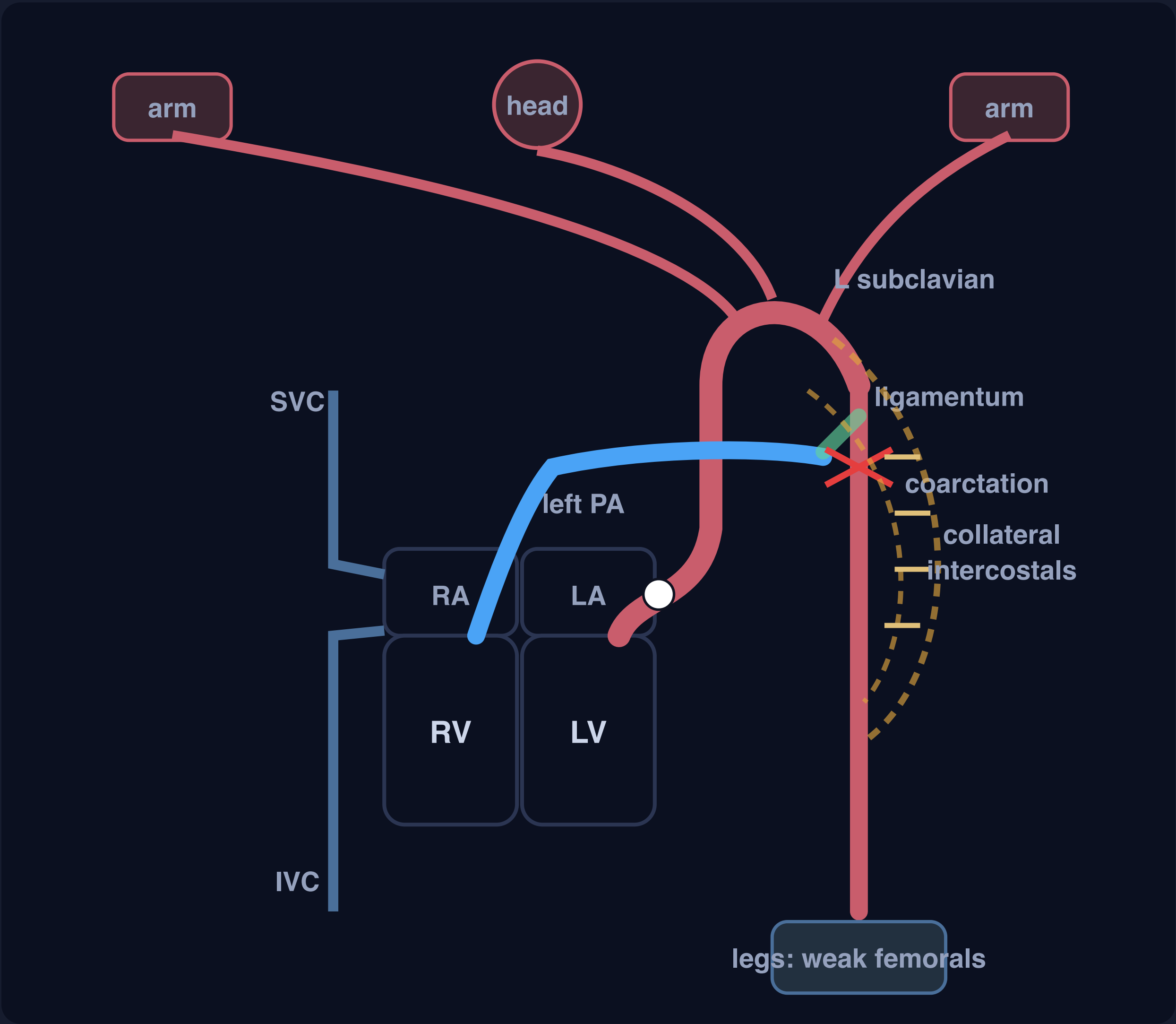
The live, toggleable version is the HTML file on the CHD hub. Below is a static render of its default view; open the hub to step through the other toggle states.

C5 - Obstructive lesions and the "other" group

The leftover pile of the CHD module: pure obstructions (coarctation on the LEFT outflow, pulmonary stenosis on the RIGHT outflow), odd valve / connection shapes (Ebstein, ccTGA), and extracardiac / laterality problems (vascular ring, isomerism). Anchors: Q11 (vascular ring, echo is LIMITED = answer D), Q13 (Noonan is pulmonary NOT mitral stenosis = answer E), Q15 (right isomerism = ASPLENIA; key says B/polysplenia, flag it), Q16 (Ebstein: ASD IS strongly associated, so the defensible EXCEPT is paradoxical S2 splitting = B; key says A, flag it).

1. COARCTATION OF THE AORTA: A NARROWING AT THE ISTHMUS, NEXT TO THE DUCT - TOGGLE PREDUCTAL VS POSTDUCTAL

Postductal (adult): pinch BELOW the duct | Preductal (infantile): pinch ABOVE the duct



Aorta off the LV; the 3 arch branches end in the left subclavian; the ductus joins the left PA to the descending aorta just distal to it. The coarctation sits next to the duct - toggle to move it above (preductal) or below (postductal).

Postductal (adult)

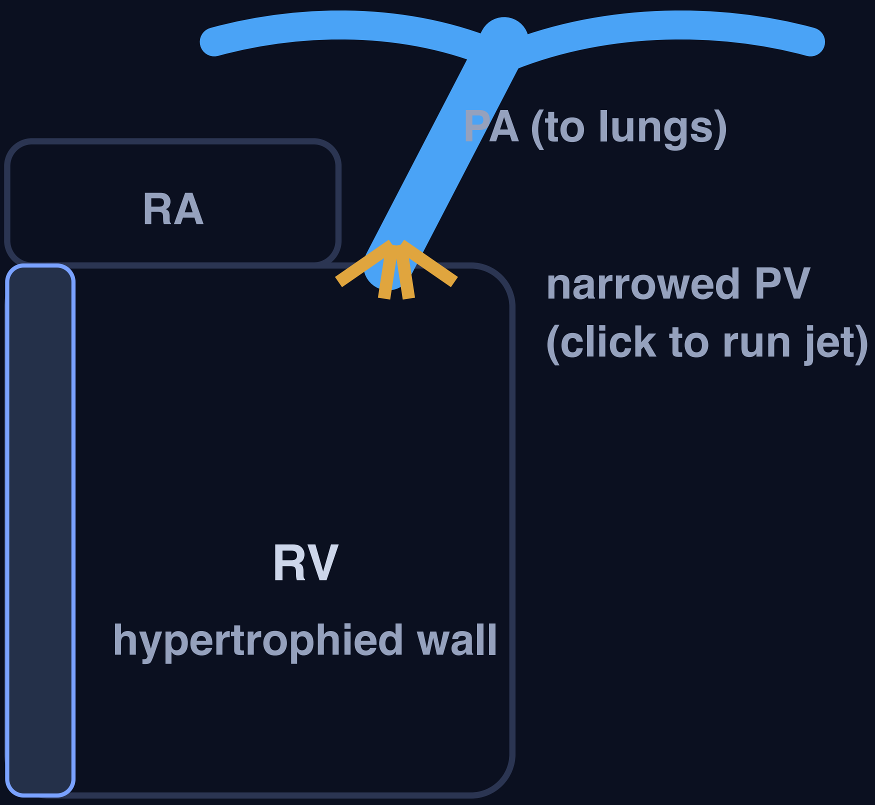
The narrowing is **DISTAL to the duct** (now closed to the ligamentum). Blood finds a way around the dam: **collateral arteries (internal mammary to intercostals) enlarge** and their pulsation **erodes the undersurface of the ribs = rib notching**. Presents later with **upper-limb hypertension**, a murmur over the back, and weak, delayed femorals.

Associations + sign

- **Upper-limb hypertension + weak, DELAYED femorals** (radio-femoral delay); murmur over the back
- **Bicuspid aortic valve** (very common)
- **Turner syndrome (45,X)**: short female, webbed neck

The duct joins the left PA to the descending aorta, just distal to the left subclavian. Postductal (adult): pinch BELOW the duct to collaterals to rib notching, upper-limb hypertension. Preductal (infantile): pinch ABOVE the duct to the lower body is fed right-to-left through the open duct to duct-dependent-SYSTEMIC: when the duct closes the baby collapses, so give PGE1 to reopen it.

2. VALVAR PULMONARY STENOSIS: THE RIGHT-SIDED OUTFLOW OBSTRUCTION, AND THE NOONAN LESION (CLICK THE VALVE)



The RV pushes through a narrowed pulmonary valve, so it hypertrophies (thick wall). The pulmonary trunk arises from the RV outflow and heads up toward the patient's left. Click the valve to run the turbulent jet; the ejection click and murmur (upper LEFT sternal border) are in the notes at right.

Pulmonary stenosis (valvar)

Chain: narrowed pulmonary valve to the RV pushes harder to **ejection click + systolic murmur at the upper LEFT sternal border** to **RV hypertrophy**. Usually acyanotic (no shunt); severe PS can right-to-left shunt through a PFO. Fix = **balloon valvuloplasty**.

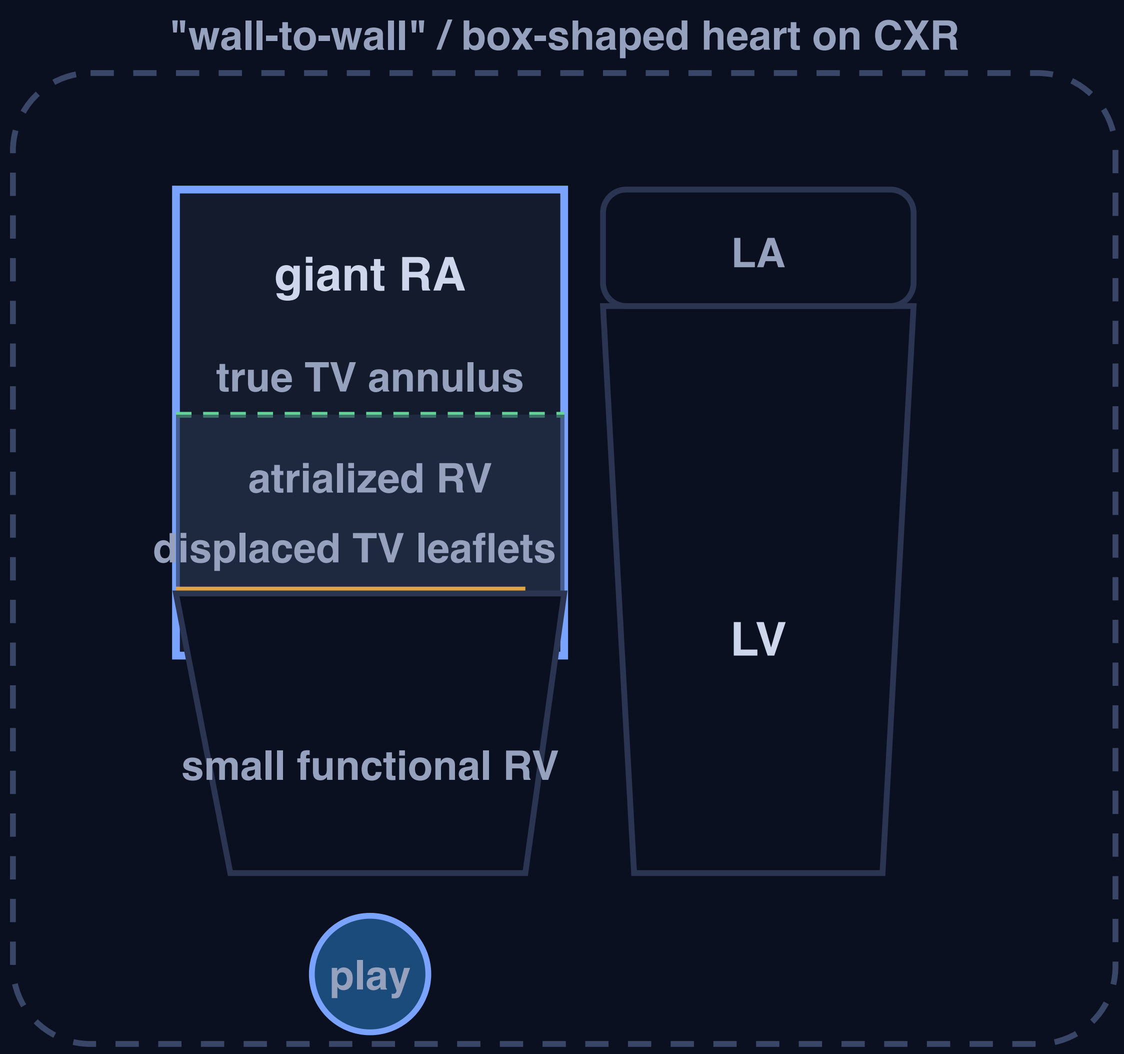
Noonan syndrome

- "Male Turner-like" phenotype, either sex, **normal karyotype**
- **Dysplastic pulmonary valve** (thick, balloon-resistant)
- Also linked to hypertrophic cardiomyopathy

Q13: CHD associations, all true EXCEPT "Noonan syndrome and MITRAL stenosis" (answer E). Noonan is associated with valvar PULMONARY stenosis (a dysplastic valve), NOT mitral stenosis, the exam swaps the valve. The true pairs: Holt-Oram +

ASD / thumb, lithium + Ebstein, rubella + PDA, antiepileptics + interrupted aortic arch.

3. EBSTEIN ANOMALY: THE TRICUSPID VALVE FALLS DOWN INTO THE RV (CLICK TO RUN THE TR JET)



The septal and posterior tricuspid leaflets are displaced DOWN into the RV, so part of the RV is "atrialized," the working RV is tiny, and the valve leaks (severe TR). Click play to run the regurgitant jet.

Ebstein anomaly

Chain: septal + posterior TV leaflets displaced apically to atrialized RV (huge functional RA) + small working RV + severe TR to a **massive wall-to-wall heart** and (via the ASD) a right-to-left shunt with cyanosis.

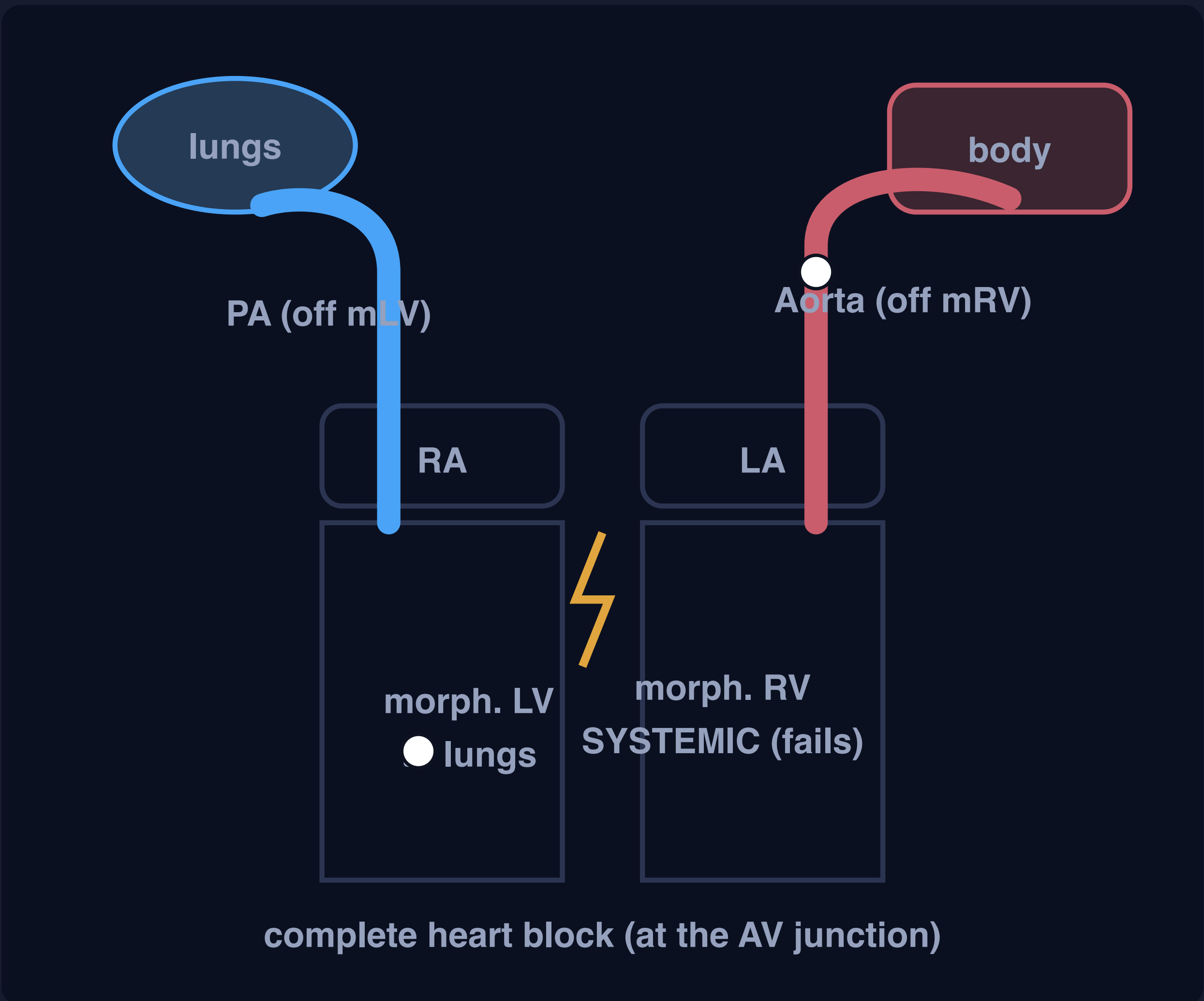
Associations you MUST own (Q16)

- **ASD / PFO** (STRONGLY associated to right-to-left shunt)

- **Ventricular pre-excitation / WPW** (accessory pathways to atrial flutter / SVT)
- **A widely split S1** (the "sail sound")
- A **widely split S2** (from RBBB), NOT paradoxical
- Maternal **lithium** (the classic teratogen)

Q16 (key = A / ASD). ⚠ CONFLICT: ASD / PFO is one of the STRONGEST Ebstein associations, so it is a poor EXCEPT. Ebstein's RBBB gives a WIDELY split S2, while paradoxical (reversed) splitting is a LEFT-sided sign (aortic stenosis, LBBB), so the defensible odd-one-out is B, paradoxical splitting of S2. Teach the real associations and confirm the intended answer with the course. Do not just assert A.

4. CONGENITALLY CORRECTED TGA (CCTGA / L-TGA): A DOUBLE DISCORDANCE THAT CANCELS OUT



Both connections are swapped: RA to morphologic LV to PA (blue path), and LA to morphologic RV to aorta (red path). Two errors in series cancel, so flow is correct and the patient is PINK, but the morphologic RV now does the systemic work.

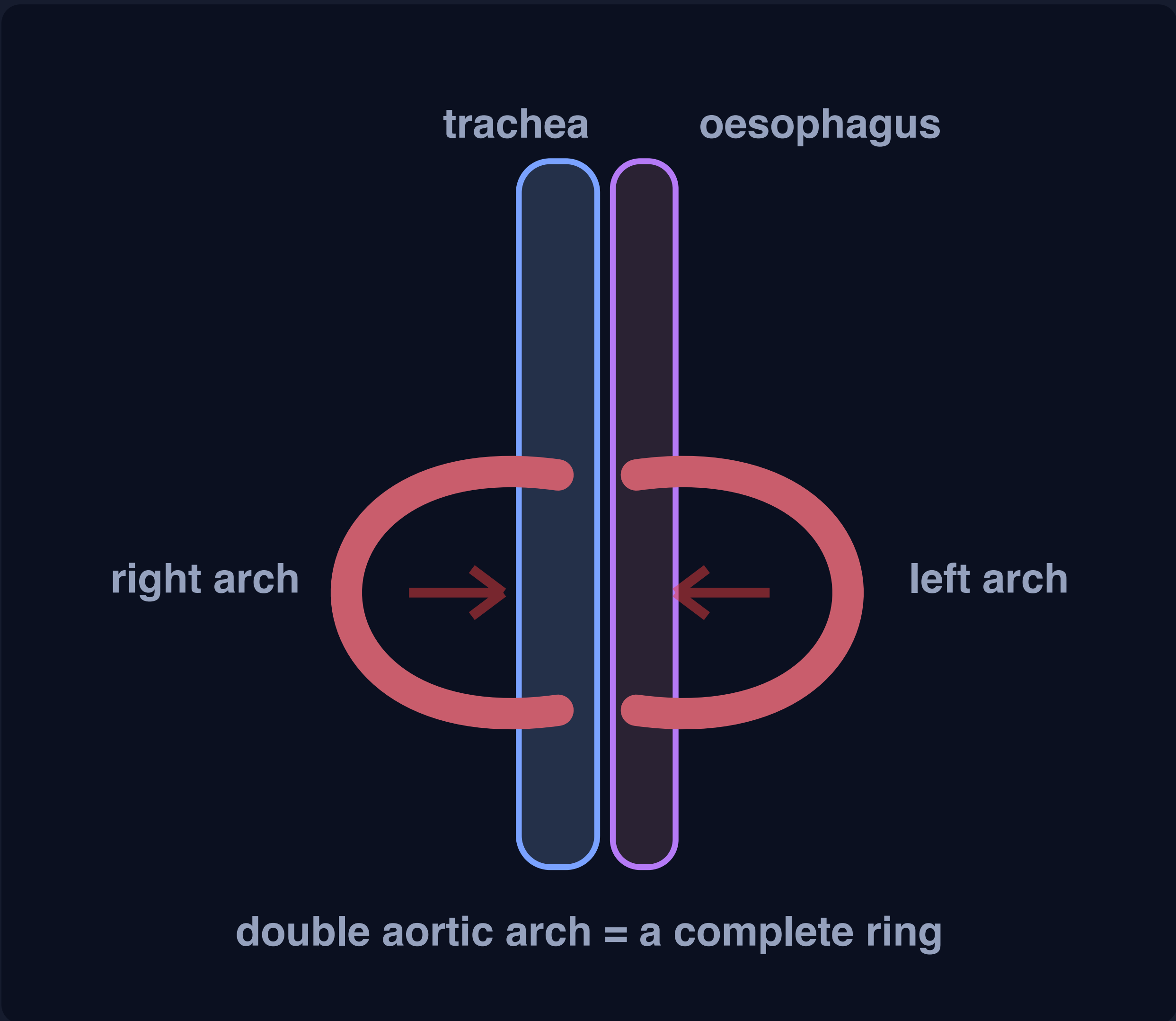
Double discordance = corrected

Chain: the atrio-ventricular AND the ventriculo-arterial connections are BOTH swapped, so the two wrongs cancel to blue blood still reaches the lungs, red blood still reaches the body to **physiologically corrected, ACYANOTIC**.

The catch: the morphologic RIGHT ventricle is the systemic ventricle, so it **dilates and fails long-term**, and there is a **high rate of complete (third-degree) heart block**.

> Contrast with C3's **D-TGA** = a SINGLE discordance = two parallel circuits = deeply **cyanotic** newborn emergency. ccTGA = double discordance = **pink**, presents late.

5. VASCULAR RING: AN ARCH ANOMALY THAT STRANGLES THE TRACHEA AND OESOPHAGUS (CLICK TO TIGHTEN THE RING)



The aorta forms two limbs that pass on either side of the trachea and oesophagus and rejoin, trapping both tubes inside a vascular noose. Click to tighten the ring.

Vascular ring (double aortic arch)

Chain: an arch anomaly forms a complete ring to it squeezes the trachea (to **stridor, noisy / obstructed breathing**) AND the oesophagus (to **feeding and swallowing difficulty**) to an infant with breathing + feeding trouble, worse with feeds.

Work-up (the whole of Q11)

- **Barium oesophagogram** = the MOST useful noninvasive tool (oesophageal indentation)
- **CT / MRI** = very accurate (define the anatomy)
- **Echo = LIMITED**: it cannot trace the arch / ligamentum through the mediastinum

Q11: regarding the diagnosis of a vascular ring, all are true EXCEPT D, "echocardiography can provide detailed anatomic information about the vascular anomaly". Echo is LIMITED for a ring; the tools are the barium swallow + CT / MRI. Options A (breathing + feeding difficulty), B (barium most useful) and C (CT / MRI very accurate) are all TRUE.

6. HETEROTAXY / ATRIAL ISOMERISM: THE BODY LOSES ITS SIDEDNESS (RIGHT = ASPLENIA, LEFT = POLYSPLENIA)

RIGHT atrial isomerism

Two RIGHT sides. Bilateral tri-lobed lungs (eparterial bronchi), **severe complex cyanotic CHD**, and, because the spleen is a left organ, **NO spleen**. Sepsis risk, **Howell-Jolly bodies** on the film.

ASPLENIA (Ivemark)

LEFT atrial isomerism

Two LEFT sides. Bilateral bi-lobed lungs (hyparterial bronchi), **multiple small spleens (POLYSPLENIA)**, an **interrupted IVC** (azygos continuation) and **conduction disease / heart**

block. Milder cardiac disease.

POLYSPLENIA

Lock the split: RIGHT isomerism = two RIGHT sides = A-splenia; LEFT isomerism = two LEFT sides = POLY-splenia. The spleen is the fastest discriminator.

Q15 (key = B / polysplenia). ⚠ CONFLICT: the girl has RIGHT atrial isomerism, and by standard teaching right isomerism = ASPLENIA (answer A). Polysplenia is the LEFT-isomerism feature, so the key contradicts the standard teaching. Teach asplenia and confirm the intended answer with the course. (Distractors: "two functional left lungs" = left isomerism, "T-cell deficiency" = DiGeorge, "trisomy 21" = AVSD.) Do not just assert B.

C5 visual - Module 5 CHD. Hand-drawn inline SVG, self-contained.

C6 - The single-ventricle pathway: when you cannot build two ventricles, you stage the heart down to a Fontan (Glenn prerequisites and complications, the whole reason the lungs must offer no resistance)

> Visual: `study/visuals/C6-single-ventricle.html` (a staged-palliation card that animates the three operations in order, Norwood then bidirectional Glenn then Fontan, each drawn per the anatomy rule with the SVC and IVC on the correct sides; a "passive flow" card that shows there is NO pump between the systemic veins and the lungs, so blood only moves if the pulmonary resistance is low; a Glenn-prerequisites checklist card built straight from Q2 with the low-PVR item flagged as the answer; and a Glenn-versus-Fontan complications card that puts protein-losing enteropathy on the Fontan side to nail Q10). This is the pathway lesson, not a single defect. It is what you do when the heart has only ONE usable pumping ventricle (hypoplastic left heart, tricuspid atresia, double-inlet left ventricle, unbalanced AVSD, pulmonary atresia with an intact septum). You cannot give each circulation its own ventricle, so instead of a two-ventricle (biventricular) repair you commit the child to a STAGED single-ventricle palliation that ends in the Fontan circulation: the one good ventricle pumps only to the body, and the systemic venous blood is plumbed DIRECTLY into the lungs to flow there passively, with no ventricle driving it. C6 owns two Master-Exam questions. **Q2** Glenn shunt prerequisites, all EXCEPT (answer **B**: the pulmonary vascular resistance must be LOW, so ">3 Wood units / high PVR" is the disqualifier, not a prerequisite). **Q10** Glenn shunt postoperative complications, all EXCEPT (answer **E**: protein-losing enteropathy is a FONTAN complication, not a Glenn complication).

Step 0. The big theory: one ventricle, so you re-plumb the veins into the lungs and pray the resistance is low

Every lesson before this one assumed the heart has two ventricles and the fix is to reconnect them correctly. C6 is the exception. Some hearts are born with only **one ventricle that actually works** (the other is too small, absent, or unusable): hypoplastic left heart syndrome (no usable LV), tricuspid atresia (no inlet to the RV, so a small RV), double-inlet left ventricle, an unbalanced complete AVSD, or pulmonary atresia with an intact ventricular septum. You cannot separate the blue and red circulations by giving each a ventricle, because there is only one. So the entire strategy changes: you accept that **the single good ventricle will pump the BODY**, and you find another way to get blood through the lungs.

The trick is that the lungs of a healthy child are a very-low-resistance circuit. If the pulmonary resistance is low enough, blood will flow through the lungs **passively**, pushed only by the small pressure in the great veins, with no ventricle behind it at all. So the plan is to **disconnect the systemic veins from the heart and sew them straight onto the pulmonary artery**, so venous blood drains directly into the lungs, gets oxygenated, returns to the single ventricle, and is pumped out to the body. That final arrangement is the **Fontan circulation**, and you get there in stages because a newborn's pulmonary resistance is still high and must be allowed to fall first.

The three stages, in order:

- **Stage 1, the neonatal setup (Norwood, for hypoplastic left heart).** In the first days of life the single ventricle must supply BOTH the body and the lungs, so you rebuild the aorta from the pulmonary trunk (so the single ventricle can reach the body), open the atrial septum for free mixing, and add a controlled source of pulmonary blood flow (a Blalock-Taussig shunt or a Sano RV-to-PA conduit). This just keeps the baby alive and balanced.

- **Stage 2, the bidirectional Glenn (about 4 to 6 months).** You take the **superior vena cava** off the heart and sew it end-to-side onto the **right pulmonary artery**, so all the UPPER-body venous blood now flows passively into the lungs. You take down the earlier shunt. This offloads the single ventricle (it no longer has to pump the extra pulmonary volume) and is the first step of the passive-flow circuit.
- **Stage 3, the Fontan / total cavopulmonary connection (about 2 to 4 years).** You now route the **inferior vena cava** into the pulmonary artery as well (a lateral tunnel inside the atrium or an extracardiac conduit). Now ALL systemic venous blood, upper and lower, flows passively to the lungs, and the single ventricle pumps ONLY to the body. The child is finally acyanotic.

The one idea that makes the whole pathway make sense, and that both exam questions test: **because there is no ventricle pushing blood through the lungs, the whole thing only works if the lungs offer almost no resistance.** That is why the prerequisites for a Glenn (Q2) are all about a low pulmonary resistance and pressure and a good ventricle, and why the late complications of a Fontan (the endpoint) all come from the chronically HIGH pressure you have to keep in the systemic veins to push blood through the lungs without a pump.

Hook: **"One working ventricle, so it pumps the BODY and you sew the systemic veins straight onto the lungs to flow PASSIVELY (no pump). You stage it: Norwood sets it up, the bidirectional Glenn sends the SVC to the pulmonary artery, the Fontan adds the IVC so ALL venous blood goes to the lungs. It only works if the pulmonary resistance is LOW (that is Q2), and the price of driving blood through the lungs with no pump is a chronically high venous pressure that later causes the Fontan complications (that is Q10, and protein-losing enteropathy is a Fontan problem, not a Glenn one)."**

Step 1. The map: the five ideas of C6

These numbers are the SAME as the concept numbers in Step 2 (map row 1 = concept 1, and so on).

#	Idea	The one-line mechanism	Exam anchor
1	**What "single ventricle" means**	Only ONE usable pumping ventricle (HLHS, tricuspid atresia, double-inlet LV, unbalanced AVSD, PA with intact septum), so a two-ventricle repair is impossible and you commit to staged palliation	The setup for the whole pathway. The single ventricle will end up pumping only the systemic circulation
2	**The passive-flow principle (the key to everything)**	You sew the systemic VEINS directly onto the pulmonary artery so venous blood flows through the lungs with NO ventricle behind it, driven only by the low venous pressure	This is WHY low pulmonary resistance is mandatory. Both Q2 and Q10 fall out of this one fact
3	**The three stages (Norwood then Glenn then Fontan)**	Norwood keeps the neonate balanced; the **bidirectional Glenn** connects **SVC to the right pulmonary artery** (upper-body venous blood to the lungs); the **Fontan** adds the **IVC to the PA** so ALL venous blood goes to the lungs and the ventricle pumps only the body	Know the order and what each stage connects. Glenn = SVC only; Fontan = SVC + IVC (total cavopulmonary connection)
4	**Glenn prerequisites (owns Q2)**	For passive flow to work you need a **LOW pulmonary vascular resistance and low PA pressure**, **good ventricular function (low end-diastolic pressure)**, **adequate non-distorted pulmonary arteries**, and a **competent AV valve**	**Q2**: all are prerequisites EXCEPT a HIGH PVR. The valve-key line is "PVR must be LOW; >3 Wood units is a contraindication" (answer **B**)
5	**Glenn vs Fontan complications (owns Q10)**	Glenn: **upper-body / SVC venous congestion, cyanosis (only the upper body drains to the lungs, and hepatic factor bypasses the lungs so pulmonary AV malformations develop), effusions**. Fontan: the high-systemic-venous-pressure diseases, including **protein-losing enteropathy, Fontan liver disease, plastic bronchitis, atrial arrhythmias**	**Q10**: all are Glenn complications EXCEPT **protein-losing enteropathy** (that is a Fontan complication) (answer **E**)

The one rule that ties it together: there is no pump between the systemic veins and the lungs, so the pathway lives or dies on a LOW pulmonary resistance. Select patients for a Glenn by proving the resistance and pressure are low and the ventricle and valve are good (Q2), and understand that the endpoint Fontan buys its acyanotic circulation at the cost of a permanently elevated systemic venous pressure, which is what eventually damages the gut (protein-losing enteropathy) and the liver, distinguishing Fontan complications from the simpler upper-body congestion of a Glenn (Q10).

Step 2. The concepts you must know (numbered to match the map in Step 1: concept N = map row N)

1) What "single ventricle" actually means, and why you cannot do a normal repair

- **Synonym:** single-ventricle physiology; univentricular heart; functionally univentricular circulation.
- **The core fact first:** a single-ventricle heart has **only one ventricle capable of doing useful work**. The classic examples are **hypoplastic left heart syndrome** (the LV, mitral valve and ascending aorta are all too small, so only the RV works), **tricuspid atresia** (no tricuspid valve, so no inlet to the RV, leaving a dominant LV), **double-inlet left ventricle** (both atria empty into one LV), an **unbalanced complete AVSD** (the common AV valve commits mostly to one ventricle), and **pulmonary atresia with an intact ventricular septum** (a hypoplastic RV). In all of them you have one good pump and one useless one.
- **Why it forces the pathway:** a normal "biventricular" repair reconnects two working ventricles so that one drives the body and the other drives the lungs. With only one ventricle you cannot do that. So you accept the loss of a pulmonary pump and commit the child to a **staged single-ventricle palliation** whose endpoint is the Fontan, where the single ventricle drives the body and the lungs are fed passively.
- **Hook:** "Single ventricle = one usable pump (HLHS, tricuspid atresia, double-inlet LV, unbalanced AVSD, PA-intact-septum). No two-ventricle repair is possible, so you stage down to a Fontan."

2) The passive-flow principle: no pump between the veins and the lungs, so the resistance must be low (the key to the whole lesson)

- **The core fact first:** in the finished (Fontan) circuit, the **systemic veins are connected DIRECTLY to the pulmonary artery**. There is **no ventricle between them and the lungs**. Blood is pushed through the lungs only by the **residual pressure in the great veins** (a systemic venous pressure of roughly 10 to 15 mmHg), which must overcome the resistance of the lung circuit and still have enough head of pressure to fill the single ventricle on the far side.
- **Reads as (the causal chain):** systemic veins to pulmonary artery directly to blood crosses the lungs only if pulmonary resistance is LOW to oxygenated blood returns to the single ventricle to ventricle pumps it to the body. Break the "low resistance" link and the whole circuit stalls: nothing fills the ventricle, cardiac output falls, and the veins engorge.
- **Why it matters (it generates BOTH exam questions):** because the driver is a feeble venous pressure and not a ventricle, the lungs must offer almost no resistance. That single requirement is the **entire content of the Glenn prerequisites (Q2)**, and the price of running a chronically raised systemic venous pressure to push blood through the lungs is the **entire content of the Fontan complications (Q10)**. If you understand this one principle you can reconstruct both answers without memorising lists.
- **Hook:** "No pump between the veins and the lungs, so blood only flows if the pulmonary resistance is LOW. That one fact IS the exam: low resistance to select for a Glenn (Q2); high venous pressure as the late price in a Fontan (Q10)."

3) The three stages: Norwood, then the bidirectional Glenn, then the Fontan

- **Stage 1 is TAILORED to the lesion (the balanced-circulation goal).** Every single-ventricle newborn needs the same Stage-1 goal: a **balanced circulation** = the body perfused + the lungs getting the RIGHT amount of flow (not too much, not too little) + free atrial mixing. But the actual **operation is chosen by what is wrong** with that particular heart, so the Norwood is NOT the Stage 1 for every single ventricle:
- **Systemic (aortic) outflow obstructed = the HLHS group to the NORWOOD.** In hypoplastic left heart the **ascending aorta is a hypoplastic thread (about 2 to 3 mm)** and the working RV has no outlet into it, so before surgery the baby survives only because the **open duct** carries RV / pulmonary blood backward into the aorta (duct-dependent systemic, it dies when the duct closes). The Norwood gives the RV a permanent wide road to the body: it **rebuilds the big main pulmonary artery (which already arises from the RV) into a "neo-aorta"**, joined to the small native aorta, so **RV to neo-aorta to body, with no duct needed**. That step uses up the RV's outlet, so the lungs are now cut off, and you add a **controlled (deliberately small) shunt** to feed them: a **modified Blalock-Taussig shunt** (systemic artery to PA) or a **Sano** (a conduit from the RV to the PA). "Controlled" is the key word: the lungs are a low-resistance circuit, so an unrestricted flow would **flood the lungs AND steal from the body** (pulmonary overcirculation + systemic hypoperfusion); the small shunt limits lung flow to keep Qp:Qs balanced. An **atrial septectomy** lets the oxygenated pulmonary-venous blood cross from the tiny LA to the RA to reach the pump.
- **Too MUCH pulmonary flow (a single ventricle with NO pulmonary stenosis) to a PULMONARY ARTERY BAND.** Here the low-resistance lungs would be flooded (overcirculation, heart failure) and the high pulmonary pressure would damage the lung vessels before the Glenn. Fix = a **band around the pulmonary artery that mechanically restricts the excess flow**. The aorta is normal, so **no Norwood**.
- **Too LITTLE pulmonary flow (a single ventricle WITH pulmonary stenosis or atresia, cyanotic, e.g. tricuspid atresia + PS) to a BT SHUNT (or ductal stent)** to **add** controlled flow to the lungs. Again the aorta is normal, so **no Norwood**.
- If the heart is already balanced, no Stage-1 surgery is needed, and you go straight to the Glenn.

So the **Norwood (the aortic reconstruction) is specific to the obstructed-systemic-outflow group (HLHS and variants)**; other single ventricles get the Stage-1 tool matched to their pulmonary flow. Whatever the Stage 1, they ALL converge to the Glenn (Stage 2) then the Fontan (Stage 3). - **Stage 2, the bidirectional Glenn (bidirectional cavopulmonary anastomosis, about 4 to 6 months):** the **superior vena cava is divided and sewn end-to-side onto the right pulmonary artery**, so **all upper-body venous blood now drains passively into both lungs** ("bidirectional" = it flows into the right AND left pulmonary arteries). The Stage-1 shunt is taken down. This is done at 4 to 6 months because by then the newborn's high pulmonary resistance has fallen enough for passive flow to work. It **offloads the single ventricle**, which no longer has to pump the recirculated pulmonary volume, so the ventricle remodels favourably. The IVC blood still returns to the heart, so the child is still mildly cyanotic (upper-body blue blood goes to the lungs, but lower-body blue blood still mixes into the systemic output). - **Stage 3, the Fontan (total cavopulmonary connection, about 2 to 4 years):** the **inferior vena cava is now also routed to the pulmonary artery**, either as a **lateral tunnel** built inside the atrium or an **extracardiac conduit** (a tube from the IVC to the PA). Now **every systemic vein, upper and lower, drains to the lungs**, the single ventricle pumps ONLY to the body, and the child becomes **acyanotic**. A small **fenestration** is often left: a deliberate hole (about 4 mm) in the wall between the **high-pressure Fontan pathway** and the **lower-pressure atrium**. Because the Fontan pathway carries systemic venous blood at a high pressure (there is no pump behind it), the hole lets a little blood shunt **right-to-left, bypassing the lungs**, which does two jobs: (1) it **pops off** any pressure build-up in the early post-operative period, so blood does not back up into the veins / liver / gut (less congestion and effusions), and (2) it **preserves ventricular filling and cardiac output** when the lungs will not yet accept enough passive flow. The price is a **mild fall in oxygen saturation (~90%)**, because some deoxygenated blood reaches the body. Once the circulation settles, the fenestration often **closes on its own or is closed later with a catheter device** (after test-occluding it to confirm the sats and output stay good). - **The clean summary: Glenn connects the SVC to the PA; Fontan adds the IVC to the PA.** Glenn = one cava; Fontan

= both cavae (total cavopulmonary connection). That distinction is exactly what Q2 and Q10 hang on. - **Hook:** "Norwood sets up the neonate. Bidirectional Glenn = SVC to the right PA (upper-body blood to the lungs). Fontan = add the IVC (total cavopulmonary connection), so ALL venous blood goes to the lungs and the ventricle pumps only the body."

4) Glenn prerequisites: everything must favour passive flow (this is Q2)

- **The core fact first:** before you commit a child to a bidirectional Glenn (and, later, a Fontan), you must prove the passive circuit will actually work. Because the driving force is a low venous pressure and not a ventricle, **every prerequisite is a condition that keeps the lung circuit easy to push blood through and the downstream ventricle able to receive it**:
- **A LOW pulmonary vascular resistance** (the single most important prerequisite; typically < 3 to 4 Wood units). A HIGH PVR is the classic contraindication.
- **A low mean pulmonary artery pressure** (roughly < 15 mmHg).
- **Good ventricular function with a low ventricular end-diastolic pressure** (a stiff or failing ventricle raises the pressure the venous blood must be pushed UP TO, and stalls the circuit).
- **Adequately sized, non-distorted, confluent pulmonary arteries** (branch stenoses or hypoplasia raise the resistance mechanically).
- **A competent AV valve** (significant AV-valve regurgitation raises filling pressures and volume-loads the single ventricle).
- Sinus rhythm and no significant pulmonary vein / inflow obstruction help too.
- **Example / exam anchor (the whole of Q2): Q2** lists the prerequisites for a Glenn shunt and asks which is NOT a prerequisite. The true prerequisites are a low PVR, a low PA pressure, good ventricular function, adequate pulmonary arteries and a competent AV valve. The **odd one out (answer B)** is the item stating that the **pulmonary vascular resistance must be HIGH (or "> 3 Wood units")**: that is not a prerequisite, it is a **contraindication**. The PVR must be **LOW** for passive flow to work. This is the direct, testable payoff of concept 2.
- **Hook:** "Glenn prerequisites all mean 'make passive flow easy': LOW PVR, LOW PA pressure, GOOD ventricle (low end-diastolic pressure), GOOD pulmonary arteries, COMPETENT AV valve. Q2: the EXCEPT is a HIGH PVR (answer B), because a high PVR is a contraindication, not a prerequisite."

5) Glenn vs Fontan complications: know which problem belongs to which stage (this is Q10)

- **The core fact first:** the complications split cleanly by stage, and Q10 tests exactly that split.
- **Bidirectional Glenn complications: upper-body / superior vena cava venous congestion** (facial and head swelling, headaches, because only the SVC is under the new higher pressure); **persistent cyanosis / desaturation** (the IVC blood still reaches the systemic circulation, and over time **pulmonary arteriovenous malformations** develop in the lungs because the "hepatic factor" in inferior-caval blood is diverted away from the lungs in a classic Glenn, so deoxygenated blood shunts through the lungs); **pleural effusions and chylothorax**; **venovenous collaterals** that decompress the high SVC pressure and worsen cyanosis. Glenn does NOT run the whole body at a high venous pressure, so the "high-systemic-venous-pressure" diseases below are NOT Glenn complications.
- **Fontan complications (for contrast, because the endpoint runs the WHOLE systemic venous system at a chronically high pressure): protein-losing enteropathy** (chronic gut venous and lymphatic congestion leaks albumin into the bowel, giving oedema and low albumin); **Fontan-associated liver disease** (chronic hepatic congestion to fibrosis / cirrhosis); **plastic bronchitis**; **atrial arrhythmias** (especially with the older atrial-tunnel Fontans); **thromboembolism**; **exercise intolerance and Fontan failure**.
- **Why each complication happens (mechanisms, so you can reconstruct them, not memorise a list):**

- **Glenn to pulmonary AVMs (the hepatic-factor story):** the liver makes a substance carried in its venous (hepatic-vein to IVC) blood, the **"hepatic factor"**, that the lung vessels need to stay normal. In a classic Glenn **only SVC blood reaches the lungs**, so the **IVC / hepatic blood carrying the factor is diverted away from the lungs**; deprived of it the lungs form **pulmonary arteriovenous malformations** (artery-to-vein shortcuts that bypass the alveoli), so blood crosses **without being oxygenated** and the patient **desaturates over time**. The **Fontan cures it**: once the IVC is plumbed to the lungs, the factor returns and the AVMs regress.
- **Every Fontan complication traces to two facts:** (1) the WHOLE systemic venous system runs at a chronically **high pressure** (there is no pump driving the lungs, so the venous pressure IS the driving force), and (2) pulmonary flow is **passive, slow, and cannot rise on demand**. From those:
  - **Protein-losing enteropathy + plastic bronchitis** = **lymph leaking** under the high venous / lymphatic pressure (into the gut = PLE with low albumin and oedema; into the airways = thick rubbery casts shaped like the bronchial tree that are coughed up = plastic bronchitis). Same mechanism, two sites.
  - **Fontan-associated liver disease** = chronic **hepatic venous congestion** to fibrosis / cirrhosis.
  - **Atrial arrhythmias** = the older atrial-tunnel Fontans have long **suture lines in a dilated, high-pressure atrium** (scar + stretch = a re-entry substrate); the modern **extracardiac conduit avoids atrial suture lines**, so far fewer arrhythmias.
  - **Thromboembolism** = **slow, non-pulsatile, stagnant venous flow** (stasis) plus abnormal clotting factors from the congested liver to clots in the Fontan pathway that embolise (hence lifelong anticoagulation in many).
  - **Exercise intolerance** = to exercise you must push MORE blood through the lungs, but there is **no pump to do it**, so the single ventricle cannot raise its preload / output on demand and the patient tires fast.
  - **Fontan failure** = the end-stage decompensation when the high-pressure, single-ventricle, passive-flow circulation runs out of reserve (worsening congestion, low output, and the complications above); the options are limited and ultimately transplant.
- **Example / exam anchor (the whole of Q10): Q10** lists postoperative complications of a Glenn shunt and asks which is NOT one. The genuine Glenn complications are SVC / upper-body venous congestion, cyanosis / desaturation, pulmonary AV malformations, and effusions. The **odd one out (answer E)** is **protein-losing enteropathy**, which is a **Fontan complication** (it needs the chronically high WHOLE-body systemic venous pressure of the completed Fontan, which a Glenn does not create). Same logic underlies "Fontan liver disease" or "plastic bronchitis" if they appear as distractors: those are Fontan, not Glenn.
- **Hook:** "Glenn complications = UPPER-body problems: SVC congestion, cyanosis, pulmonary AV malformations (hepatic factor bypasses the lungs), effusions. Fontan complications = WHOLE-body high-venous-pressure problems: protein-losing enteropathy, liver disease, plastic bronchitis, atrial arrhythmias. Q10: the EXCEPT is protein-losing enteropathy (answer E), because that is a FONTAN, not a Glenn, complication."

Step 3. EXAMINE yourself (questions only; answers are sealed below, do not peek)

1. A child is being worked up for a bidirectional Glenn. Which single haemodynamic measurement, if HIGH, would contraindicate the operation, and why?
2. What exactly does a bidirectional Glenn connect, and what does the Fontan add to it?
3. Why is the single ventricle, after a completed Fontan, pumping to only ONE circulation, and which one?
4. Q2 logic: list four genuine prerequisites for a Glenn shunt, then the one item that is NOT a prerequisite.
5. Q10 logic: name two real Glenn complications and the one complication that actually belongs to the Fontan, not the Glenn.

6. Mechanistically, why does protein-losing enteropathy belong to the Fontan and not the Glenn?
7. Why do pulmonary arteriovenous malformations develop after a classic Glenn, and what is the missing ingredient called?

▶ 🔒 Sealed answer key (open only after you have answered all seven)

C6 owns Q2 (Glenn prerequisites, answer B = a high PVR is the odd-one-out) and Q10 (Glenn complications, answer E = protein-losing enteropathy is a Fontan complication). No answer-key conflicts in this lesson. Next: C7, adult CHD (repaired TOF + PVR, adult D-TGA options, Eisenmenger), which owns Q1, Q14 and Q17.

C6 - Interactive visual (rendered): diagrams & schematics

The live, toggleable version is the HTML file on the CHD hub. Below is a static render of its default view; open the hub to step through the other toggle states.

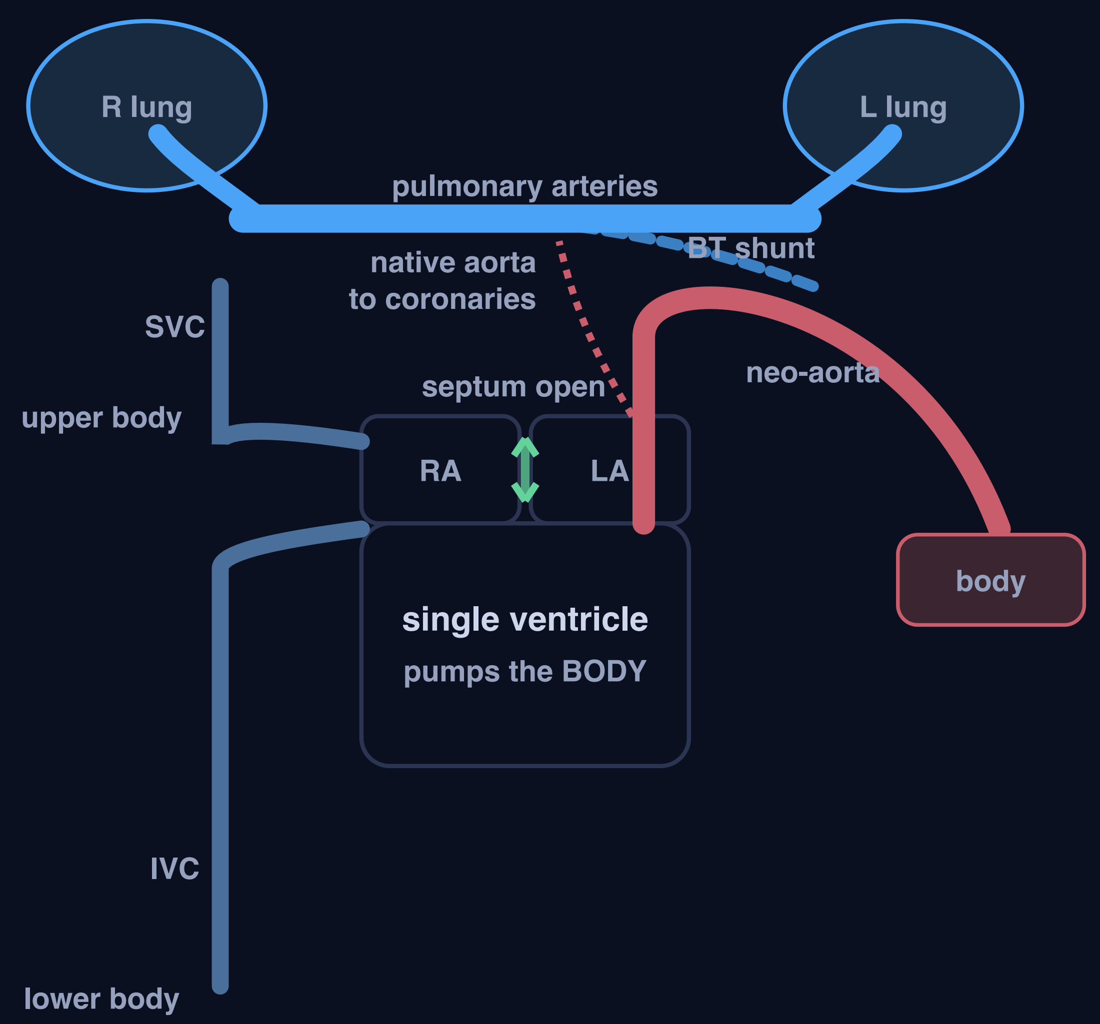
C6 - The single-ventricle pathway (Glenn and Fontan)

One usable pumping ventricle, so it drives the BODY and you plumb the systemic veins straight onto the pulmonary artery to flow through the lungs PASSIVELY, with no pump. Anchors: Q2 (Glenn prerequisites, the odd-one-out is a HIGH PVR = answer B) and Q10 (Glenn complications, the odd-one-out is protein-losing enteropathy, a FONTAN problem = answer E).

1. THE STAGED PATHWAY: STAGE 1 IS TAILORED TO THE LESION, THEN EVERYONE GOES GLENN THEN FONTAN

> **The goal at every stage is a BALANCED circulation: the body perfused + the lungs getting the RIGHT amount of flow (not too much, not too little) + free atrial mixing. Stage 1 depends on WHAT is wrong; Stages 2 and 3 are the same for everyone.**

Stage 1: first-stage palliation	**Stage 2: bidirectional Glenn**	**Stage 3: Fontan**
Obstructed aorta (HLHS): Norwood	**Too MUCH lung flow: PA band**	**Too LITTLE lung flow: BT shunt**
Norwood lung supply: BT shunt	**Norwood lung supply: Sano**	



Hand-drawn inline SVG, self-contained. The cavopulmonary connections sit on the patient's right (screen-left), where the Glenn and Fontan are actually built.

Stage 1 - Norwood (HLHS), BT shunt

In HLHS the **ascending aorta is a thread** and the working RV has no outlet into it. The Norwood rebuilds the big **main PA into a NEO-AORTA** off the RV (RV to body), while the native aorta now only fills the **coronaries retrograde**. An **atrial septectomy** opens mixing.

Lung supply = a **modified BT shunt** (a small tube, systemic artery to PA). **Small on purpose**: an unrestricted flow would flood the lungs and starve the body.

All three Stage-1 routes CONVERGE: everyone funnels into the Glenn (Stage 2) then the Fontan (Stage 3). And the one idea underneath it all: after a Glenn/Fontan there is NO pump between the systemic veins and the lungs, so blood crosses only if the pulmonary resistance is LOW (that drives Q2 and Q10).

2. GLENN PREREQUISITES: EVERYTHING MUST FAVOUR PASSIVE FLOW (Q2)

TRUE prerequisites (make passive flow easy)

- **LOW pulmonary vascular resistance** (the key one; typically < 3 to 4 Wood units)
- **Low mean PA pressure** (about < 15 mmHg)
- **Good ventricular function** with a **low end-diastolic pressure**
- **Adequate, non-distorted pulmonary arteries** (no branch stenosis / hypoplasia)
- **Competent AV valve** (no significant regurgitation)

The Q2 trap (the EXCEPT)

A HIGH pulmonary vascular resistance (for example "> 3 Wood units") is **NOT a prerequisite - it is a CONTRAINDICATION.** With no pump behind the lungs, a high resistance stalls the whole passive circuit.

Q2 answer = B (the PVR must be LOW, so the "high PVR" option is the odd-one-out).

3. GLENN VS FONTAN COMPLICATIONS: KNOW WHICH PROBLEM BELONGS TO WHICH STAGE (Q10)

Bidirectional Glenn (only the SVC / upper body is under pressure)

- **SVC / upper-body venous congestion** (facial swelling, headaches)
- **Cyanosis / desaturation** (the IVC still returns to the heart)
- **Pulmonary arteriovenous malformations** (the hepatic "factor" in IVC blood bypasses the lungs, so AVMs form and worsen the cyanosis)
- **Pleural effusions / chylothorax**, venovenous collaterals

Fontan (the WHOLE systemic venous system runs at a high pressure)

- **Protein-losing enteropathy** (gut venous/lymphatic congestion leaks albumin)
- **Fontan-associated liver disease** (chronic hepatic congestion to fibrosis)
- **Plastic bronchitis**
- **Atrial arrhythmias**, thromboembolism, exercise intolerance, Fontan failure

The WHY (so you can reconstruct them): Glenn to AVMs = the lungs get only SVC blood, so the liver's "hepatic factor" (in IVC blood) never reaches them and artery-to-vein shortcuts form (the Fontan cures it). **Every Fontan complication** = the whole venous system at HIGH pressure (lymph leaks: **PLE** into the gut, **plastic bronchitis** into the airway; liver congestion) +

SLOW stagnant flow (**clots**) + NO pump for the lungs (**exercise intolerance**, and eventual **Fontan failure**); atrial suture lines in the old tunnel Fontans give the **arrhythmias**.

Q10: postoperative complications of a Glenn shunt, all EXCEPT protein-losing enteropathy (answer E). PLE needs the chronically high WHOLE-body systemic venous pressure of the completed Fontan; a Glenn only raises the SVC (upper body).

C6 visual - Module 5 CHD. Hand-drawn inline SVG, self-contained.

C7 - Adult congenital heart disease: the patient survived childhood, so now you manage the LATE lesion the repair left behind (repaired TOF and its pulmonary regurgitation, the adult D-TGA repair options, and Eisenmenger, the shunt that reversed)

> Visual: `study/visuals/C7-adult-chd.html` (a repaired-TOF card drawn per the anatomy rule showing the old transannular patch across the RV outflow, the free PULMONARY REGURGITATION it leaves, and the progressive RV dilation and dysfunction that follow, with a "timed pulmonary valve replacement" flag and the Q1 "earlier repair is better" trap; an adult D-TGA card laying out the three operations side by side (the atrial switch / Mustard-Senning with the morphologic RV left as the systemic ventricle, the modern arterial switch, and the Rastelli for TGA + VSD + LVOTO/PS) with the Q17 answer flagged; an Eisenmenger card showing a large left-to-right shunt driving pulmonary vascular disease until PVR climbs above systemic and the shunt REVERSES to right-to-left, giving cyanosis, clubbing and erythrocytosis, marked INOPERABLE; and a sign-to-lesion matching card for Q14 with the ambiguous holosystolic / click / hepatomegaly row flagged). This is the ADULT chapter of the CHD module, and it is a different kind of thinking. C2 to C6 asked "what is the defect and how do you fix it in a baby." C7 asks "the patient is now an adult, either never repaired or (far more often) repaired years ago, so what is the RESIDUAL or LATE lesion, and what do you do about it now." The single unifying idea is that a childhood repair is rarely a cure; it trades the original defect for a slower, later problem (a leaking valve, a failing systemic right ventricle, a reversed shunt), and adult CHD is the management of that trade. C7 owns three Master-Exam questions. **Q1** TOF repair in adults, all true EXCEPT (answer **A**: "the later the repair the better" is FALSE; EARLIER repair gives better long-term outcomes, and the dominant late lesion is chronic pulmonary regurgitation). **Q14** physical-exam-to-lesion matching, all EXCEPT (answer **E** per the key; it is a match-the-column question and the holosystolic / click / hepatomegaly item is the one flagged as not cleanly matching, so confirm the exact wording with the course). **Q17** adult D-TGA, all true EXCEPT (answer **B**: for D-TGA with severe pulmonary stenosis + a large VSD the operation is the **Rastelli** procedure, NOT a univentricular / Fontan-type repair).

Step 0. The big theory: a childhood repair is a trade, not a cure, so the adult presents with the LATE lesion

Everything before this lesson was paediatric: a cyanotic or acyanotic baby, a defect, and an operation. C7 picks the story up decades later. Two things bring a congenital heart to an adult clinic. A few patients arrive **unoperated** (a lesion that was mild or missed in childhood, like a secundum ASD or a small PDA, presenting for the first time in adult life). But the large and growing majority are **survivors of a childhood repair**, and for them the exam question is never "what was the original defect," it is "what did the repair leave behind, and is it time to act on it now." That reframing is the whole of adult CHD.

Three archetypes carry the chapter, and each is a clean example of "the repair traded the defect for a later problem":

- **Repaired tetralogy of Fallot (the model, owns Q1).** The childhood repair closes the VSD and relieves the right-ventricular outflow obstruction, often by sewing a **transannular patch** across the pulmonary valve ring to widen it. That patch cures the obstruction but destroys the pulmonary valve, so the price is **chronic, free PULMONARY REGURGITATION**: blood leaks back into the right ventricle with every diastole for the rest of the patient's life. For years it is silent, but the volume overload slowly **dilates the right ventricle, then impairs it**, and a dilated scarred RV generates **ventricular arrhythmia and a risk of sudden cardiac death**. The management is a **timed pulmonary valve replacement (PVR)** before the RV dilation becomes irreversible. And the specific exam trap: **earlier primary repair gives better long-term outcomes**, so "the later the repair the better" is the false statement.

- **Adult D-TGA (owns Q17).** A D-transposition patient in an adult clinic has ALWAYS been operated on (unoperated D-TGA is fatal in infancy, straight back to C3). What matters is WHICH operation, because each leaves a different adult problem. The **historic atrial switch (Mustard or Senning)** reroutes blood at the atrial level with a baffle, which corrects the circulation but leaves the **morphologic RIGHT ventricle pumping to the body**; decades later that systemic RV fails, the baffles leak or obstruct, and the atria develop arrhythmia. The modern **arterial switch (ASO)** is the true anatomic repair (the great arteries are swapped back and the coronaries reimplanted, so the left ventricle becomes the systemic pump). And when D-TGA comes with a **large VSD and left-ventricular outflow obstruction / pulmonary stenosis**, the operation is the **Rastelli** procedure. The Q17 trap replaces Rastelli with a univentricular (Fontan-type) repair, which is wrong: this is a two-ventricle heart.

- **Eisenmenger syndrome (the shunt that reversed).** Take any large, long-standing left-to-right shunt (a big VSD, ASD, PDA, or AVSD) and leave it unrepaired. The chronically high pulmonary blood flow damages the pulmonary arterioles, the **pulmonary vascular resistance climbs**, and when the **PVR finally rises above the systemic resistance the shunt REVERSES to right-to-left**. Now deoxygenated blood crosses into the systemic side and the once-pink patient turns **cyanotic, with clubbing and secondary erythrocytosis**. The cruel rule: once the shunt has reversed, closing it is FATAL (you remove the only pop-off for the fixed high-resistance right heart), so Eisenmenger is **INOPERABLE**, pregnancy is contraindicated, and management is supportive plus pulmonary vasodilators plus, ultimately, transplant. This is the adult-CHD payoff of the [[C1]] PVR concept: the number you calculate in C1 is the number that decides whether a shunt can still be closed.

Hook: **"A childhood repair is a TRADE, not a cure, so the adult shows up with the LATE lesion. Repaired TOF trades the obstruction for chronic PULMONARY REGURGITATION to RV dilation, arrhythmia, sudden death, so you time a pulmonary valve replacement (and EARLIER repair is better, that is Q1). Adult D-TGA is defined by WHICH operation: atrial switch (Mustard/Senning) leaves a failing systemic RV, arterial switch is the modern anatomic fix, and TGA + VSD + PS is repaired by RASTELLI, not a Fontan (that is Q17). And a big unrepaired left-to-right shunt eventually drives the PVR above systemic so the shunt REVERSES: Eisenmenger, cyanotic and INOPERABLE."**

Step 1. The map: the five ideas of C7

These numbers are the SAME as the concept numbers in Step 2 (map row 1 = concept 1, and so on).

#	Idea	The one-line mechanism	Exam anchor
1	**What "adult CHD" means (the reframe)**	A few adults are unoperated, but MOST are repair survivors; the question shifts from "what was the defect" to "what LATE / residual lesion did the repair leave, and is it time to act"	The mindset for the whole lesson. A repair is a trade, not a cure. Underlies Q1, Q17 and the Eisenmenger story
2	**Repaired TOF to chronic pulmonary regurgitation (owns Q1)**	The transannular patch cures the RV-outflow obstruction but destroys the pulmonary valve, leaving **free pulmonary regurgitation to progressive RV dilation then dysfunction to ventricular arrhythmia + sudden-death risk**; treat with a **timed pulmonary valve replacement**	**Q1**: all true EXCEPT **A**. "The later the repair the better" is FALSE; EARLIER primary repair gives better outcomes. The dominant late lesion is chronic pulmonary regurgitation

#	Idea	The one-line mechanism	Exam anchor
3	**Adult D-TGA: which operation (owns Q17)**	**Atrial switch (Mustard/Senning)** = baffle at atrial level, leaves the **morphologic RV as the systemic ventricle** (late systemic-RV failure, baffle leak/obstruction, atrial arrhythmia). **Arterial switch (ASO)** = the modern anatomic repair. **Rastelli** = for TGA + VSD + LVOTO/PS	**Q17**: all true EXCEPT **B**. For D-TGA with severe PS + a large VSD the operation is **Rastelli**, NOT a univentricular / Fontan-type repair (this is a two-ventricle heart)
4	**Eisenmenger syndrome (shunt reversal)**	A large long-standing left-to-right shunt (VSD/ASD/PDA/AVSD) drives pulmonary vascular disease until **PVR rises ABOVE systemic and the shunt REVERSES to right-to-left**: cyanosis, clubbing, erythrocytosis. **INOPERABLE** (closing it is fatal), pregnancy contraindicated	Not a numbered Q here, but core adult-CHD teaching and the payoff of the [[C1]] PVR concept. Management = supportive + pulmonary vasodilators + transplant
5	**Sign-to-lesion matching (owns Q14)**	Each classic lesion advertises itself with one clean sign: **fixed split S2 = ASD; continuous machinery murmur = PDA; holosystolic at LLSB = VSD; ejection click at upper-left sternal border = pulmonary stenosis; radio-femoral delay = coarctation; pulsatile liver = severe TR**	**Q14**: a match-the-column question, all true EXCEPT **E** per the key (the holosystolic / click / hepatomegaly item is the one flagged as not cleanly matching). Confirm the exact wording with the course; do not overstate certainty

The one rule that ties it together: in adult CHD you are almost never treating the original defect; you are treating the shadow it or its repair casts years later. The repaired-TOF patient is treating a leaking pulmonary valve (Q1). The D-TGA patient is defined by which operation they had, and the sickest late problem (a failing systemic RV) belongs to the OLD atrial switch, while the modern arterial switch and the Rastelli-for-TGA-plus-VSD-plus-PS are the other two options (Q17). The unrepaired big shunt is treating a reversed, cyanotic circulation whose PVR has crossed the systemic line and can no longer be closed (Eisenmenger). And underneath all of it, Q14 tests whether you can still read a single physical sign back to its lesion.

Step 2. The concepts you must know (numbered to match the map in Step 1: concept N = map row N)

1) What "adult CHD" actually means: mostly repair survivors, so you manage the LATE lesion

- **Synonym:** adult / grown-up congenital heart disease (ACHD, GUCH); congenital heart disease in the adult; late follow-up of repaired CHD.
- **The core fact first:** an adult with congenital heart disease falls into two groups. A minority are **truly unoperated**, because their lesion was mild enough to survive childhood undetected (a secundum ASD found on an incidental murmur or a stroke work-up, a small restrictive VSD, a quiet PDA). The majority are **survivors of a childhood operation**, and for them the original defect is history; what remains is a **residual, recurrent, or entirely new LATE lesion created by the repair itself**. So the clinical question moves from "what is the defect and how do I fix it" (the paediatric question of C2 to C6) to "what did the repair leave behind, and has it reached the threshold where I must act again."
- **Reads as (the causal chain):** childhood repair relieves the original defect to but leaves a residual burden (a leaking valve, a patch, a baffle, a conduit, a scar) to that burden slowly loads or scars a chamber over years to the chamber dilates, then fails, then arrhythmias to a SECOND, timed intervention in adult life. The whole specialty lives in the gap between "repaired" and "cured."
- **Why it matters (it frames all three questions):** repaired TOF (concept 2) is this idea in its purest form (the patch that cured the obstruction is the same patch that causes the late regurgitation). Adult D-TGA (concept 3) is the same idea sorted by WHICH operation created the late problem. Eisenmenger (concept 4) is the mirror image, the price of NOT repairing in time. Hold "a repair is a trade, not a cure" and the lesson coheres.

- **Hook:** "Adult CHD = mostly REPAIR SURVIVORS, so stop asking 'what was the defect' and ask 'what LATE lesion did the repair leave, and is it time to act.' A repair is a trade, not a cure."

2) Repaired TOF and chronic pulmonary regurgitation: the patch that cured the obstruction is the valve you now have to replace (this is Q1)

- **Synonym:** repaired tetralogy of Fallot; post-repair TOF; the transannular-patch / free-pulmonary-regurgitation problem.
- **The core fact first:** the childhood TOF repair does two things, closes the VSD and relieves the right-ventricular outflow tract obstruction. To relieve severe obstruction the surgeon often has to cut across the pulmonary valve annulus and sew in a **transannular patch** to widen the outflow. That patch abolishes the obstruction but **destroys the competence of the pulmonary valve**, so the patient is left with **free, chronic PULMONARY REGURGITATION**: with every diastole, blood falls back from the pulmonary artery into the right ventricle. For years this is well tolerated and clinically silent, which is exactly why it is dangerous.
- **Reads as (the causal chain):** transannular patch relieves the obstruction BUT wrecks the pulmonary valve to chronic free pulmonary regurgitation to the RV takes a diastolic volume overload every beat to over years the **RV dilates**, then its function drops (**RV dysfunction**) to a dilated, scarred, patched outflow becomes an arrhythmia substrate to **ventricular tachycardia, a widened QRS, and a risk of SUDDEN CARDIAC DEATH**. The late enemy of repaired TOF is not the old obstruction, it is the regurgitation the repair introduced.
- **Why it matters (the management + the exam trap):** because the RV can dilate a long way before it declares itself, you **survey the patient with serial imaging (cardiac MRI for RV volumes) and act BEFORE the dilation becomes irreversible**, by performing a **timed pulmonary valve replacement (PVR)** (surgical or transcatheter). PVR restores a competent valve, stops the volume overload, and allows the RV to remodel if you have not left it too late. Other late issues you may see: residual VSD, residual RV outflow or branch pulmonary stenosis, aortic root dilation with aortic regurgitation, and atrial as well as ventricular arrhythmia. And the specific Q1 point: **the timing of the ORIGINAL repair matters, and EARLIER primary repair gives BETTER long-term outcomes** (less cyanosis time, less RV hypertrophy and fibrosis, better late function). So the statement "the later the repair the better" is FALSE.
- **Example / exam anchor (the whole of Q1): Q1** = "regarding TOF repair in adults, all are true EXCEPT," and the false statement is **A, that the later the repair the better**. The true statements describe the reality of the repaired-TOF adult: the dominant late lesion is **chronic pulmonary regurgitation** from the transannular patch, it causes **progressive RV dilation and dysfunction with arrhythmia and a risk of sudden death**, and the management is a **timed pulmonary valve replacement**. Earlier repair, not later, gives the better outcome, so **A is the odd one out (answer A)**.
- **Hook:** "Repaired TOF: the transannular patch CURES the obstruction but leaves free PULMONARY REGURGITATION to silent RV dilation to RV dysfunction to arrhythmia and SUDDEN DEATH, so you TIME a pulmonary valve replacement before the RV is too far gone. Q1: EARLIER repair is better, so 'the later the repair the better' is the FALSE statement (answer A)."

3) Adult D-TGA: the patient is defined by WHICH operation, and TGA + VSD + PS is a Rastelli (this is Q17)

- **Synonym:** adult D-transposition of the great arteries; post-atrial-switch (Mustard / Senning) TGA; post-arterial-switch (ASO) TGA; Rastelli repair.
- **The core fact first:** an adult with D-TGA has ALWAYS had surgery (untreated D-TGA, the parallel-circulation cyanotic newborn emergency of C3, does not survive infancy). So the exam is not about the

diagnosis, it is about **which of three operations the patient had, because each leaves a different adult problem**:

- **The atrial switch (Mustard or Senning), the historic repair.** Instead of moving the arteries, the surgeon builds a **baffle inside the atria** that redirects venous return: systemic venous blood is channelled to the left ventricle and out to the lungs, and pulmonary venous blood to the right ventricle and out to the aorta. The circulation is corrected and the patient is pink, BUT the **morphologic RIGHT ventricle is left pumping to the body (the systemic ventricle)**. The RV was never built for a lifetime of systemic pressure, so the late problems are **systemic (morphologic-RV) failure, baffle leaks and baffle obstruction, and atrial arrhythmias / sinus node dysfunction** (the atrial suture lines scar the conduction tissue). If the exam shows you a middle-aged D-TGA patient with a failing systemic RV, they had an atrial switch.
- **The arterial switch (ASO), the modern anatomic repair.** The great arteries are transected and **switched back to their correct positions, and the coronary arteries are reimplanted** onto the neo-aorta. Now the **left ventricle is the systemic pump** (anatomically correct), so it avoids the systemic-RV problem. Late issues are different and milder: neo-aortic root dilation and regurgitation, coronary ostial stenosis, and supravalvar / branch pulmonary stenosis at the suture lines.
- **The Rastelli procedure, for D-TGA with a VSD and LV outflow obstruction / pulmonary stenosis.** When D-TGA comes packaged with a **large VSD plus severe LVOTO / pulmonary stenosis**, the surgeon uses the VSD as the route: an **intracardiac baffle tunnels the left ventricle through the VSD to the aorta** (so the LV becomes the systemic pump), and a **valved conduit connects the right ventricle to the pulmonary artery** (bypassing the obstructed native outflow). This is a **two-ventricle (biventricular) repair**.
- **Reads as (the causal chain, for the trap):** D-TGA + a large VSD + severe PS = there ARE two adequate ventricles and a VSD to route the LV to the aorta, so you can build a two-ventricle repair (Rastelli), NOT a single-ventricle Fontan. A univentricular / Fontan pathway (C6) is only for hearts with ONE usable ventricle, which this is not.
- **Why it matters (the whole of Q17): Q17** = "regarding adult D-TGA, all are true EXCEPT," and the false statement is **B**, which claims that D-TGA with severe pulmonary stenosis and a large VSD is treated by a **univentricular (Fontan-type) repair**. It is NOT: the correct operation is the **Rastelli procedure** (LV-to-aorta baffle through the VSD + an RV-to-PA valved conduit), a two-ventricle repair. The true statements describe the atrial switch leaving a systemic morphologic RV that fails late (with baffle and arrhythmia problems) and the arterial switch as the modern anatomic repair. So **B is the odd one out (answer B)**.
- **Hook:** "Adult D-TGA is defined by WHICH operation: atrial switch (Mustard/Senning) = a baffle that leaves the morphologic RV as the systemic ventricle, so it FAILS late (baffle leak/obstruction, atrial arrhythmia); arterial switch (ASO) = the modern anatomic fix (LV becomes systemic, coronaries reimplanted); and TGA + VSD + LVOTO/PS = the RASTELLI (LV-to-aorta baffle through the VSD + an RV-to-PA conduit), a TWO-ventricle repair. Q17: it is Rastelli, NOT a univentricular / Fontan repair (answer B)."

4) Eisenmenger syndrome: the big shunt that raised the PVR above systemic and then reversed (the payoff of the C1 PVR concept)

- **Synonym:** Eisenmenger syndrome / reaction / physiology; pulmonary vascular obstructive disease with shunt reversal.
- **The core fact first:** take any **large, long-standing left-to-right shunt** (a big non-restrictive **VSD**, a large **ASD**, a big **PDA**, or a complete **AVSD**) and leave it unrepaired. The shunt drives an abnormally **high pulmonary blood flow at high pressure** through the lungs, and the pulmonary arterioles respond by remodelling and constricting. Over years the **pulmonary vascular resistance (PVR) rises progressively and becomes fixed**, until it finally climbs **ABOVE the systemic vascular resistance**. At that moment the pressure gradient across the defect flips, and the **shunt REVERSES from left-to-right to right-to-left**.

Deoxygenated blood now crosses to the systemic side, and the patient who was pink for years becomes **cyanotic**. That end-stage is Eisenmenger syndrome.

- **Reads as (the causal chain):** a large left-to-right shunt to chronic high pulmonary flow and pressure to pulmonary arteriolar remodelling to PVR rises and fixes to **PVR climbs above systemic resistance** to the shunt REVERSES to right-to-left to systemic desaturation to **central cyanosis, digital clubbing, and secondary erythrocytosis** (the marrow makes more red cells to carry oxygen, thickening the blood and adding a hyperviscosity and thrombosis risk). This is the direct clinical destination of the [[C1]] PVR concept: the transpulmonary-gradient-over-flow number you learn to calculate in C1 is exactly the number that, once it crosses the systemic line, defines Eisenmenger and decides operability.
- **Why it matters (the management rules, which are the exam points):**
- **Eisenmenger is INOPERABLE.** Once the shunt has reversed, the defect is the only **pressure pop-off** for the fixed, high-resistance right heart. **Closing the shunt is FATAL** (acute right-ventricular failure), so surgical or device closure is contraindicated. The operability window closed when the PVR became fixed and supra-systemic; this is precisely why large shunts must be repaired EARLY, before the pulmonary vascular disease becomes irreversible.
- **Pregnancy is contraindicated** (very high maternal mortality; the fall in systemic resistance in pregnancy worsens the right-to-left shunt and deepens cyanosis).
- **Management is supportive plus pulmonary vasodilators plus, ultimately, transplant.** Supportive care (avoid dehydration, high altitude, and vasodilating triggers; careful management of the erythrocytosis; endocarditis and thrombosis precautions), **pulmonary arterial vasodilator / hypertension therapy** (endothelin receptor antagonists, PDE-5 inhibitors, prostacyclins) to lower PVR and improve function, and in the end stage a **heart-lung (or lung + defect-repair) transplant**.
- **Example / exam anchor:** Eisenmenger is not tagged to a numbered question here, but it is core adult-CHD teaching and the reason the whole module hammers PVR. The testable spine is: **large unrepaired left-to-right shunt to fixed pulmonary vascular disease to shunt reversal to cyanosis + clubbing + erythrocytosis to INOPERABLE (closing it is fatal), no pregnancy, treat with vasodilators / transplant.**
- **Hook:** "Eisenmenger = a big unrepaired left-to-right shunt (VSD/ASD/PDA/AVSD) that drove the PVR ABOVE systemic, so the shunt REVERSED to right-to-left: cyanosis, clubbing, erythrocytosis. Now the defect is the right heart's only pop-off, so it is INOPERABLE (closing it kills the patient), pregnancy is contraindicated, and you can only support, give pulmonary vasodilators, and ultimately transplant. It is the [[C1]] PVR number crossing the systemic line."

5) Sign-to-lesion matching: read one physical sign back to its lesion (this is Q14)

- **Synonym:** physical-exam-to-diagnosis matching; the "match the auscultation / peripheral sign to the congenital lesion" question.
- **The core fact first:** most classic congenital lesions announce themselves with **one signature physical finding**, and Q14 is a match-the-column question that tests whether you can pair the sign with its lesion. The clean, defensible pairs to lock in:
- **A FIXED, widely split second heart sound (S2) = a secundum ASD** (the left-to-right shunt keeps the right ventricle volume-loaded through the whole respiratory cycle, so the split does not vary with breathing).
- **A continuous "machinery" murmur (loudest under the left clavicle) = a PDA** (flow crosses the duct in both systole and diastole).
- **A harsh HOLOSYSTOLIC (pansystolic) murmur at the lower-left sternal border = a VSD** (also the site of tricuspid regurgitation).

- **A systolic ejection murmur with an ejection CLICK at the upper-LEFT sternal border = valvar pulmonary stenosis** (from C5; the click is the domed mobile valve popping open).
- **Upper-limb hypertension with weak, delayed femoral pulses (radio-femoral delay), plus a murmur over the back = coarctation of the aorta** (from C5).
- **A pulsatile, enlarged liver (hepatomegaly with systolic pulsation) + raised JVP with a large v wave = severe tricuspid regurgitation / right-heart failure** (the regurgitant wave is transmitted back into the systemic veins and liver).
- **Central cyanosis with digital clubbing = a chronic right-to-left shunt / Eisenmenger** (concept 4).
- **Reads as (the logic of a matching question):** each row pairs a sign with a lesion; four rows pair correctly and ONE row deliberately mismatches (a sign attached to the wrong lesion). Your job is to find the mismatched row.
- **Why it matters, and the honest limit on this one (the whole of Q14): Q14** = "regarding the matching of physical-exam findings to congenital lesions, all are true EXCEPT," and the provided **course key says E**. Per the key, the flagged mismatch involves the **holosystolic murmur / ejection click / hepatomegaly set**, which does not cleanly pair with the lesion the option attaches it to. Because Q14 is a **match-the-column item whose exact option wording is not reproduced verbatim in the extracted signal**, I can tell you the answer key is **E** and that the intended trap is a mispaired sign, but I **cannot reconstruct the precise mismatched pair with full certainty from the condensed text**. So: **learn the clean sign-to-lesion pairs above (they are what the question rewards), take the keyed answer as E, and confirm the exact wording of option E with the course** before relying on the specific mismatch. Do not overstate certainty on the internal wording of this one.
- **Hook:** "Sign-to-lesion: FIXED split S2 = ASD; machinery murmur = PDA; HOLOSYSTOLIC at the lower-left sternal border = VSD; ejection CLICK at the upper-left sternal border = pulmonary stenosis; radio-femoral delay = coarctation; PULSATILE liver + big v wave = severe TR. Q14 is a match-the-column question; the key says the mismatched (EXCEPT) row is E, built around the holosystolic / click / hepatomegaly set, but confirm the exact wording with the course (answer E per key)."

Step 3. EXAMINE yourself (questions only; answers are sealed below, do not peek)

1. In a repaired-TOF adult, what is the dominant LATE lesion, what specifically caused it during the original operation, and what does it do to the right ventricle over time?
2. Q1 logic: why is "the later the repair the better" FALSE, and what is the timed intervention that treats the late lesion?
3. An adult with D-TGA presents with a failing SYSTEMIC ventricle that is a morphologic RIGHT ventricle. Which operation did they have, and name two other late problems it leaves.
4. Q17 logic: for D-TGA with a large VSD and severe pulmonary stenosis, what is the correct operation, and why is a univentricular (Fontan) repair the WRONG answer?
5. Eisenmenger: walk the chain from a large left-to-right shunt to a cyanotic patient, and state the single haemodynamic event that defines the syndrome.
6. Why is Eisenmenger INOPERABLE, and what happens if you close the shunt anyway?
7. Q14 logic: give the clean sign-to-lesion pair for a FIXED split S2, a continuous machinery murmur, and a pulsatile liver, and say why you would still confirm option E's exact wording with the course.

▶ 🔒 Sealed answer key (open only after you have answered all seven)

C7 owns Q1 (repaired-TOF adult, answer A = "the later the repair the better" is the false statement; the late lesion is chronic pulmonary regurgitation from the transannular patch, treated by a timed pulmonary valve *replacement), Q14 (physical-exam-to-lesion matching, answer E per the key, a match-the-column item whose exact option wording should be confirmed with the course), and Q17 (adult D-TGA, answer B = TGA + VSD + severe PS is a Rastelli, NOT a univentricular / Fontan repair). Eisenmenger is taught here as core adult-CHD material and the clinical payoff of the C1 PVR concept, though it is not tagged to a numbered question. This is the final CHD lesson (C1 to C7).*

C7 - Interactive visual (rendered): diagrams & schematics

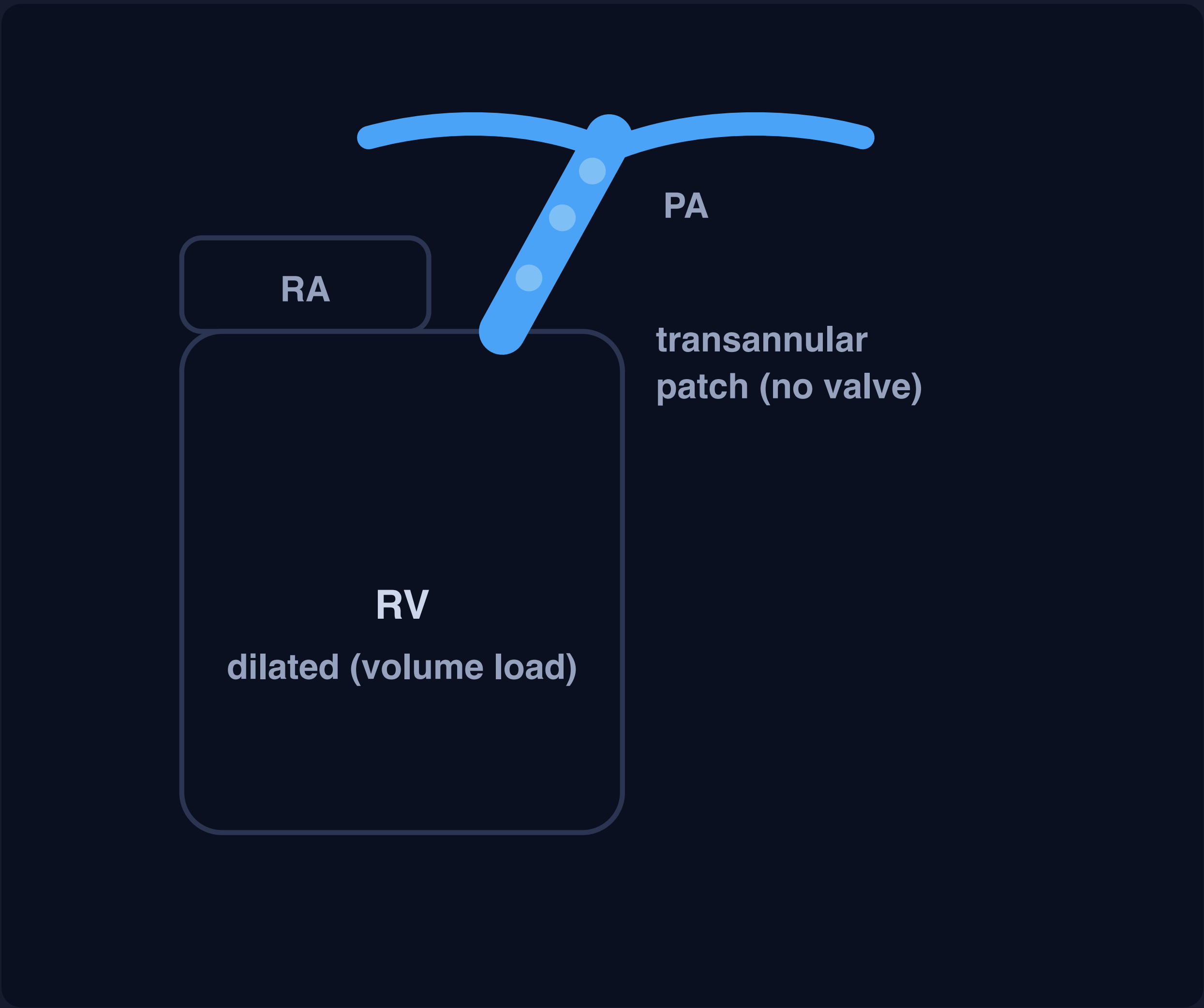
The live, toggleable version is the HTML file on the CHD hub. Below is a static render of its default view; open the hub to step through the other toggle states.

C7 - Adult congenital heart disease (the LATE lesion the repair left behind)

A childhood repair is a trade, not a cure: the adult presents with the residual or late lesion. Three archetypes carry the chapter. Repaired TOF to chronic pulmonary regurgitation (Q1). Adult D-TGA defined by WHICH operation, and TGA + VSD + PS = Rastelli (Q17). Eisenmenger = a big shunt that raised the PVR above systemic and reversed. Plus the sign-to-lesion matching set (Q14).

1. REPAIRED TOF: THE TRANSANNULAR PATCH CURES THE OBSTRUCTION BUT LEAVES FREE PULMONARY REGURGITATION (Q1)

Systole (RV to PA) | Diastole (free PR: PA to RV)



The patch widened the outflow but removed the valve, so in diastole blood falls back into the RV every beat. Over years that volume load dilates then fails the RV.

Systole: RV to PA

In systole the RV ejects forward through the widened, valveless outflow into the pulmonary artery. The patch did its job: **no more obstruction**.

The problem only shows in diastole (toggle it).

Q1: about TOF repair in adults, all true EXCEPT A ("the later the repair the better"). EARLIER primary repair gives better outcomes. The dominant late lesion is chronic pulmonary regurgitation to RV dilation to dysfunction to arrhythmia / sudden-death risk; treat with a timed pulmonary valve replacement (PVR) before the RV is too far gone.

2. ADULT D-TGA: WHICH OPERATION DEFINES THE ADULT PROBLEM (Q17)

Atrial switch
(Mustard / Senning)

systemic = morph RV

- Baffle redirects blood at the **atrial** level
- The **morphologic RV pumps to the body** to **fails late**
- Baffle **leak / obstruction**, atrial arrhythmia / sinus node disease
- The historic repair

Arterial switch
(ASO)

systemic = LV (anatomic)

- Great arteries **switched back**, coronaries **reimplanted**
- The **LV is the systemic pump** (correct)
- Late: neo-aortic root dilation / regurgitation, coronary ostial or branch-PA stenosis
- The modern repair

Rastelli
(TGA + VSD + PS)

two-ventricle repair

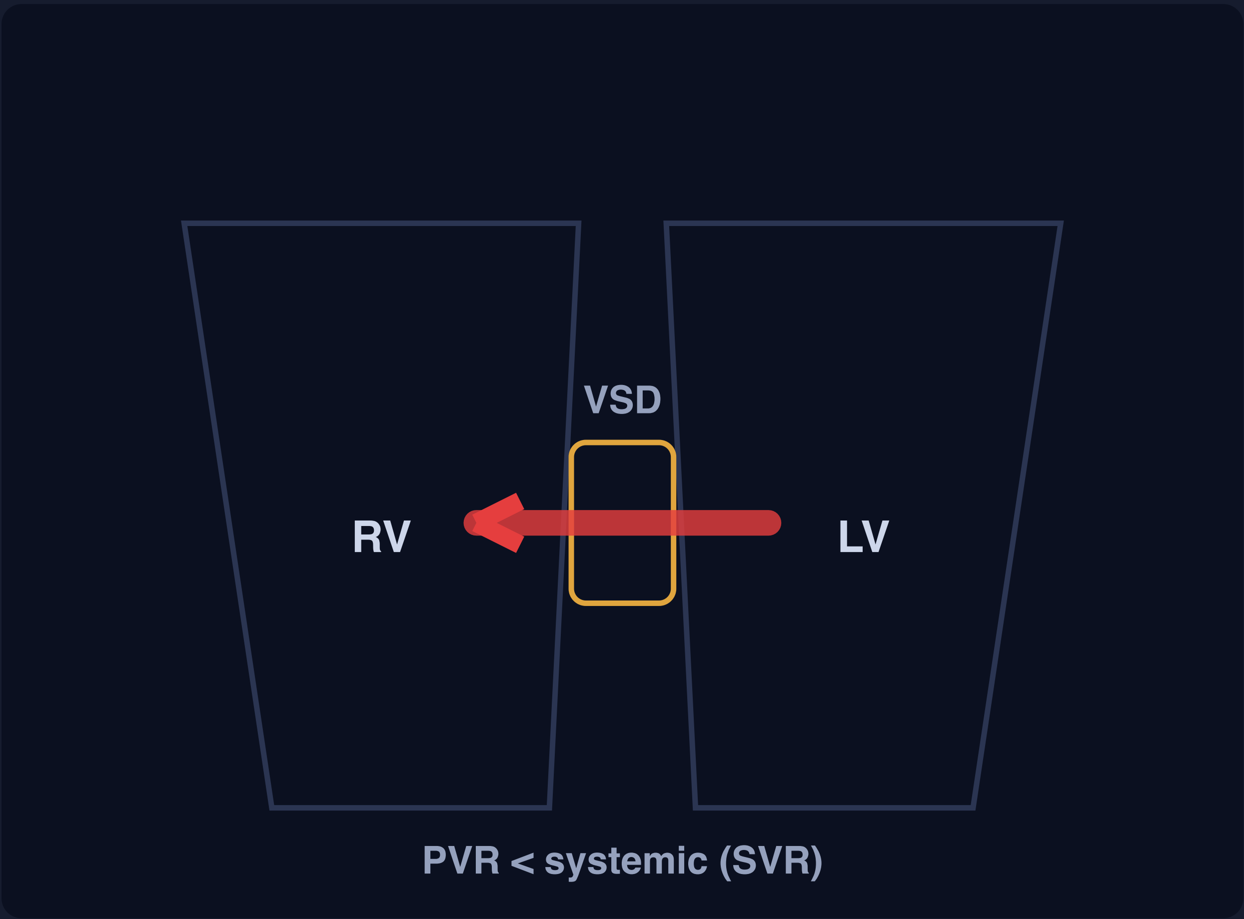
- For D-TGA with a **large VSD + LVOTO / PS**
- **LV to aorta** baffle through the VSD
- **RV to PA** valved conduit
- A **biventricular** repair (two good ventricles)

Q17: about adult D-TGA, all true EXCEPT B. D-TGA with severe PS + a large VSD is repaired by the Rastelli procedure (LV-to-aorta baffle through the VSD + an RV-to-

PA conduit), NOT a univentricular / Fontan repair: this heart has two adequate ventricles, so a Fontan (single-ventricle pathway, C6) does not apply.

3. EISENMENGER: A BIG SHUNT RAISES THE PVR ABOVE SYSTEMIC, SO IT REVERSES (TOGGLE THE TWO STAGES)

Early: L to R shunt (pink) | Eisenmenger: R to L (cyanotic)



The direction of flow across the VSD is decided by which resistance is higher. When the pulmonary vascular disease pushes PVR above systemic, the shunt flips to right-to-left.

Early: left-to-right shunt (pink)

While the **PVR is still below systemic**, blood shunts **left-to-right** (LV to RV through the VSD). The extra pulmonary flow keeps the patient **acyanotic (pink)**, but it is slowly damaging the pulmonary arterioles.

Once the shunt has REVERSED, the defect is the right heart's only pressure pop-off, so Eisenmenger is INOPERABLE (closing it is fatal), pregnancy is contraindicated, and management is supportive + pulmonary vasodilators +

transplant. This is the C1 PVR number crossing the systemic line. Repair big shunts EARLY, before the PVR fixes.

4. SIGN-TO-LESION MATCHING: READ ONE PHYSICAL SIGN BACK TO ITS LESION (Q14)

PHYSICAL SIGN	LESION
Fixed, widely split S2	Secundum **ASD** (RV volume-loaded through the whole respiratory cycle)
Continuous "machinery" murmur	**PDA** (flow across the duct in systole and diastole)
Holosystolic murmur at the LLSB	**VSD** (also the site of tricuspid regurgitation)
Ejection click, upper-LEFT sternal border	Valvar **pulmonary stenosis** (domed mobile valve, from C5)
Radio-femoral delay + murmur over the back	**Coarctation** of the aorta (from C5)
Pulsatile liver + large JVP v wave	Severe **tricuspid regurgitation** / right-heart failure
Central cyanosis + clubbing	Chronic right-to-left shunt / **Eisenmenger**

Q14 is a match-the-column question: all pairs true EXCEPT E per the key (the flagged mismatch is built around the holosystolic / click / hepatomegaly set). The exact option wording is not in the extracted signal, so learn the clean pairs above and confirm the precise wording of option E with the course rather than overstate certainty.

C7 visual - Module 5 CHD. Hand-drawn inline SVG + tables, self-contained. Final CHD lesson (C1 to C7).

Mock Exam - Module 5 Master Exam (CHD, 20 Q)

Answer all, then check the sealed key on the last page. Note: the 2018 Congenital paper swapped Q20 for a written question, "components required to calculate PVR" (mean PA pressure, mean LA pressure / PCWP, and pulmonary blood flow Qp; PVR = transpulmonary gradient / Qp, in Wood units).

1.Regarding TOF repair in adults all is true except :
A)the later the repair the better the clinical outcome
B)Determining coronary anatomy is important before repair
C)RV dilatation and hypertrophy may not regress after repair
D)If they complain of palpitations after the repair it should be taken seriously and further investigations to diagnose the type of arrhythmia is important

2. All the following are prerequisites for performing a
glenn
shunt EXCEPT
A. A mean pulmonary artery pressure of 14 mmHg
B. PVR more than 3 woods unit
C. A LV end-diastolic pressure less than 12 mmHg
D. Confluent sizable pulmonary arteries

3. A 52 year old woman presents with shortness of breath , her ECG shows RAD,RVH her echo revealed dilated RV and a high normal PAP, the most probable diagnosis is
A)primum ASD
B)Secundum ASD
C)TOF
D)Truncus arteriosus

4.Regarding PDA in adults all is true except :
A)a restrictive PDA indicates a normal PAP
B)Infective endarteritis is a dreadful complication
C)All PDAs should be closed even silent ducts
D)Small PDAs should be closed
percutanously

5. Congenital cyanotic heart diseases with increased pulmonary blood flow include all the following except:
A. Tricuspid atresia with ASD and VSD
B. D-TGA with VSD and severe pulmonary stenosis
C. Single ventricle with malposed great vessels
D. Truncus arteriosus type 1

6. An infant with a marked cyanotic congenital heart defect with decreased pulmonary vascularity should be initially treated with:
A. Digoxin
B. Indomethacin
C. Prostaglandin E1
D. Epinephrine

7. A 3-month-old girl with trisomy 21 exhibits poor weight gain, tachypnea and a low-pitched grade 2 systolic murmur over the lower left sternal border. Her CXR shows cardiomegaly and her ECG shows left axis deviation. The most likely diagnosis is:
A. Coarctation of aorta
B. Complete AV septal defect
C. PDA
D. VSD
E. Secundum ASD
8-A "
tet
spell" or "blue" spell of tetralogy of
fallot
is treated with all of the following except:
a. oxygen
b. knee chest position
c. morphine
d. digoxin
e. propranolol
f. sodium bicarbonate

9.All of the following are indications for VSD closure except :
A.Dilated
left ventricle
B.Aortic
valve prolapse and aortic regurgitation
C.A peak pressure gradient across the VSD of 100
mmHG
D.Failure
to thrive
10 .All of the following are possible postoperative complications for a
glenn
shunt except?
A.SVC syndrome
B.Thrombus
formation in the SVC
C.Veno
-venous collaterals
D.Pulmonary
AV malformations
E.Protein
losing enteropathy

11. Regarding the diagnosis of a vascular ring, all are true EXCEPT:
A. Vascular ring can be suspected when the child shows breathing and feeding difficulties
B. Barium
esophageogram
is the most useful noninvasive tool
C. CT and MRI are very accurate for the diagnosis
D. Echocardiography can provide detailed anatomic information about the vascular anomaly

12. In fetal circulation, which vessels carry the highest oxygenated blood?
A. aorta
B. umbilical arteries
C. vitelline veins
D. umbilical veins
E. cardinal veins

13. All of the following associations with congenital heart diseases are true, EXCEPT:
a) ASD and
triphalyngeal
thumb
b) Infant of a psychiatric mother -
Ebstein
anomaly
c) Maternal rubella and infant PDA
d) Epileptic mother and interrupted aortic arch in the neonate
e) Noonan's syndrome and mitral stenosis

14. All the following physical examination findings match with the corresponding adult congenital heart disorder, EXCEPT?
a) Right ventricular (RV) lift with a loud systolic ejection murmur along the left sternal border, with a single S2
□
Fallot's tetralogy
b) Systolic ejection click, loud S1, holosystolic murmur, split S2 and hepatomegaly
□
VSD
c) Weak or delayed femoral arterial pulses, harsh systolic ejection murmur in the back, and a systolic ejection click in the aortic area
□
Coarctation of aorta
d) Cyanosis, digital clubbing, loud P2, and a variable Graham
Steell
murmur
□
Eisenminger
syndrome
e) Wide pulse pressure, prominent left ventricular (LV) impulse, and a continuous machinery murmur enveloping S2
□
PDA

15. You are asked to see in clinic a 6-year-old girl with a diagnosis of right atrial isomerism. Which one of the following features would you expect her to have?
A- Asplenia
B-
Polysplenia
C- Two functional left lungs
D- T- cell deficiency
E- Trisomy 21

16. Each of the following is associated with
Ebstein
anomaly of the tricuspid valve except
A- Atrial septal defect
B- Paradoxical splitting of S2
C- Ventricular
preexitation
D- A widely split S1
E- Atrial Flutter

17. Regarding Patients with DTGA who present in the adult life, all is true except
A) May present with heart failure symptoms
B) Univentricular repair may be an option if there is severe PS and a large VSD
C) Rastelli operation can be done for DTGA -VSD -severe PS patients
D) Arterial switch can be done for the adult DTGA

18. You are called to perform an echocardiogram on a neonate with cyanosis. The patient is mildly tachypneic with retractions, O2 saturation is 65%. Chest x- ray shows no increase in cardiothoracic ratio with increase pulmonary vascular markings. On examination, you appreciate a loud S2 with no significant murmurs. Echocardiography shows side by side great arteries. Which of the following diagnosis is most likely?
a. DORV with subpulmonic VSD and no pulmonary stenosis
b. DORV with subpulmonic VSD and pulmonary stenosis
c. DORV with subaortic VSD and no pulmonary stenosis
d. DORV with subaortic VSD and pulmonary stenosis
e. DORV with subaortic VSD and
suprasystemic
pulmonary hypertension
19- A 6-hour-old neonate is seen by the neonatologist in the normal newborn nursery. The newborn has developed cyanosis which worsens with crying. The neonate was born at 37 weeks' gestation by cesarean section for non reassuring heart tones to a 33-year-old woman whose pregnancy was complicated by preterm labor. Apgar scores were and 8 and 9 at 1 and 5 minutes, respectively. He was placed skin-to-skin and breast-fed well shortly after birth. On physical examination vital signs and growth parameters are normal. Head is normocephalic. Fontanelles are flat. Chest has good air movement without distress. Heart has normal S1 and S2. No murmur is heard. Distal pulses are normal. Capillary refill is less than 3 seconds. Abdomen is soft and nontender. No hepatosplenomegaly is noted. An ECG shows an axis of 120° and right ventricular prominence. The chest radiograph is shown in the photograph. Which of the following is the most likely diagnosis?
A Tetralogy of Fallot
B Transposition of the great vessels
C Tricuspid atresia
D Pulmonary atresia with intact ventricular septum
E Total anomalous pulmonary venous return below the diaphragm

20. A distressed 3 months old infant with ASD shunting right to left and markedly dilated right side, this was his TTE (suprasternal view), what is your diagnosis and management?
a Patient has persistent left SVC with ASD for medical treatment including diuretics and follow up till age of 2 years
b Patient has large PDA causing severe pulmonary hypertension, for urgent surgical ligation
c Patient has TAPVD (vertical vein) for hospitalization, medical treatment and MSCT before surgical repair
d Patient has severe pulmonary hypertension and should undergo hemodynamic study before ASD surgical closure
e Patient has severe aortic coarctation complicated by severe pulmonary hypertension, for MSCT and palliative balloon dilatation of aortic coarctation

Mock Exam - Answer Key

Try every question first.

`1. A 2. B 3. B 4. C 5. B 6. C 7. B 8. D 9. C 10. E 11. D 12. D 13. E 14. B 15. A 16. B 17. D 18. A 19. B 20. C`

Key-vs-clinical conflicts to confirm with the course: Q15 (right atrial isomerism: key says polysplenia, standard teaching = ASPLENIA) and Q16 (Ebstein EXCEPT: key says ASD, but ASD is strongly associated, so the defensible odd-one-out is paradoxical S2 splitting).